Department of the Air Force

Hazardous Waste Management Plan

Minot_HWMP

Installation Supplement



About This Plan 4

Document Control 4

HWMP Approval (Signature Page) 5

1 Overview and Scope 5

2 Installation Profile 6

3 Environmental Management System 7

4 Roles and Responsibilities 7

5 Training 13

6 Recordkeeping and Reporting 15

7 Procedures 20

7.1 Waste Inventory 20

7.2 Waste Identification 22

7.3 Container Management 22

7.4 Labeling and Marking 25

7.5 Accumulation Area Management 30

7.6 Transportation 36

7.7 Turn In/Disposal 37

7.8 Inspection 39

7.9 Waste Minimization/HW Recycling 39

7.10 Preparedness and Prevention 39

7.11 Waste Specific Procedures 40

8 References 48

9 Acronyms 48

10 Definitions 48

11 Installation-Specific Content 48

Appendix A Waste Analysis Plan 50

ABOUT THIS PLAN

This installation-specific Environmental Management Plan (EMP) uses the U.S. Air Force's (AF) standardized Hazardous Waste Management Plan (HWMP) template. This plan is not an inventory of all hazardous waste (HW) requirements and practices. Where applicable, external resources and associated document links, including Department of the Air Force Instructions (DAFIs); Department of the Air Force Manuals (DAFMANs); AF Playbooks; and federal, state, local, and permit requirements, are referenced to ensure current content.

Each section of this plan begins with standard language that addresses AF and Department of Defense (DoD) policy and federal requirements. The standard language is restricted from editing to ensure consistent application across the AF enterprise. Standard language is maintained by the Air Force Civil Engineer Center (AFCEC) designated Subject Matter Expert (SME) for this plan.

Immediately following the standard text are installation-specific sections that address state, local, and installation-specific requirements and processes. Installation sections are maintained and updated by the installation HW Program Manager and/or the AFCEC Section appointed to support this installation.

This document is optimized to be accessed and viewed electronically on the installation and AF eDASH website, the primary communication tool for AF EMPs. Installations should refer to the AF T-EMP Repository on eDASH to ensure the Approved/Published Plan is referenced.

DOCUMENT CONTROL

Standardized HWMP Template

In accordance with (IAW) the AFCEC Environmental (CZ) Business Rule (BR) 08, *EMP Review, Update, and Maintenance,* the standard content in this HWMP template is reviewed periodically, updated as appropriate, and approved by the HW SME.

This version of the template is current as of 12/23/2025 and supersedes the 10/02/2024 version.

NOTE: When installations update their HWMPs, installations should adopt the most recent version of this template available. Installations are not required to change HWMPs after template updates.

Installation HWMP

The initial HWMP must be approved and signed by the Installation Commander (at the time of publication) as the legal HW Generator for the Environmental Protection Agency (EPA) ID assigned to this installation. The approval can be via signature or as documented in appropriate Environment, Safety, and Occupational Health Council (ESOHC) minutes (DAFMAN 32-7002, 5.3.2.). NOTE: This is not a Wing Contingency Plan and is not governed by installation or Wing plans or readiness format or coordination requirements of DAFI 10-401, *Operations Planning and Execution*. It is governed by DAFMAN 32-7002, *Environmental Compliance and Pollution Prevention*, Section 5.2. *Hazardous Waste Management Plan (HWMP)*. Because this is not a contingency plan, coordination should be with EMS CFT and applicable units with roles and responsibilities in the HWMP. The plan does not require a new signature after each change in wing command.

Record of Updates – The HWMP is updated as changes to waste generation and management practices occur, including those driven by changes in applicable regulations and approved by the installation HW Program Manager as the plan Office of Primary Responsibility (OPR). Formatting and administrative changes such as the incorporation of updated regulatory language, or other minor updates are documented below.

Record of Annual Review – IAW DAFMAN 32-7002, this plan is reviewed annually by the EMS CFT. Formatting and administrative changes should be noted in the previous paragraph for record of updates as approved by the EMS CFT Chair and do not require Installation Commander or Environmental, Safety, and Occupational Health Council (ESOHC) approval. Substantive revisions require coordination and approval by the Installation Commander as determined by the EMS CFT Chair and/or IAW host installation procedures. Substantive changes include manpower or resource requirements changes that impact installation hazardous waste generating organizations.

Installation Supplement

Record of Updates

Change No.	Nature of Change	Date of Change	Approved By:
1	Administrative updates	9 Sep 2021	Thomas Filkins
2	Administrative updates	22 Aug 2021	Marie Gagnier
3	Administrative updates	1 Sep 2022	Marie Gagnier
4	Administrative updates	1 Sep 2023	Marie Gagnier
5	Administrative updates	16 Aug 2024	Naomi Winburn/Jennifer Hensley
6	Addition of the QRG and other LQG corrections	16 Aug 2024	Naomi Winburn

Record of EMS CFT Annual Review

Review Date	EMS CFT Chair	Notes/Remarks
19 OCT 2018	ESOHC	Accepted/No update needed
18 SEP 2019	EMS CFT	Accepted/No update needed
16 SEP 2020	EMS CFT	Accepted/No update needed
15 Sep 2021	EMS CFT	Accepted/No update needed
14 Sep 2022	EMS CFT	Accepted/No update needed
20 Sep 2023	EMS CFT	Accepted/No update needed
18 Sep 2024	EMS CFT	Accepted/No update needed

HWMP APPROVAL (SIGNATURE PAGE)

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Approved by the CFT on Sept 18, 2024

Digital Signature

WINBURN, NAOMI R CIV USAF AFGSC 5 CES/CEIEC	WINBURN, NAOMI R CIV USAF AFGSC 5 CES/CEIEC Date: 09/30/2025 3:20:35 pm
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1 OVERVIEW AND SCOPE

This HWMP contains procedures for management of HW. In lieu of federal, or state requirements, DAFMAN 32-7002, acts as the main driver for the HWMP. The HW Playbook serves as supplemental guidance to this plan. Where applicable, DoD, AF, and Federal Resource Conservation and Recovery Act (RCRA) requirements are included. Each installation must include supplements as warranted to address applicable State and/or local HW requirements/procedures.

Installation Supplement

Instructions

- The long title of this plan is the Minot AFB Hazardous Waste Management Plan; the short title is HWMP.
- This document is unclassified.
- This document does not contain information affecting the national defense of the United States within the meaning of the Espionage Laws, Title 18, U.S. Code, Section 793 and 794.

Purpose

- The purpose of this Hazardous Waste Management Plan (HWMP) is to provide base personnel with a comprehensive reference relating to the policies and procedures implemented on Minot Air Force Base (AFB) to ensure hazardous wastes are characterized, handled, stored, and managed in a manner that is safe, cost effective, responsive to mission requirements, and compliant with applicable regulations. Because many specific procedures are identified, organizations should supplement this HWMP with more detailed instructions and checklists to assist shop-level operations as long as the shop instructions are as stringent as the HWMP and do not conflict with applicable regulations. References to the most pertinent regulations are included in this plan.
- This HWMP is the primary document for managing hazardous waste activities on Minot AFB and its associated geographically separated units (GSUs). The HWMP references federal, state, local, and U.S. Air Force requirements that establish compliance requirements. If there is a conflict between the HWMP and regulations, the regulatory requirements shall take precedence. In some instances, the HWMP directs Minot AFB waste management requirements that are more stringent than federal and state regulations.
- The HWMP shall be used by any tenant organization or other entity that generates or uses hazardous or potentially hazardous waste on Minot AFB and its GSUs. The HWMP shall be used by all activities, and tenant organizations that generate, and use hazardous waste or potentially hazardous waste on Minot AFB and its GSUs. Procedures and policies outlined in this plan are designed to meet the needs of hazardous waste generators and to assist compliance with applicable federal, state, and local laws and regulations.
- Compliance with this plan is essential. Failure to comply with this plan and associated regulations results in the base being Enforcement Action Vulnerable (EAV), which can result in a Notice of Violation, fines, and/or criminal prosecution of the negligent individual(s) responsible for the non-compliance.
- The success of Minot AFB's hazardous waste management program depends on a team effort and the total commitment of all involved units and individuals. Accordingly, anyone who observes improper hazardous waste management practices or who requires clarification regarding proper hazardous waste management and handling should notify the installation Hazardous Waste Program Manager, Environmental Management Element, 5 CES/CEIEC, at 723-4825/6380.
- A list of emergency notification phone numbers is provided in the appendices of this plan.

Conditions for Implementation

- **Statement.** This summary provides a brief overview of the scope and objectives of the HWMP, which is based on the federal, state, and local regulations and directives available at the time of preparation. It is subject to modification as those requirements change. Accordingly, the HWMP will be reviewed annually and updated.
- **Political-Military Situation.** Compliance with this plan as it relates to the management of hazardous waste is required for all readiness conditions.
- **Legal Considerations.** Management of hazardous waste on Minot AFB will adhere to applicable federal, state, and local regulations and Air Force directives without exception.
- **Environmental Considerations.** The key purpose of this plan is to ensure compliance with environmental laws and regulations.

Operations to be Conducted

Operations to be conducted encompass the management of hazardous waste in a manner that is responsive to mission requirements while being consistent with the crucial need to ensure personnel safety, cost effectiveness, and strict compliance with applicable laws and regulations.

To ensure strict compliance with applicable federal, state, and local regulations, hazardous waste on Minot AFB is managed in accordance with large quantity generator (LQG) rules. Specifically, RCRA hazardous waste is shipped off the installation in less than 90 days, shop personnel are trained annually, and emergency spill response procedures are the those of an LQG.

2 INSTALLATION PROFILE

Installation Supplement

Installation Profile

Scope of Plan	The HWMP shall be used by all activities, and tenant organizations that generate, and use hazardous waste or potentially hazardous waste on Minot AFB and its GSUs. This plan does not apply to municipal solid waste and radioactive waste management, but it does apply to mixed radioactive/hazardous wastes.
OPR (Civil Engineer or other designated office)	5 CES/CEIE has overall responsibility for implementing the HW management program and is the lead organization for monitoring compliance with applicable federal, state, and local regulations.
HW Program Manager	Naomi Winburn
Alternate HW Program Manager	Thomas Filkins
Emergency contacts	911
Waste registration numbers	EPA Identification Number ND4571924758
HW generator status	Large Quantity Generator (LQG)
Universal waste handler status	
Permitted HW operations	
Federal regulatory references	
State and local regulatory agencies	North Dakota Department of Environmental Quality 1-701-328-5150 Local Emergency Planning Committee 1-701-857-6534 State Emergency Response Commission 1-701-328-8100 North Dakota State Department of Health 1-701-328-2310
State and local regulatory references	North Dakota Century Code, Chapter 23.1-04 North Dakota Administrative Code (NDAC), Article 33.1-24
DLA-DS area office and/or approved HW disposal contractors	DLA Disposition Services
HW accumulation sites	82
HW accumulation time limits	90 Days
HW generator reporting frequency	Annually

3 ENVIRONMENTAL MANAGEMENT SYSTEM

The DAF environmental program adheres to the Environmental Management System (EMS) framework and its "Plan, Do, Check, Act" cycle for ensuring mission success. DoDI 4715.17, *Environmental Management Systems*, DAFI 32-7001, *Environmental Management;* and International Organization for Standardization (ISO) 14001 standard, *Environmental Management Systems - Requirements with guidance for use*, provide guidance on how environmental programs should be established, implemented, and maintained to operate under the EMS framework.

The HW management program employs EMS-based processes to achieve compliance with all legal obligations and current policy drivers, effectively manage associated risks, and instill a culture of continuous improvement. The HWMP serves as an administrative operational control that defines compliance-related activities and processes.

Environmental Action Plans (EAPs) are developed and maintained as part of the overall EMS on eDASH. EAPs are tools that translate environmental requirements and targets into actionable plans for the HWMP and responsible personnel. HW requirements should be incorporated in installation EAPs to record monitoring and conformance for inspections and audits.

4 ROLES AND RESPONSIBILITIES

The major roles/organizations involved in supporting the HW program include:

Wing/Installation Commander

- ESOHC
- EMS CFT
- HW Program Manager/Alternate(s)

- Shop/HW Generator Personnel
- Initial Accumulation Point (IAP) and Hazardous Waste Accumulation Site (HWAS) Supervisors/Managers
- Unit Commanders
- Unit Environmental Coordinators (UECs), see DAFI 32-7001 for role description
- Contracting Officer
- Defense Logistics Agency (DLA) Disposition Services
- Tenant Organizations
- AFCEC

Detailed information about typical responsibilities for these and other roles is available in DAFMAN 32-7002, applicable installation supplements, and the HW Playbook. This plan implements these responsibilities for this installation. Additional HW management-related roles and responsibilities are described throughout this plan and in referenced documents.

Installation Supplement

Implementation

Implementation of the hazardous waste management program requires the full support and cooperation of all activities on Minot AFB. Each shop and activity that generates hazardous and/or potentially hazardous waste including industrial waste — hereafter referred to as the generator — is responsible for managing its own hazardous waste. In particular, each generator is accountable for complying with waste management regulations, coordinating disposal, and adhering to the policies and procedures prescribed by this plan. If questions arise that cannot be resolved at the shop level, the matter should be forwarded to Environmental Management for resolution. **When in doubt, ask!**

Roles and Responsibilities

The following details specific roles and responsibilities associated with the management of hazardous waste on Minot AFB and its associated GSUs.

Installation Commander (5 BW/CC). In accordance with AFMAN 32-7002, the Installation Commander:

- Is responsible for ensuring the installation is complying with hazardous waste disposal laws. Disposal of hazardous waste by the Defense Logistics Agency Disposition Services (DLA-DS), or by Air Force contractors does not relieve the Installation Commander of this responsibility.
- Monitors base compliance with applicable federal, North Dakota, and Air Force regulations through the ESOHC, which will periodically review the status of the hazardous waste management program on Minot AFB as it relates to planning, funding, prioritization of requirements, and the resolution of any outstanding issues.
- Is responsible for signing hazardous waste manifests, unless that responsibility has been specifically delegated.
- Ensures that a hazardous waste minimization program in place to reduce the volume and toxicity of waste generated, as required by RCRA Section 3002(b), 42 U.S. Code 6922 (b), and Executive Orders 13423, *Strengthening Federal Environmental, Energy, and Transportation Management*, and 13514, *Leadership in Environmental, Energy, and Economic Performance*.
- Has the authority to take whatever action is necessary to require installation personnel, tenant organizations and all contractors on Minot AFB to comply with the base's hazardous waste management program as detailed in this plan

Environmental, Safety, and Occupational Health Council (ESOHC). In accordance with AFI 32-7002 and AFI 90-801, the ESOHC will:

- Review and approve solid waste and hazardous waste policies that affect the installation's generator status and the Environmental Management System (EMS) significant aspects for hazardous waste generation.
- Review and sign the installation solid waste and hazardous waste management plans.
- Review solid waste and hazardous waste program goals to ensure they are appropriate and produce actions that provide necessary mission capability and capacity.
- Conduct management reviews to ensure waste management goals are met, implementation and operation procedures are followed, and Environment Inspection Process (EIP) assessment findings are tracked and completed.

Environmental Management Element (5 CES/CEIE). As the OPR for the Minot AFB hazardous waste management program, the Environmental Management Element will:

- Develop and maintain this HWMP in accordance with AFI 32-7002. Specifically, Environmental Management will update the plan as required to (1) reflect changes in applicable hazardous waste management regulations; (2) accommodate evolving mission requirements; (3) implement initiatives that based on experience will enhance efficiency and/or regulatory compliance; and (4) ensure plan accuracy and currency. At a minimum, Environmental Management will review this plan annually.

- Serve as the liaison for Minot AFB with regulatory agencies for matters relating to the management of hazardous waste on the installation and its associated GSUs. Environmental Management will communicate directly with EPA Region 8, the NDDH, DOT, and local government agencies to resolve issues, apply for hazardous waste permits, coordinate the preparation of reports, and report spill incidents as required by applicable regulations.
- Monitor hazardous waste disposal, recycling, and/or treatment to ensure such activities are in full compliance with applicable regulations. Accordingly, Environmental Management will periodically inspect hazardous waste operations to confirm shop hazardous waste is being properly characterized, managed, labeled, packaged, and stored in accordance with RCRA and Minot AFB waste minimization policies.
- Develop and maintain a Waste Analysis Plan (WAP) in accordance with AFI 32-7002 that describes the procedures used to (1) identify hazardous waste streams on Minot AFB and its associated GSUs and (2) designate those waste streams that require further analysis. The WAP is incorporated into this plan at Appendix A.
- Supervise development and maintenance of the Waste Stream Inventory (WSI) for Minot AFB and its associated GSUs by the Central Accumulation Point (CAP) contractor. The WSI is an integral part of the WAP and is essential to attaining waste minimization goals.
- Supervise hazardous waste sample collection as required and coordinate transportation of the samples to an approved analytical laboratory for analysis. In particular, Environmental Management will ensure the sampled waste containers are properly labeled with the suspected contents, the collection date, and a sample number. Upon receipt and review of the analytical results, Environmental Management will forward a copy of the analytical results to the generator directly or through the CAP contractor.
- Compile and archive documentation relating to the management of hazardous waste consistent with regulatory requirements, Air Force Records Information Management System (AFRIMS), and prudent management practice. This process includes preserving electronic and/or paper file copies of the following:
    - Manifests;
    - Hazardous waste profile sheets (Defense Reutilization and Marketing Services [DRMS] Form 1930);
    - Issue Release/Receipt Documents (Defense Department [DD]-1348-1A); o WSIs;
    - Inspection reports from EPA; NDDH, Waste Management Division; Air Force Global Strike Command (AFGSC), and other agencies;
    - Disposal documentation; and
    - Applicable hazardous waste management reference materials.
- Provide technical expertise, guidance, and training to base agencies concerning the operation and maintenance of the CAP and Satellite Accumulation Points (SAPs) to include hazardous waste profiling, marking, labeling, storage, disposal, documentation, record keeping, and the completion of waste turn-in documentation. Training requirements are further detailed in Section 5.0 of this plan.
- Budget for hazardous waste analyses, training, and equipment requirements. In particular, Environmental Management shall oversee expenditures and budgeting related to the disposal of hazardous waste. This extends to submitting funding requests to the Accounting and Finance Office (AFO), certifying funds availability for waste transfer to DLA-DS, reviewing billings received by the AFO for accuracy, tracking waste disposal costs to facilitate budgeting for future requirements, and associated record keeping.
- Coordinate hazardous waste disposal requirements with DLA-DS and/or affected contractors.
- Maintain generation and disposal records required to develop hazardous waste metrics as may be requested by higher headquarters and regulatory agencies.

Bioenvironmental Engineering (5 MDOS/SGPB). In support of the hazardous waste management program on Minot AFB, Bioenvironmental Engineering will:

- Provide industrial hygiene and occupational health consultant services to all hazardous waste generating facilities as required.
- Conduct health risk assessments at the CAP and SAPs in conjunction with hazardous waste spill response operations as required.
- Assist the Environmental Management Element in performing an annual review of the WAP per Appendix A of this plan as required.
- Provide technical expertise relating to the identification of hazardous waste as needed.
- As requested by the Civil Engineer Squadron, review plans to build or modify facilities used to treat, store, or dispose of hazardous waste including all proposals to establish or relocate hazardous waste collection points.

DLA Disposition Services (DLA-DS). The DLA-DS, formerly known as the Defense Reutilization and Marketing Office (DRMO), implements a three-tiered Operational Compliance Review (OCR) Program that includes evaluation of the hazardous waste disposal support provided by DLA-DS to Minot AFB. The OCR Program includes quarterly self-inspections conducted by DLA-DS, annual inspections by the DLA-DS Zones, and periodic, no-notice inspections by DLA. Results of the OCR inspections pertaining to hazardous waste management will be provided to Environmental Management to ensure any required corrective actions are implemented. In accordance with support agreements relating to the management of hazardous waste on Minot AFB, the DLA-DS will:

- Provide hazardous waste disposal services for Minot AFB and administer associated contracts. Specifically, the DLA-DS will arrange for a qualified contractor to transport hazardous waste to an EPA-permitted TSDF. The DLA-DS will monitor compliance with associated hazardous waste disposal contracts, ensure contractor compliance with the EPA Land Disposal Restrictions (LDR) per associated hazardous waste profile sheets and manifests, and ensure waste is properly prepared for transport in accordance with EPA and DOT regulations.
- Prepare and sign hazardous waste manifest shipping papers that accompany the hazardous waste shipments.

Distribution Flight (5 LRS/LGRM). In support of the hazardous waste management program, the Distribution Flight will:

- Support hazardous waste minimization efforts on Minot AFB by monitoring and controlling the procurement of hazardous materials through the hazardous materials pharmacy (HAZMART) and the regular Supply System.
- In accordance with AFI 32-7002, *Hazardous Materials Management*, ensure MSDSs or SDSs are provided for all hazardous materials issued by the HAZMART and the regular Supply System to facilitate hazardous material disposal and subsequent hazardous waste profiling and analysis.

Contracting Squadron (5 CONS). In support of the hazardous waste management program on Minot AFB, the Contracting Squadron will:

- Ensure only qualified contractors are awarded contracts involving the transportation and/or disposal of hazardous waste. The Contracting Squadron will coordinate hazardous waste disposal contracts with Environmental Management, the Staff Judge Advocate, and DLA-DS as needed to ensure hazardous waste disposal contracts are only awarded to contractors possessing the required federal and North Dakota qualifications and certifications.
- Ensure all contracts resulting in the generation or management of hazardous waste include language requiring contractor compliance with the provisions of this plan.
- Support the solicitation and issue of contracts for hazardous waste collection, transportation, analyses, and recycling as required by base agencies.
- Incorporate appropriate contract language to require contractors that generate hazardous waste on Minot AFB and its associated GSUs to comply with all applicable federal, state, and Air Force regulations and the provisions of this plan. In particular, it should be noted that the disposal of contractor-generated hazardous waste must be in accordance with the contract and coordinated through Environmental Management before removal from the installation. The disposal of non-hazardous waste is the responsibility of the contractor.

Fire Protection Flight (5 CES/CEF). In support of the hazardous waste management program, the Fire Protection Flight will:

- As required by base agencies, provide technical advice and training on the fire protection aspects of hazardous waste management.
- Periodically inspect all facilities where hazardous waste is generated and/or stored to ensure compliance with fire protection regulations.
- Retain a copy of the Site-Specific Contingency Plan for each shop containing a waste accumulation point as required.
- Obtain initial EMS Awareness training from Environmental Management as required for flight personnel and annual refresher training thereafter.

Safety Office (5 BW/SE). In support of the hazardous waste management program on Minot AFB, the Safety Office will:

- As requested, provide technical advice to base agencies on the safety aspects of hazardous waste management and storage.
- Provide adequate time during ESOHC meetings for Environmental Management to brief hazardous waste issues and compliance to attendees, and summarize the hazardous waste briefing in the ESOHC meeting minutes.

Staff Judge Advocate (5 BW/JA). In support of the hazardous waste management program on Minot AFB, the Staff Judge Advocate will:

- Provide interpretation of laws and regulations relating to the handling, storage, transport, management, and disposal of hazardous waste.
- Review environmental plans and procedures affecting the management of hazardous waste to ensure the plans and procedures are consistent with applicable federal, state, and Air Force regulations.

- As requested by Environmental Management, the Materiel Management Flight, the Contracting Squadron, and other affected base agencies, review contracts, purchase orders, registrations, permits, licenses, and requests for variances applicable to the management of hazardous waste. This support will extend to participating in meetings with regulatory agencies and providing legal advice to help resolve issues relating to hazardous waste management.

Public Affairs Office (5 BW/PA). In accordance with AFI 32-7002, Public Affairs Office Support the installation EMS as liaison between the installation and external communities by assisting with procedures for communicating environmental aspects of the installation's EMS and environmental programs, and input for media publications.

Fulfill the environmental responsibilities in AFI 35-108, Environmental Public Affairs, and notify the installation environmental function of any changes and/or updates to the Instruction.

Designate an individual as the Public Affairs representative on the CFT.

Commanders and Tenant Equivalents. A compliant hazardous waste management program is only possible if generators are proactive in maintaining an awareness of required hazardous waste procedures *and* exercise due diligence in carrying out their responsibilities as outlined in this plan. Accordingly, Commanders and tenant equivalents whose units generate hazardous waste shall:

- Ensure all hazardous waste is properly managed, stored, and disposed of. Above all, hazardous wastes must be under the control of a Hazardous Waste Manager in an approved hazardous waste accumulation site as soon as it is generated.
- Appoint a Unit Environmental Coordinator (UEC) as the Commander's agent to manage the unit environmental program.
    - The UEC may be appointed at the Squadron level, and will be the main point of contact between the organization and Environmental Management for all environmental issues.
    - The UEC must have adequate knowledge of waste generating processes to direct the proper disposal of generated wastes, and sufficient rank to rectify discrepancies found during assessments and inspections.
    - The UEC shall be responsible for the implementation of an internal environmental audit program that includes active participation in ESOHCAMP, EMS, and other environmental compliance inspections.
    - An additional UEC must be appointed and trained when the primary UEC is away from the shop longer than 3 weeks. **The UEC delegation letter shall be updated and sent to Environmental Management within 1 week of any change in assigned personnel.**
- Appoint at least two Hazardous Waste Managers in writing for each shop that generates hazardous and/or industrial waste and forward the delegation letter to Environmental Management within 2 weeks of appointment. An example of the delegation letter is provided in Appendix G.
    - Although two Hazardous Waste Managers are required, more than one primary and alternate shall be appointed in shops with high deployment rates or frequent personnel changes.
    - Additional Hazardous Waste Managers must be appointed and trained when both the primary and alternate managers are away from the shop longer than 2 weeks. **The delegation letter shall be updated and sent to Environmental Management within 1 week of any change in assigned personnel.**
    - Appoint Hazardous Waste Managers who are in direct control of the wastes generated. That is, Hazardous Waste Managers must work in the same building where the waste collection point is located. In a multistory building, the Hazardous Waste Manager must be on the same floor as the collection point. The installation Hazardous Waste Program Manager (5 CES/CEIEC) may allow exceptions on a case-by-case basis.
    - Ensure at least one Hazardous Waste Manager is assigned to each work shift. Primary managers working night shifts shall ensure all alternate managers are thoroughly familiar with their responsibilities for properly managing hazardous waste. Exceptions may be made on a case-by-case basis by the Hazardous Waste Program Manager.
- The Commander or tenant equivalent may delegate appointment of Hazardous Waste Managers and UECs to a lower level as long as the subordinate has direct knowledge of the unit's hazardous waste generation and personnel management. A squad ron UEC can delegate authority to appoint shop Hazardous Waste Managers. A delegation letter for the appointee must be signed by the commander or tenant equivalent and forwarded to Environmental Management within 2 weeks of appointment. The appointee then has to delegate authority to appoint shop Hazardous Waste Managers or UECs as appropriate.

Unit Environmental Coordinator (UEC)

- The UEC is the focal point for environmental activities at the squadron level. While the UEC is not necessarily an environmental expert, the UEC is responsible for successful implementation of the unit's environmental program. UECs are tasked with communicating the requirements of Minot AFB's environmental policies to their units and coordinating environmental activities as directed by the unit commander. Because environmental regulations are numerous and complex, UECs are strongly encouraged to obtain support from Environmental Management.
- Although the extent of a UEC's responsibilities will vary depending on the number and type of facilities within the unit, the number and type of processes performed by the unit, and the associated environmental requirements, a UEC's responsibilities

encompass the following: o Serve as the unit focal point for environmental compliance.

- Promote compliance with Minot AFB policies relating to the management of hazardous materials and hazardous waste.
- Provide guidance to shop personnel, including unit supply/material acquisition managers and supervisors, relating to those unit activities or process changes that have potential impact on Minot AFB environmental programs.
- Promote waste minimization efforts through product substitution, the Free- Issue Program, and recycling.
- Ensures quarterly inspections are completed using the Environmental Self Inspection Checklist (Stage I Assessment) currently available under the Management Internal Control Toolset (MICT) Program.
- Conduct periodic inspections of unit collection sites to ensure compliance with environmental regulations and this plan. Assist Environmental Management and federal, state, and local regulators with inspections of squadron/unit waste collection sites. o As the environmental focal point, encourage successful communication among unit personnel, Environmental Management, Bioenvironmental Engineering, and Wing Safety personnel to facilitate implementation of Air Force, AFGSC, base, and unit environmental management programs.
- Distribute information from Environmental Management to unit personnel to keep them informed of their individual responsibilities for ensuring compliance with environmental regulations and policies.
- Provide the installation Hazardous Waste Program Manager with unit- specific information as may be needed to satisfy environmental regulations; in particular, alert the Hazardous Waste Program Manager of any new waste streams, changes in shop Hazardous Waste Managers, and designation/deletion of waste accumulation points.
- If delegated by the unit commander, appoint shop Hazardous Waste Managers and forward delegation letters to the installation Hazardous Waste Program Manager.
- Ensure newly appointed Hazardous Waste Managers complete initial unit and shop specific training and annual refresher training.
- Attend ESOHC subcommittee and working group meetings as requested.
- Obtain UEC environmental management training and complete annual refresher training thereafter.

Hazardous Waste Managers. Hazardous Waste Managers are essential to assuring compliant management of hazardous waste on Minot AFB and its associated GSUs. They oversee waste generating and handling activities at their respective shops and are directly responsible for the success of the Minot AFB hazardous waste management program. It is crucial that Hazardous Waste Managers be knowledgeable, proactive, and empowered by their respective commanders to strictly and completely implement the policies and procedures outlined in this plan. As the individuals directly responsible for ensuring hazardous waste compliance, Hazardous Waste Managers will:

- Directly manage all hazardous waste generated by their shops activities.
- Properly segregate, store, characterize, label, mark, package, and transport hazardous waste in accordance with federal, state, and Air Force regulations and as detailed in this plan.
- Ensure that RCRA-hazardous and large-volume wastes are only accumulated in compatible DOT containers that are in good condition with a clean exterior and have no pitting, no rusting, no sharp dents/creases, and no bulges. Small waste containers (for example, tubes, ampoules, and single aerosol cans), industrial waste, and nonhazardous waste may be placed in non-DOT containers, but the containers must also be in good condition and compatible with the product placed in them.
- Conduct regular inspections of waste accumulation points as described in Section 7.5 of this plan. SAP must be inspected monthly with inspections not more than 30 days apart.
- Complete all required documentation and ensure a waste container is full prior to scheduling a pickup and transfer of hazardous waste to the CAP. Documentation includes container labels and the correct MSDSs or SDSs.
- Maintain a Hazardous Waste Management Binder as detailed in Appendix D of this plan.
- Maintain a current delegation letter signed by the squadron commander or delegated representative.
- Maintain a current waste stream inventory and provide an estimate of waste generation for future years to Environmental Management as requested.
- Notify Environmental Management of all changes in waste activities including waste stream process modifications, product substitutions, and relocation of hazardous waste accumulation sites, including SAPs.
- Train shop personnel on the proper handling of hazardous materials and waste as detailed in Section 5.0 of this plan.
- Promptly notify Environmental Management and Bioenvironmental Engineering of chemicals or processes that may result in either the generation or elimination of hazardous waste.
- Notify Environmental Management of any pending change in Hazardous Waste Managers, including but not limited to deployments, extended TDYs or leave (that is, greater than 2 weeks), and PCS. Managers shall notify the Hazardous Waste Program Manager at least 30 days prior to leaving Minot AFB.
- Obtain initial Hazardous Waste Manager training in accordance with Section 5.0 of this plan and promptly notify the installation Hazardous Waste Program Manager that training is complete. Notification must be made via e-mail. All managers must thereafter complete annual refresher training.

- Create and implement a shop-specific training plan for hazardous waste handlers assigned to the shop in accordance with Section 5.0 of this plan. A template is available from the Hazardous Waste page on the Minot AFB Installation Management Flight website at https://usaf.dps.mil/teams/10624/Minot/WPP/ProgramPage/Hazardous+Waste.aspx
- Ensure appropriate hazardous waste handler training is accomplished annually by shop personnel in accordance with Section 5.0 of this plan. When completed, Hazardous Waste Managers must document handler training on AF Form 1098 as described in Section 5.0.
- Prepare and keep current a Site-Specific Contingency Plan in accordance with Section 7.10 of this plan and report all spills to the installation Water Program Manager at 723-3436.

CAP Support

- The following base agencies will support the CAP as described.
    - **Environmental Management Element (5 CES/CEIE)**. Environmental Management will:
        - Provide technical assistance as needed.
        - Act as a liaison for the base with all federal and state agencies.
        - Keep records of hazardous waste activities on Minot AFB in accordance with 40 CFR 261 including, but not limited to, hazardous waste turn-in documentation, accounts of any situation requiring implementation of the Hazardous Materials Emergency Response Plan, and the Hazardous Waste Biennial Report.
    - **Fire Protection Flight (5 CES/CEF).** The Fire Protection Flight will:
        - Provide technical assistance for storage and compatibility concerns as needed.
        - Provide technical assistance for spills greater than 55 gallons and spills that are immediately hazardous to CAP personnel and the environment.
        - Provide fire protection services.
    - **Bioenvironmental Engineering (5 MDOS/SGPB).** The Bioenvironmental Engineering will:
        - Coordinate as needed on the characterization of all unknown wastes.
- Provide technical assistance to the CAP contractor as requested.

5 TRAINING

5.1. Hazardous waste training is required by law. All training requirements and sources are provided in the Environmental Training Matrix on eDASH. Specific local training procedures are provided in the Installation Supplement to this plan.

5.2. Hazardous waste training is provided only by qualified authorized personnel. Training records are maintained IAW the Recordkeeping and Reporting section of this plan.

5.3. Hazardous waste program managers and alternates appointed in writing by the installation Commander to sign manifests, require specific training to include HW Management Compliance Training and Department of Transportation training to sign HW manifests. Installation HW Program Managers should complete the following in-residence courses (or equivalent): Air Force Institute of Technology (AFIT) 521, Hazardous Waste Management; and DLA - DCPSO00510, Transportation of Hazardous Material/Hazardous Waste (Interservice Environmental Education Review Board [ISEERB] approved). After initial course completion, refresher training is required annually for RCRA, and every 3-years for the DoT course. See training matrix for more information.

5.4. Function-Specific Training (Local In-House/On-Line Training Sources): (Group A) Organizational (non-central) Hazardous Waste Storage Area Managers, Satellite Accumulation Area (Initial Accumulation Point) Managers, and their immediate supervisor; and (Group B) all shop personnel, and their immediate supervisors, who generate HW.

Installation Supplement

To help ensure this training is administered effectively to individual requirements, hazardous waste management training on Minot AFB is customized for hazardous waste managers and hazardous waste handlers. The Unit Environmental Coordinator, Environmental Management, and shop managers are responsible for ensuring required hazardous waste management training is accomplished and documented on AF Form 1098.

Hazardous Waste Manager Training

- Personnel who are appointed as Hazardous Waste Managers must receive training on hazardous waste management if they perform the following tasks:
    - Manage hazardous waste at a SAP or the CAP.
    - Make determinations whether a waste is hazardous or not.
    - Inspect SAPs and/or Accumulation Sites.

- Prepare hazardous waste manifests, annual reports, and exception reports.
- Make decisions or communicate information concerning personnel and assets involved with hazardous waste.
- Supervise and train hazardous waste handlers.
- Perform as a Unit Environmental Coordinator.

- Initial and annual refresher hazardous waste management training shall include the following topics:
  - Introduction to the Resource Conservation and Recovery Act (RCRA).
  - Identification, evaluation, and characterization of hazardous waste.
  - Container use, marking, and labeling.
  - SAP and Accumulation Site management as appropriate to the level of responsibility.
  - Waste turn-in procedures.
  - Manifesting and transport of hazardous waste.
  - Spill prevention and emergency response.
  - Waste minimization.
  - Personnel safety.
  - Documentation and record keeping.
- Initial training for all Hazardous Waste (HW) Managers is provided on-line through TEACH. Hazardous Waste Managers will complete this training prior to starting their waste management responsibilities, and must complete annual refresher training thereafter.
  - At a minimum, Hazardous Waste Managers must complete the following modules in the TEACH training plan: EMS Awareness Level training, Green Procurement Program (GPP), Hazardous Waste Satellite Accumulation Point (SAP) Manager. Hazardous Waste Managers do not need to complete the Recycling and universal waste modules because that information is contained in the SAP Manager training. Additional training modules must be completed if they apply to the career field. For example, Entomology personnel will complete the "Pesticide" training module, and Vehicle Maintenance personnel will complete the "Waste Management Guidelines for Vehicle Maintenance Waste."
  - Additional duties, such as GPC holder and facility manager, have their own TEACH training modules, which must be completed before a Hazardous Waste Manager assumes those duties.
  - The Hazardous Waste Manager shall also obtain classroom training from Environmental Management. This two-hour class emphasizes applicable EPA and DOT regulations that address hazardous waste identification, characterization, containerization, handling, accumulation, and minimization.
- The Hazardous Waste Manager training is tracked by the respective UEC, HAZMART, or Environmental Management. When a Hazardous Waste Manager is finished with the CBT, that individual must contact his or her UEC to receive a signed AF Form 1098. If a UEC is unavailable, the installation Hazardous Waste Program Manager will need to sign the Hazardous Waste Managers AF Form 1098. Hazardous Waste Managers and Unit Environmental Coordinators (UEC) may not sign for each other. The UEC's AF Form 1098 must be signed by the installation Hazardous Waste Program Manager.
  - The Hazardous Waste Manager is responsible for training shop HW Handlers, and must implement a training plan that explains hazards and wastes specific to the shop. A template for the HW Handler training plan is available from the installation Hazardous Waste Program Manager.
  - Hazardous Waste Managers are responsible for completing their annual refresher training without prompting, and ensuring shop personnel complete annual HW Handler training.

Hazardous Waste Handler Training

- Hazardous waste handler training is required for all personnel on Minot AFB who accomplish the following:
  - Add or remove hazardous waste to or from an accumulation container.
  - Respond to spills, fires, or explosions involving hazardous waste, as required.
  - Conduct any tasks involving occupational exposure to hazardous waste.
- Initial and annual refresher hazardous waste handler training shall include the following topics:
  - Introduction to the Resource Conservation and Recovery Act.
  - Identification, evaluation, and characterization of hazardous waste to include (1) recognition of those materials and processes that will generate a hazardous waste and (2) identification of high-risk materials such as those containing MEK, benzene, chlorinated solvents, lead, mercury, chromium, and cadmium.
  - Container use, marking, and labeling.
  - Spill prevention and response to emergencies.
  - Waste minimization.
  - Personal safety.

- At Minot AFB, this training is provided on-line through TEACH. At a minimum, the HW handler should accomplish all modules in his or her TEACH training plan including recommended training modules.
- The Hazardous Waste Manager is responsible for supplementing the CBT training of their hazardous waste handlers with shop specific policies, procedures, and items of interest, such as highly hazardous chemicals, radiation hazards, and extreme temperatures. This training must encompass (a) compliance with EPA/DOT regulations; (b) waste identification, containerization, handling, accumulation, and minimization; and (c) emergency procedures and the use of emergency equipment and systems.
- Initial training for newly assigned personnel must be completed within three months of assignment to the shop, and refresher training must be provided at least once every 12 months thereafter.
- Hazardous Waste Managers will develop a shop-specific training plan that shall be reviewed by Environmental Management. The plan must specify the length of the training, and detail how initial training will be provided to newly assigned personnel within 3 months of their assignment to the shop. The plan will also include details on how every handler will be monitored to ensure he/she receives training at least once every twelve months. New personnel must not work unsupervised with hazardous waste until this training is completed.
- When the training plan is approved by the installation Hazardous Waste Program Manager, the shop Hazardous Waste Manager will file the plan in the Hazardous Waste Management binder to fulfill the requirements in 40 CFR 265.16(a)(2-3).
- The installation Hazardous Waste Program Manager will work with a SAP Hazardous Waste Manager to address unique training situations. Variances on the Hazardous Waste Handler training requirements can be granted by the installation Hazardous Waste Program Manager as deemed appropriate.

Training Documentation

- On Minot AFB, AF Form 1098, *Special Task Certification*, and *Recurring Training*, is used to track recurring environmental training. The AF Form 1098 (or equivalent) will be used to document all hazardous waste training for managers and handlers. Copies of AF Forms 1098 training forms will be kept in the training section of the Hazardous Waste Management Binder for each currently assigned individual.
- Training records will include the following information for each environmental training course attended by the employee, including those received prior to his or her assignment:
    - Student's full name, duty section, and duty phone number;
    - Type of training received;
    - Date of training completion;
    - Instructor's signature (when completed, the Hazardous Waste Management Course will be certified by the UEC or installation Hazardous Waste Program Manager. The shop Hazardous Waste Manager will certify the shop's Hazardous Waste Handler Course);
    - Student's initials;
    - Frequency of training (annually, quarterly, etc.); and
    - Date next training due.
    - Job title and description. The job title and description of duties MUST be written on the bottom of the proper training form. That is, the manager's title on the managers form, and the handler's title on the handler's form. Use a new form when a handler becomes a Hazardous Waste Manager. Contact the installation Hazardous Waste Program Manager (723-4870) for the specific language.
- Training records on current personnel must be kept until three years after the closure of the accumulation point. Training records on former employees must be kept for at least three years from the date that the employee last performed hazardous waste duties at the facility. Training records should include a note of temporary or permanent assignments that may prevent Hazardous Waste Managers and/or handlers from receiving training on time. In such a situation, training must be provided as soon as possible upon their return.

1. Air Force personnel training records may accompany personnel transferred within the same organization. However, copies should be made of the records if they are transferring to another organization.

6 RECORDKEEPING AND REPORTING

Recordkeeping

The installation complies with the following U.S. Federal HW recordkeeping requirements as applicable based on generator status. NOTE: Retention time is defined by regulation as onsite and readily accessible for inspections and/or reference; after retention, the record(s) follow applicable Air Force Records Management rules.

Summary of HW Recordkeeping Requirements

Record*	Citation	Retention Time**	Citation
HW determination documentation	40 CFR 262.11(f)	3 years from the date that the waste was last sent to a TSDF	40 CFR 262.11(f)
HW Biennial/Annual Report	40 CFR 262.41	3 years from the due date of the report	40 CFR 262.40(b)
HW manifest (electronic or paper)	40 CFR 262.20	3 years from the day the waste was accepted by the initial transporter	40 CFR 262.40(a)
Small Qty HWAS inspection logs	40 CFR 262.16(b)(2)(iv)	Although records are not formally required, the best management practice is to record and retain for 3 years to demonstrate compliance	N/A
Large Qty HWAS inspection logs	40 CFR 262.17(a)(1)(v) 40 CFR 264.15(d) 40 CFR 265.15(d)	For interim and permitted operations, 3 years from the date the inspection was conducted. For all other LQGs, the best management practice is to retain for 3 years to demonstrate compliance	40 CFR 265.14(d) 40 CFR 265.15(d)
Preparedness and prevention arrangements with local authorities	40 CFR 262.16(b)(8)(vi)(B)	The federal regulations do not offer a minimum retention time, but the best management practice is to retain the plan while active and for 3 years thereafter to demonstrate compliance	N/A
Consolidation of HW received from very small quantity generators.	40 CFR 262.17(f)	3 years from the date the HW was received from the very small quantity generator	40 CFR 262.17(f)
Exception reports	40 CFR 262.42	3 years from the due date of the report	40 CFR 262.40(b)
Land restricted waste determination	40 CFR 268.7(a)(1)	3 years from date the determination was required to be conducted. If not required, 3 years from the date the waste was last sent to a TSDF	40 CFR 268.7(a)(8)
Land restriction notice and certification	40 CFR 268.7(a)(2)	3 years from the date the waste was last sent to a TSDF	40 CFR 268.7(a)(8)
Notification of intent to export waste	40 CFR 262.83(b)	3 years from the date the HW was accepted by the initial transporter	40 CFR 262.83(i)(1)(i)
Waste export confirmation of receipt and exception reports	40 CFR 262.83(h)	3 years from the date the HW was accepted by the initial transporter	40 CFR 262.83(i)(1)(iii)
Annual report (required of primary exporters of HW)	40 CFR 262.83(g)	3 years from the date the HW was accepted by the initial transporter	40 CFR 262.83(i)(1)(ii)
Employee training records (including appointment letters for key HW personnel)	40 CFR 262.16(b)(9)(iii) 40 CFR 262.17(a)(7)(iv) 40 CFR 264.16(d) 40 CFR 265.16(d)	For interim and permitted operations- current personnel: until closure of the site; Former personnel: 3 years from date the individual last worked there. For all other LQGs and SQGs, the best management practice is to retain for 3 years to demonstrate compliance	40 CFR 262.17(a)(7)(iv) 40 CFR 264.16(e) 40 CFR 265.16(e)

*Permitted Treatment, Storage, and Disposal Facilities (TSDF) comply with recordkeeping requirements established in their HW permit.

**Retention Time may be extended during the course of any unresolved enforcement action or as requested by the EPA. The AF, through the Air Force Records Information Management System (AFRIMS), requires that HW-related reports, documents, studies, HW manifests, and disposal records (including contracts) are destroyed 50 years from the date of the record. Note that the records required by law or external regulation, should be readily available on-site. Records required by AF Record Retention requirements can be archived.

Reporting

The HW Program Manager, and other designated personnel, generate needed reports from the Enterprise Environmental, Safety, and Occupational Health - Management Information System (EESOH-MIS).

Enforcement actions, spills, and inspections are reported via the Enforcement Actions, Spills, and Inspections Environmental Reporting (EASIER) database.

Installation Supplement

Biennial Report

Minot AFB ships hazardous waste off site to a TSDF located within the United States and, therefore, must prepare and submit a biennial report to NDDH by March 1 of each even- numbered year. The hazardous waste program manager ensures that the report is submitted before March 1 as required.

Waste Minimization Report

As required by the North Dakota Department of Health and as requested by the state regulator and/or higher headquarters staff, Environmental Management must submit a waste minimization report along with the Biennial Report. The following wastes are currently part of an active waste minimization effort:

- Spent antifreeze that is recycled on base and reused.
- Spent parts-washing solvent that is recycled on base per 40 CFR 261.4(a)(8) and is reused.
- Lead/acid and rechargeable batteries that are recycled off base per 40 CFR 261.6(a)(2).
- Fluorescent and other specialized light bulbs/lamps that are recycled off base per 40 CFR 273.5.
- Spent bullets and shell casings that are recycled off base per 40 CFR 261.4(a)(3)(ii).
- Waste sulfuric acid that is neutralized and discharged to the installation sanitary sewer per 40 CFR 264.1(g)(6).
- Waste electronics, including cathode-ray tubes (CRT), that are recycled off base per 40 CFR 261.4(a)(22).
- Used oil that is burned for energy recovery per 40 CFR 261.6(a)(4).
- Used aerosol cans that are punctured and recycled as scrap metal per 40 CFR 261.1(b)(6), 261.4(a)(13), and 261.7(a)(1).
- Spent oil filters that are hot-drained and recycled as scrap metal per 40 CFR 261.4(b)(13).
- Wastewater generated from washing operations that is pretreated and discharged to the sanitary sewer system under the domestic sewage exclusion per 40 CFR 261.4(a)(1) and 268.7(a)(7).

The wastes listed above are exempted hazardous wastes as long as they are managed in accordance with the cited regulation. If not managed in this manner, they must be counted as hazardous wastes generated by Minot AFB.

Hazardous Waste Determinations

All generators of solid waste are required to determine if that waste is hazardous based on generator knowledge (that is, SDSs) or laboratory analysis. If the waste is determined to be hazardous, then the RCRA regulations apply, the waste must be properly stored and disposed of, and this determination becomes part of the base's operating record. It must be kept on file for at least 3 years from the date the waste was last sent to an off-site TSDF. That retention period is automatically extended during any unresolved enforcement actions or as requested by EPA. However, Air Force records disposition services procedures require that hazardous waste documentation be stored for 50 years.

Hazardous Waste Manifests

The hazardous waste manifest is the shipping document that identifies the hazardous waste generator, transporter, and TSDF. It also describes the contents of the waste shipment. When a waste shipment leaves the base, the manifest must be completed through block 17, Transporter 1. One copy of this "open" manifest must be kept on file at the base that generated the waste, and the remaining copies must be provided to the transporter. The open manifest accompanies the transporter to the designated TSDF. The transporter may also deliver the hazardous waste to additional transporters who will move the waste to the designated TSDF. After acceptance of the waste, the owner or operator of the TSDF must sign the manifest signifying receipt of shipment. This signed (or "closed") manifest is then returned to the waste generator to complete the paper trail.

Uniform Hazardous Waste Manifests are completed by DLA-DS. The hazardous generator should ensure that sufficient copies of the manifest are made to provide the generator, each transporter, and the designated TSDF, and a final copy to be returned to the generator. Each closed manifest must be retained by the hazardous waste office on base for at least 3 years from the date the waste was shipped off site and maintained for 50 years per Air Force requirements.

Manifests must also have LDR certifications attached, as required by 40 CFR 268.7(a) (NDAC 3324- 05-256), if the wastes disposed of are considered Land Disposal Wastes and must meet treatment standards set forth in 40 CFR 268.40 (NDAC 33-24-05-280 to 289).

CAP Inspection Logs

EPA requires that all inspection logs at the CAP be kept on file at the base for at least 3 years from the date of the inspection. The retention is extended if there is an enforcement action or requested by EPA. Air Force records disposition procedures require that hazardous waste documentation be stored for 50 years.

Exception Reports

For large quantity generators, the exception report must be submitted if the closed manifest is not received within 45 days of the date the transporter accepted the waste. In this case, a legible copy of the manifest must be submitted with some indication that confirmation of delivery has not been received.

Exception reports must be retained on site at least 3 years from the due date of the report and retained for 50 years as required by Air Force document procedures.

Land Disposal Restriction Notifications/Certifications

Land disposal restriction notifications/certifications are completed by DLA-DS. This document is required to accompany the manifest for which it applies, or must be on file with the TSDF for the waste streams that are on the manifest of the shipment. This documentation must be kept on file at the base for at least 3 years from the date of the shipment, or longer if there is enforcement actions or if requested by the EPA. Air Force procedures required that hazardous waste documentation be maintained for 50 years.

Universal Waste Shipment Documentation

Large quantity handlers of universal waste must maintain documentation (a bill of lading, a manifest, or log) of their universal waste shipments. Each document must identify:

- The name and address of the handler and the destination of the shipment;
- The quantity of each type of universal waste shipped; and
- The date the shipment left the facility.

Shipment documentation must be maintained for at least 3 years according to EPA requirements. Air Force record retention procedures require that hazardous waste documentation be maintained for 50 years.

Employee Training

Training records for shop employees are maintained by the shop supervisor and are recorded on either AF Form 1098 or AF Form 55. Former employees training records must be kept for at least 3 years from the date the person last worked on the base. Air Force record retention procedures require that hazardous waste related documents be maintained for 50 years.

Discrepancy Reports Record Keeping

Discrepancy reports are initiated by the TSDF. Hazardous waste generators should document discrepancy-related telephone conversations and keep copies of all correspondence and waste analyses concerning the discrepancy. Air Force records disposition procedures require that hazardous waste documentation be stored for 50 years.

Hazardous Waste Spill Reports

RCRA regulations in 40 CFR 262, 264, and 265 require hazardous waste generators to report releases, fires, or explosions involving hazardous waste that could threaten human health or the environment outside the base to local authorities and to the National Response Center at (800) 4248802. These reports are required to be maintained on file for a period of 3 years from the date of the incident. The retention period is automatically extended if there is an enforcement action, or if requested by EPA.

An environmental release report should be completed for all environmental releases that require reporting to federal, state, or local agencies, and are caused by an Air Force activity or occur on an AFB or facility.

Base Hazardous Waste Records

Hazardous waste records are maintained in files at the hazardous waste office in Building 525. These files must contain hazardous waste manifests, hazardous waste turn-in documents, universal waste shipping documentation, land disposal notifications/certifications, hazardous waste profiles, laboratory analytical results, and similar documents. All records within this file must be maintained on site for at least 3 years. Air Force records disposition services procedures require that hazardous waste records be kept for 50 years (https://afrims.cce.af.mil/rds/rules/1925; AFRIMS T32 – 01 R 17.00). Contact the hazardous waste program manager for further records disposition instructions.

Shop Hazardous Waste Records

To ensure compliance with RCRA record keeping requirements, each Hazardous Waste Manager will maintain a Hazardous Waste Management Binder and an associated Inactive File as detailed below for as long as the shop generates hazardous waste in accordance with the requirements of the Hazardous Waste Binder section below. The binder shall be maintained in the shop generating the waste and may not consolidate information from similar shops without written permission from the installation Hazardous Waste Program Manager. If required, multiple binders may be used providing the order of documents is maintained. The binders will be moved with the shop if it moves to a new location. If a shop no longer generates hazardous waste and the associated SAP is closed in coordination with Environmental Management, all documents will be maintained for three years and then as specified below.

Hazardous Waste Management Binder

The Hazardous Waste Management binder (HW binder) shall contain the following information in the order indicated:

- Tab A, Hazardous Waste Appointment Letter. The most current Hazardous Waste Manager appointment letter(s) per Appendix G. All previous appointment letters are filed in the Inactive File for 3 years. After 3 years, previous versions may be discarded.
- Tab B, Training Documentation. Current Hazardous Waste Manager and Handler training documentation per Section 5.0 of this plan are kept in the binder until the person permanently leaves the shop. Documents for previous employees are maintained in the Inactive File for at least one year, then sent to records management and staged for one year. After that time, the records may be destroyed.
- Tab C, Handler Training Plan. The most current version of the shop Hazardous Waste Handler Training Plan per Section 5.0 of this plan is kept in the active binder. All previous versions should be dated and filed in the Inactive File for 1 year. After 1 year, previous versions should be sent to records management and staged for 1 year, then discarded.
- Tab D, Site Specific Contingency Plan. A current Site-Specific Contingency Plan for the SAP should be maintained in the binder. Previous versions should be kept in the Inactive File for 3 years and then discarded.
- Tab E, Turn-in Documentation. All Turn-in Documents should be kept for 3 years, then sent to Records Management and staged for 50 years. The documents include:
    - An Accumulation Storage Log sheet, as described on pages 4-17, for each waste stream generated by the shop. The log includes (1) the container numbers; (2) container size and type;(3) EPA waste codes assigned to the waste (for example, D001 or F003); (4) the date the container was first used at the SAP; and (5) the date and place to which the container was transferred from the SAP. o Hazardous waste turn-in documentation filed by the waste stream number (with repeat turn-ins of the same wastes filed together) to include (1) most current waste profile sheet; (2) a paper/electronic record of the documents used to identify the waste (for example, MSDS or SDS number, manufacturer MSDS or SDS, laboratory analytical report, and/or associated Technical Order); and (3) DD Form 1348-1A, the receipt and proof of delivery of the hazardous waste. This documentation should be kept for 3 years and then sent to Records Management and staged for 50 years.
- Tab F, Inspection Findings.
    - Periodic inspection findings from Environmental Management, the CAP contractor, EIP, UEI, or any other inspection must be kept for five years and then may be discarded.
    - All SAP inspection checklists/logs must be kept for at least 3 years and then may be discarded.

- Tab G, Minot AFB Policy Letters. Policy letters generated by Environmental Management must be filed in this section in chronological order. Keep the letters until rescinded. Previous versions of the same policy letter may be discarded.
- Tab H, Hazardous Waste Management Plan. A printed copy of this Hazardous Waste Management Plan, or a placeholder for an electronic copy of the Plan. An electronic copy of this plan must be kept on a share drive and the exact location printed on the placeholder.

The HW binder will not contain HAZCOM and HAZMAT documents, except for MSDS or SDS, which may be referenced in the HW binder and maintained separately.

Hazardous Waste Handbook

The Hazardous Waste Handbook was designed to assist personnel unfamiliar with the numerous requirements for managing hazardous waste on Minot AFB. The handbook addresses routine hazardous waste situations encountered throughout the base, and information is organized and written so it is easy to locate and understand.

It should be noted that the contents of the handbook provide a general overview of the hazardous waste compliance program at Minot AFB, and more detailed information on the topics covered may be found throughout this plan and associated appendices. Accordingly, each section of the handbook includes a reference where more detailed information on subject matter may be found in this Hazardous Waste Management Plan.

Inactive File

The Inactive File is used to archive older documentation still required by regulation. The Inactive File is an extension of the Hazardous Waste Management Binder, and documents should be filed chronologically in the same order as the binder. The documentation comprising the Inactive File may be kept in its own binder or in files in the Hazardous Waste Manager's desk. The Inactive File should specifically include the following:

- Previous Hazardous Waste Manager Appointment letters, which should be kept for three years and then may be discarded.
- Previous version of the shop-training plan for Hazardous Waste Handlers, which should be kept for one year, staged for one year, and then discarded.
- Hazardous Waste Manager and Handler training documentation for previous employees must be maintained in the Inactive File for 1 year, staged for 1 year, and then discarded.
- Accumulation Storage Log sheets, which must be kept for 3 years and then staged for 50 years.

1. Waste Profile Sheets, waste analysis, and associated DD Forms 1348-1A, which must be kept for 5 years, then staged for 50 years. Duplicate copies of identical Waste Profile Sheets may be discarded.

7 PROCEDURES

This section contains procedures for managing HW from identification, accumulation, offsite transportation, and disposal. The HW Program Manager ensures that appropriate procedures are properly communicated and followed by all necessary personnel.

7.1 Waste Inventory

A current waste inventory can be generated within EESOH-MIS using the Ad-Hoc Reporting Tool or by completing the following steps:

- Log into EESOH-MIS, select the "Reporting" option, and select "Hazardous Waste" to generate the Waste Site Waste Stream Summary Report.

The resulting waste inventory report can be placed into Appendix B, Hazardous Waste Streams/Profiles.

Installation Supplement

Hazardous Waste Stream Inventory

The Environmental Management Element, with the support of shop Hazardous Waste Managers, Bioenvironmental Engineering, and the CAP contractor, maintains an inventory of every waste stream generated on Minot AFB and its associated GSUs. Environmental Management ensures all wastes are properly characterized and classified as hazardous, industrial (that is, non-RCRA, state-regulated waste), universal, exempt, or excluded waste in accordance with the Minot AFB WAP (Appendix A). The DRMS Form 1930, *Waste Profile Sheet*, is used to record waste characterization and is the basis for the proper disposal of generated wastes. Descriptions of the waste types generated on Minot AFB and its associated GSUs are provided below.

RCRA Hazardous Waste

- Environmental Management is required to maintain an inventory of every hazardous waste stream generated on Minot AFB and its associated GSUs. Documentation of hazardous wastes must include all background information including MSDSs SDSs, analytical results, related correspondence, and any shipping papers or DLA-DS transfer documentation.
- The shop Hazardous Waste Managers are responsible for quantifying all hazardous waste streams for the WSI, notifying Environmental Management of changes to the waste streams, and forecasting annual generation of each hazardous waste stream.
- Unless otherwise authorized by AFGSC, Minot AFB shall not accept, handle, store, transport, dispose of, or inventory non-DoD owned and non-Minot AFB generated hazardous waste.

Non-RCRA State-Regulated Waste

- Non-RCRA state-regulated waste is classified as an industrial waste by North Dakota. Industrial waste is defined as wastes containing organic or inorganic contaminants that should not be disposed of in a municipal landfill. These types of waste include, but are not limited to, the following:
    - Wastes subject to the solid waste exemption (for example, bead media sent for reuse per 40 CFR 261.2(e)).
    - Wastes subject to the solid waste or hazardous waste exclusion (for example, recycled sulfuric acid per 40 CFR 264.1(g) and spent solvent per 40 CFR 261.4).
    - Wastes that are recycled and, therefore, not subject to the generator waste determination requirements of 40 CFR 262.11 (for example, lead acid batteries that are being reclaimed, used oil burned for energy recovery, and scrap metal for recycling per 40 CFR 261.6).
    - Latex paint, adhesives, spent antifreeze, and oily wastes.
- Industrial waste is managed and disposed of as type of solid waste. Municipal solid waste landfills may accept limited amounts of industrial waste co-disposed of with municipal solid waste. If the Municipal Solid Waste Landfills will not accept your industrial waste, it must be disposed of in one of the five state permitted industrial waste landfills in North Dakota.
- Hazardous Waste Managers must identify and document each waste stream that is classified as a non-RCRA state-regulated waste in their areas of responsibility. Records for non-RCRA state- regulated wastes will include all background information regarding the waste such as analytical results, related correspondence, associated contracts, and any shipping papers or DLA-DS transfer documentation. These documents shall be maintained in the shop Hazardous Waste Management Binder as described in Appendix D of this plan.

Universal Waste

- Universal waste is a hazardous waste that is commonplace and poses a relatively low risk to personnel safety and health and to the environment. Accordingly, both federal and North Dakota regulations provide simpler rules for handling, recycling, and disposing of universal waste. Note that a universal waste must be recycled; otherwise, the waste must be managed as a hazardous waste. At Minot AFB, universal waste is collected and recycled.
- Universal waste may include batteries, some pesticides, mercury-containing devices, and various types of vapor lamps. See Section 7.11 of this plan for additional information about universal waste types and disposal directions.

Used Oil

- Used oil is another common potentially hazardous waste that presents a relatively low risk to personnel safety and health and to the environment. Used oil that is recycled is subject to less stringent hazardous waste requirements. At Minot AFB, used oil is collected and recycled.
- Federal and North Dakota regulations define used oil as any oil that has been refined from crude oil or any synthetic oil that has been used and, because of such use, is contaminated by physical or chemical impurities. Used oil that has been mixed with a hazardous waste must be characterized. If used oil contains more than 1,000-ppm, total halogens, it is presumed to have been mixed with a listed spent solvents.

Exempted Waste

Exempted wastes are hazardous wastes or potentially hazardous wastes that are exempted from the RCRA hazardous waste regulations. These wastes include, but are not limited to, spent lead acid batteries being recycled, PCBs that fail the test for toxicity characteristic (D018 through D043 only), and residues in RCRA empty containers.

Excluded Waste

Excluded wastes are specifically excluded from RCRA hazardous waste compliance under 40 CFR part 261.4.

- Certain solvent-contaminated wipes (contaminated with F-listed solvents) that are sent for cleaning and reuse provided certain container, labeling, accumulation, recordkeeping, and other regulatory requirements are met (40 CFR 261.4).
- Wastes that are not considered solid wastes such as domestic sewage, industrial wastewater discharge under a National Pollutant Discharge Elimination System (NPDES) permit, and used cathode ray tubes that are being recycled.
- Solid wastes that are not hazardous wastes such at household wastes, and agricultural waste/manure returned to the soil as fertilizer.

Hazardous Waste Stream Inventory

- In October of each year, shop Hazardous Waste Managers must provide a forecast of types and quantities of hazardous wastes expected during the upcoming fiscal year to Environmental Management so the Minot AFB hazardous WSI can be updated.
- The most current hazardous WSI for Minot AFB and its associated GSUs is maintained by Environmental Management in a computerized waste- tracking database, which is incorporated into this plan by reference. A list of known waste streams can be found in EESOH-MIS.

7.2 Waste Identification

The HW Program Manager determines the nature of waste based on a detailed qualitative analysis of the regulated waste generating process, associated Safety Data Sheet (SDS) information, and coordination with generating activity personnel involved in the use of hazardous materials. If uncertainties about a waste stream exist, the HW Program Manager pursues waste stream sampling and analysis IAW the Waste Analysis Plan (WAP) found in Appendix A.

The WAP details the wastes that have been evaluated and analyzed, a description of the testing and analytical methods used, the HW sampling methods used, the location of samples taken for analysis and frequency, sample documentation, sample quality assurance and quality control procedures, and sample request procedures.

Generator knowledge and the results of the WAP are used to minimize waste re-characterizations to those instances where a process change has occurred or the waste stream is highly variable.

Installation Supplement

7.3 Container Management

Container management procedures are as follows:

- Containers storing HW must be in good condition and meet transportation and other applicable requirements. "Good condition" means there should be no severe rusting, no sharp-edged creases or dents, no bulging heads, and no severe structural defects.
- Ensure that the waste material will not react with the container itself.
- Use plastic or plastic-lined steel drums to safely store corrosive wastes.
- Immediately transfer the contents of a leaking container to another container or over pack into a salvage drum.
- Containers with free liquid or drum contents on top must be cleaned or over packed in the case of a leak.
- Containers must remain closed at all times except when adding or removing waste.
- Adequate headspace must be maintained at all times when filling a container to account for content expansion. The required headspace is unique to each waste stream and expected storage conditions. For liquids and volatile chemicals, a general rule is to not fill to more than 85% of container capacity to allow for temperature changes, and this equates to approximately four inches of headspace in a typical 55-gallon drum. The ultimate requirement is performance based, such that drums do not bulge or leak. A good inspection process will enable any recommendations for specific waste streams as appropriate.
- Containers holding HW must not be opened, handled, or stored in a manner which may rupture the container or cause it to leak.
- Containers of flammable liquids must be grounded when transferring flammable liquids from one container to the other.

Installation Supplement

Waste Compatibility

Do not place different hazardous wastes into the same container. Mixing incompatible materials may cause fire, explosion, toxic fumes, or other dangerous chemical reactions. Incompatible waste mixtures include acids with bases, and oxidizers with ignitable/combustibles.

To avoid the possibility of accidental mixing, a dike, wall, or other suitable barrier must separate containers of incompatible wastes are stored at the CAP or a SAP. Contact the Fire Protection Flight's Technical Services (723-6689), the Bioenvironmental Engineering (723-5151), or Occupational Safety (723-2259) for specific storage requirements and guidance.

Container Handling

All containers must be handled carefully to avoid damage. The following requirements apply:

- Hazardous waste containers must meet DOT specifications per 49 CFR 178, *Specifications for Packaging's*. The Traffic Management Flight, 5 LRS/LGRT (723-4199) can help ensure containers with the required specifications are used. As required, the Distribution Flight can provide the national stock number (NSN) to determine applicable container specifications. Shops may obtain containers from the CAP, and may request specialized containers from Environmental Management.
    - Containers must always be closed except when it is necessary to add or remove the waste. Shop personnel must ensure the following:
        - Bungs and lids on hazardous waste drums have seals and gaskets in place and are tightly fastened.
        - Open head drum lids are maintained with locking rings securely in place, either bolted tight or with the safety pin for the spring-lever lock-ring installed.
        - Funnels are configured with (1) bung fittings that are tightly secured to the drum and (2) hinged covers that are kept closed except when adding liquid waste.
        - Storage boxes must be in good condition and of sufficient size to fully contain the waste. The flaps of partially full boxes may be folded closed. Full boxes must be securely taped closed.
    - Used oil drums with drain pans on top may remain open while draining oil from filters or absorbent wringers. When draining is completed (no more than 12 hours from the start), a lid- type cover must be secured over the drain pan. If the drain pan would allow the contents to leak when the drum was turned over, a bung must be inserted into the drum opening. After absorbents are wrung out, the absorbents must be placed in a drum and the lid closed and secured
    - Maintain sufficient aisle space, at least 36 inches, to provide ready access for handling and inspection.
    - Containers with liquids will only be filled to 90 percent capacity, or not less than 4 inches of headspace. Containers with solids (like absorbents and painting debris) may be filled to 95 percent capacity or not less than 2 inches of headspace. The headspace is needed to allow for waste expansion.
    - Containers of ignitable waste must be protected from sources of ignition and be grounded. Post signs prohibiting smoking or open flames on all inside walls.
    - Ensure markings and labels are clearly visible from the aisle.
    - Immediate action shall be taken to replace or over pack hazardous waste containers that are found to be dented, rusted, corroded, leaking, missing bungs, or otherwise damaged. Spilled or leaked waste and spill residue must be managed as hazardous waste. Call the CAP (723- 4074/-4142) to obtain an over pack container.
    - If a container is found to be bulging, notify the Fire Protection Flight and Occupational Safety for assistance. **Do not attempt to open the container.**
    - If a container is found to be leaking, implement the Site-Specific Contingency Plan and/or the Minot AFB Hazardous Materials Emergency Response Plan as appropriate.

Empty Container Management

The compliant management of "empty" containers that previously contained a hazardous substance encompasses proper management of any residual substance and of the container itself. Regulations allow for the reuse or recycling of containers that previously contained a hazardous material or a hazardous waste if certain requirements are met relating to the type of hazardous substance involved, the amount of residual material, and applicable emptying and cleaning procedures. If a container can be reused or recycled, it is not subject to RCRA regulations. Otherwise, the container may have to be managed as a hazardous waste. If a hazardous material or hazardous waste container can be considered "empty" (as defined in 40 CFR 261.7, *Residues of Hazardous Waste in Empty Containers*), then any residue and the container itself are normally exempt from RCRA regulations. However, Minot AFB has stricter requirements for managing empty containers.

Hazardous Material Containers

- A container that holds a hazardous material is managed in accordance with AFMAN 32-7002, *Environmental Compliance and Pollution Prevention*. However, when a determination is made that any remaining hazardous material can no longer be used and must be disposed of, the hazardous material is normally re-categorized as a hazardous waste and the container then becomes a hazardous waste container. In particular, to avoid having to manage the container as a hazardous waste, the container must satisfy the exemption criteria for empty hazardous waste containers per 40 CFR 261.7 and as described in the Management of Empty Containers and Gas Cylinders section below.

Per 40 CFR 261.7, residual hazardous waste in an "empty" container is not subject to regulation under RCRA. A hazardous waste container that held any hazardous waste is considered to be empty if the following criteria are satisfied:

- A container that held any hazardous waste *except* a compressed gas or an acutely hazardous waste as listed in 40 CFR 261. 31, 32, or 33(e) is considered empty if all of the waste material has been removed that can be removed using the practices commonly employed to remove materials from that type of container (for example, pouring, pumping, and aspirating) *and* no more than 1 inch or 3 percent (0.3 percent if the container has a capacity of more than 110 gallons) remains. **However, on Minot AFB, removal to this standard is not sufficient to qualify the container as empty if additional removal is possible. For example, a drum that has been hand-pumped must also have the residual poured out by inverting the drum, if possible.**
- A container that contained a hazardous compressed gas is considered empty when the pressure inside the gas container approaches atmospheric pressure.
- A container that contained an acutely hazardous waste as listed in 40 CFR 271 .31, 32, or 33(e) is considered empty if (1) the container has been triple rinsed using a solvent capable of removing the acutely hazardous waste; (2) the container has been cleaned using another method that can achieve equivalent results as evidenced by tests performed by the generator or as reported in the scientific literature; *or* (3) the liner that contained the acute hazardous waste has been removed. **Given the complexity and associated cost of emptying a container that previously contained an acute hazardous waste, such containers will be managed as hazardous waste on Minot AFB.**

Cleaning Empty Containers

- Triple Rinsing Empty Containers. A container that meets the definition of empty as described above may be triple rinsed for reuse/recycling if the residual content of the container is soluble in water or a soap/water solution. Empty petroleum containers (that is, fuel, oil, hydraulic fluid) may be effectively rinsed with a warm soap/water solution, and do not need to be triple rinsed if reused. Empty organic solvent (for example, MEK, acetone, alcohol, and antifreeze) containers may be effectively rinsed with warm water. Contact Environmental Management or Bioenvironmental Engineering for assistance in selecting the proper rinsing solution. Triple rinse the container as follows:
    - Add cleaning solution into the container equivalent to 10 to 25 percent of the container volume and rinse the entire inner surface of the container. Discard the cleaning solution into the sanitary sewer at the location designated for your organization. Contact your Unit Environmental Coordinator or Environmental Management for the location of an approved discharge location. **Do not dump rinse solution into an oil water separator, storm water drain, or on the ground.**
    - Repeat the previous step two additional times using an equivalent volume of the rinse solution.
    - Stencil the container with the words "EMPTY" and "TRIPLE RINSED" on the top and side of the container. Writing, "3X rinsed" is not acceptable.
- Cleaning Containers that Cannot Be Triple Rinsed
    - Empty containers that previously contained a hazardous substance that cannot be removed by rinsing (for example, putty, joint compound, paint) may be wiped or scraped clean of residual commodity inside and out. Before starting the cleaning process, the contents should be used up to the maximum extent possible.
    - Recyclable rags used in this process may be laundered as described above for contaminated rags, as long as the rags are not saturated with product. Rags saturated with product must be managed, characterized, and disposed of as waste.
    - Residues of the container cleaning process (for example, paper towels, paint chips, and sludge) shall be handled in the same manner as the waste stream that comprised the original contents.
    - Containers cleaned in this manner will be marked "EMPTY" and "CLEANED" and set aside for reuse/recycling.
- Management of Empty Containers and Gas Cylinders. Hazardous material/waste containers meeting the definition of "empty" as detailed above and gas cylinders are to be managed as follows:
    - Empty containers having a volume of 1 gallon or less of non-RCRA or non- hazardous materials may be disposed of in the base refuse; however, metal containers should be turned in for recycling.
    - Note that hazardous material containers that include an internal application device (for example, dauber for polyvinyl chloride (PVC) cement or PVC cleaner) may need special handling and should be returned to CAP for evaluation and proper disposal.
    - Containers greater than 1 gallon in volume may be reused at the shop for the same material or a compatible waste stream. For example, an empty antifreeze container may be used to accumulate used antifreeze or an empty paint bucket may be used for paint mixing. Empty containers greater than 1 gallon in volume will be stenciled "EMPTY" while being stored for reuse or recycling. Remember to contact HAZMART to close out the barcode for the original contents of the container as applicable.

- When an empty container is returned to service, delete the word “EMPTY” and mark/label the container as appropriate for its planned use.
- All containers having a capacity of greater than 1 gallon that cannot be reused in the shop will be returned to the CAP for proper disposal. Prepare the container for reuse/recycling by cleaning and labeling as described above. Coordinate transportation with the CAP as required. The CAP will place reusable containers in the empty container inventory and arrange periodic deliveries of recyclable containers to DLA-DS.
- Expended propane/MAPP™ gas cylinders with a residual internal pressure of 2 psi or less must be completely emptied by holding the stem down and then removing the stem. These cylinders can then be recycled as scrap.

- Various contracts are in place at Minot AFB for the management of larger gas cylinders. If a gas cylinder cannot be disposed of through one of these contracts, it should be disposed of through the CAP. Large, unserviceable cylinders are purged by contractor personnel using a gas recovery system. The valve stem is then removed and the cylinder is recycled. Note that DLA-DS requires purged cylinders to be rendered unusable by cutting or drilling a hole into the cylinder or by destroying the inlet threads such that a valve cannot be re-installed. In addition, the words “CONDEMNED SCRAP” must be metal stamped into the shoulder of the cylinder.

7.4 Labeling and Marking

Containers used for the accumulation and transportation of HW are properly labeled IAW applicable laws and regulations.

Each container is properly marked and labeled from an IAP to HWAS to disposal/turn-in. The waste-generating activity ensures that the label on each waste container is clearly visible for inspection. During accumulation at an IAP, HW containers are marked with the following:

- The words “Hazardous Waste”
- A description of the contents of the container
- The hazards associated with the waste

Once an IAP accumulates more than 55 gallons of HW (or 1 quart of acute HW), the IAP site manager marks the container with the date on which 55 gallons (or 1 quart of acute HW) is exceeded and removes the excess of 55 gallons (or 1 quart of acute HW) within three days.

HW containers 110 gallons or less that are shipped offsite are marked with the following:

- “Hazardous Waste – Federal Law prohibits Improper Disposal. If found, contact the nearest police or public safety authority or the U.S. Environmental Protection Agency”
- Generator’s name and address
- Generator’s EPA ID Number
- Manifest tracking

These markings are:

- Durable
- In English
- Printed on or affixed to the surface of a package or on a label, tag, or sign displayed on a background of sharply contrasting color
- Unobscured by labels or other attachments
- Located away from any other markings that might substantially reduce visibility or effectiveness

Universal waste (UW), or a container in which a UW is contained, is labeled and marked clearly with the date the material became a waste and the name of the waste, as described below:

- UW batteries must be labeled with any one of the following phrases: "Universal Waste—Battery(ies)," or "Waste Battery(ies)," or "Used Battery(ies)"
- UW thermostats must be labeled with any of the following phrases: "Universal Waste-Mercury Thermostat(s)," or "Waste Mercury Thermostat(s)," or "Used Mercury Thermostat(s)"
- UW pesticides must be labeled with one of the following phrases: "Universal Waste—Pesticide(s)" or "Waste-Pesticide(s)"
- UW lamps must be labeled with one of the following phrases: "Universal Waste—Lamp(s)," or "Waste Lamp(s)," or "Used Lamp(s)"
- UW aerosol cans must be labeled with one of the following phrases: "Universal Waste—Aerosol Can(s)," or "Waste Aerosol Can(s)," or "Used Aerosol Can(s)"

Waste Stream and Container Number Designation

Regulations require that waste be tracked from "cradle to grave." Minot AFB uses unique container numbers to aid the tracking requirement. The container number is a code that includes the waste stream identification number and a unique 3-digit sequence number added by the shop Hazardous Waste Manager. To create a waste stream and container number, accomplish the following four steps:

- Select the code from the Hazardous Waste Process Code Table that most closely matches the process generating the waste. If the process does not fit any of the categories listed, use "MS" for miscellaneous. For example, BA-1901-001 refers to a container holding batteries.
- The two digits after the process code are the accumulation area shop identification number. In this example, BA-1901-001 refers to the CAP. If your shop is not listed, contact the Hazardous Waste Program Manager (723-4870) for direction.
- The next two digits refer to the waste stream identifier, which must be unique to the waste stream. Thus, BA-1901-001 may be alkaline batteries, BA- 1902-001 may be lithium batteries, and BA- 1903-001 may be lead acid batteries. In this example, each battery waste stream has its own waste stream identifier (01, 02, 03), but the same process code (BA).
- The last three digits in the sequence denote a specific container. Since each container must be uniquely identified, shop Hazardous Waste Managers are responsible for preventing duplication of container numbers and accurate recordkeeping by using an Accumulation Storage Log. Thus, the first container of alkaline batteries is BA-1901-001. The next one is BA-1901-002, then BA- 1901-003, and so on. When you get to container -999, **DO NOT START OVER!** Go to the next waste stream available under that process code. In this example above, when the CAP gets to BA- 1901-999, the next container will be BA-1904-001 because waste streams BA-1902 and BA-1903 are already used.

RCRA Hazardous Waste Container Markings

Each hazardous waste container must be marked in a specific manner. The shop Hazardous Waste Manager must use a label. The following directions describe the process for either choice.

Hazardous Waste Label

The Hazardous Waste Label must conform to the label shown in Appendix C. This label is only placed on containers used to store RCRA hazardous wastes. Information on the label must be legibly written in permanent ink using a fine-point marker.

Hazardous Waste Process Code

Process Source	Examples	Process Code
Abrasive Blasting	Grit, paint chips, expended media	**AB**
Aircraft Cleaning	Cleaning compounds	**AC**
Battery Shops	Battery acid, lead	**BA**
Biological Operations	Pesticides, such as insecticides, herbicides, rodenticides, etc.	**BO**
Boiler Operations	Morpholine, nitrates	**BL**
Chemical Paint Stripping	Paint strippers, paint sludge	**CP**
Decarbonizers	Nitric acid, sulfuric acid, others	**DE**
Avionics/Electronics	Solvents	**EE**
Electroplating	Cleaning compounds, chromium, cyanide, tank sludges contaminated with heavy metals	**EP**
Expired Shelf Life	Expired shelf life wastes such as paints, solvents, cleaning materials, etc.	**ES**
Fire-Fighting Operations	AFFF (Aqueous Film Forming Foam)	**FF**
Fluids Change out/Purging	Oily waste, turbine oil, lube oil, hydraulic fluid, contaminated fuel, brake fluid, antifreeze, fluid containing filters	**FC**
Industrial and Facility Maintenance	Cleaning supplies, mercury vapor lamp bulbs, PCBs	**IM**
Industrial Operations	Tool and machine waste, cutting oils	**IO**
Industrial Waste Treatment	Wastewater treatment sludge, chlorine	**IW**
Installation Restoration Program (IRP)	Waste from cleanup of IRP sites	**IR**
Laboratory/NDI	Samples, test chemicals, penetrant	**LA**
Medical	Mercury, test chemicals, pharmaceutical, chemotherapeutic drugs, infectious waste	**ME**

Miscellaneous	Only wastes that cannot possibly be attributed to any other process listed	**MS**
Ordnance, Demolition/Disposal	Explosive, pyrotechnic, propellant, lead- contaminated water, soil, or dust	**OD**
One-Time Only	Purging underground storage tank prior to replacement	**OO**
Oil Water Separators	Contaminated sludge, floating product	**OS**
Painting Operations	Paint, paint sludge, filters, surface preparation (solvents)	**PO**
Preservation and Packaging	Pentachlorophenol, copper arsenite	**PP**
Photo/X-ray	Fixer, developer	**PX**
Research and Development	Chemical testing, equipment testing	**RD**
Recycled Material	Material recycled on or off base and returned to original use	**RE**
Spill Cleanup	Absorbents, rags, contaminated soil	**SC**
Solvents/Degreasing/Cleaning	Cold cleaning solvents, PD 680, MEK	**SO**

Notes:
IRP Installation Restoration Program
MEK methyl ethyl ketone
PCB polychlorinated biphenyl

The container number will be written on the label and logged onto the Accumulation Storage Log sheet when the first drop of waste is put into the container. Each container number must be unique and created in accordance with the directions this section. The container number is written above the Generator information as shown.

- Fill in the generator name, address and phone number as shown. Do not use an office symbol for the generator name.
- The EPA identification number is the one used for Minot AFB: ND4571924758. If a missile flight must manifest waste from the field to Minot AFB, the Hazardous Waste Label will have the EPA identification number that matches the missile flight. See the EPA Identification Numbers for 91st Missile Wing Missile Flights table in Section 7.11 for a list of all the missile field EPA numbers.
- The space for the Manifest Document Number is left blank
- The ASD is the date a container in a SAP is determined to be full. Containers filled at the CAP have an ASD of the day the first drop of waste is put into the container. For expired shelf-life items, the ASD is the date the container is determined to be a waste. **When the ASD is put on the label, the waste must be moved to the CAP in no more than 3 calendar days.** Curbside pickup can be arranged by calling the CAP at 723-4074 or 723-4733.
- The label will have all EPA waste codes that apply. The EPA waste codes can be found on the associated Hazardous Waste Profile Sheet. The codes must be put on the label when the first drop of waste is put into the container. If the waste stream is new and/or the hazards are unknown, contact the installation Hazardous Waste Program Manager (723-4870) for direction.
- Add the name of the waste to the appropriate area on the label. The name of the waste (or nomenclature) always starts with the word "Waste." The nomenclature should describe the waste and easily identify the contents of the container to prevent unintended or incompatible materials being put into the container.
    - If the waste is a mix of various process debris (for example. mixing cups for paint, latex gloves, masking paper, and stir sticks), it is not necessary to list all items. A generic statement, such as "dried paint and paint debris," is acceptable. Do not use the abbreviation "etc." in the waste description.
    - Although the DOT Proper Shipping Name is supposed to be written in this section, Minot AFB allows SAP Managers to use a more descriptive waste name, which more clearly identifies the container contents and reduces the potential for mixing dissimilar or incompatible wastes. At the time the waste is manifested for shipment off base, DLA- DS affixes a new label that contains all required information and the correct DOT Proper Shipping Name.

Example Accumulation Storage Log

Accumulation Storage Log

Waste Stream Number: ***MS-4301*** **Waste Stream Description:** ***Sealant***

Container Number	Container Size/Type	EPA Code(s)	Waste Start Date	Date Container is Full	Date Container is taken to CAP
001	*30 gal plastic, open head*	*D001, D007, D035*	*1 June 2008*	*3 Dec 2008*	*4 Dec 2008*
002	*30 gal plastic, open head*	*D001, D007, D035*	*4 Dec 2008*	*8 July 2009*	*8 July 2009*
003	*30 gal plastic, open head*	*D001, D007, D035*	*8 July 2009*	*5 Jan 2010*	*6 Jan 2010*
004	*30 gal plastic, open head*	*D001, D007, D035*	*6 Jan 2010*		

005
006
007
008
009
010
011
012
013
014

The container must be marked with the words "Hazardous Waste" and the name of the waste, such as "spent solvent" or "spent paint and paint debris." The name of the waste must be written just below the words "Hazardous Waste."

- Next to, or just under, the name of the waste, write in the appropriate EPA identification code(s). Include all that apply. For example, a waste may be marked as "Spent sealant and debris D001 D035."
- Below the waste name is the container number. Each container number must be unique, recorded on an Accumulation Storage Log sheet, and created in the format shown in this section.

Marking Full Containers

When the container is considered full, write in the Accumulation Start Date below the container number. The format for this date is "ASD: 22 Jun 2009" or "22 June 2009." Do not use another format for the date. When the Accumulation Start Date is put on the container, it must be moved to the CAP within 3 calendar days. For expired shelf-life items, the ASD is the date the item is determined to be a waste.

Industrial Waste Container Markings

Industrial waste is not subject to RCRA regulations, but is regulated by North Dakota law. Industrial waste containers must be marked with the waste name, container number, and an Accumulation Start Date. SAP Managers must use a label and adhere to the following procedures for marking industrial waste containers:

- Labels must conform to the "Non-RCRA Regulated Waste" label shown in Appendix C. This label is used when the waste is not a RCRA waste, but is regulated by North Dakota. Information on the label must be legibly written in permanent ink using a fine point marker.
- The container number must be written on the label and logged onto the Accumulation Storage Log sheet when the first bit of waste is put into the container. Each container number must be unique and created in accordance with the directions in this section. The container number is written above the Generator information as shown.
- Fill in the generator name, address and phone number as shown. Do not use an office symbol for the generator name.
- Add the name of the waste to the space for the "Proper DOT Shipping Name." The name of the waste (or nomenclature) always starts with the word "Waste." The nomenclature should describe the waste and easily identify the contents of the container to prevent unintended or incompatible materials being put into the container.
    - If the waste is a mix of various process debris (for example, mixing cups for paint, latex gloves, and stir sticks), it is not necessary to list all items. A generic statement, such as "paint and paint debris" is acceptable. Do not use the abbreviation "etc." in the waste description.
    - Although the DOT Proper Shipping Name is supposed to be written in this section, Minot AFB allows SAP Managers to use a more descriptive waste name, which more clearly identifies the container contents, and reduces the potential for mixing dissimilar or incompatible wastes. At the time the waste is manifested for shipment off base, DLA- DS affixes a new label that contains all required information and the correct DOT Proper Shipping Name.
- The Accumulation Start Date (ASD) is the date a container in a SAP is determined to be full. Containers filled in the CAP have an ASD of the day the first drop of waste is put into the container. For expired shelf-life items, the ASD is the date the container is determined to be a waste. When the ASD is put on the label, the waste must be moved to the CAP in no more than 3 calendar days. Curbside pickup can be arranged by calling the CAP at 723-4074 or 723-4733.
- The label in Non-RCRA Regulated Waste Container has space for the UN or NA number. Leave this line blank. It will be filled in by DLA-DS on their shipping label before the waste is removed from base.
- Enter the shop Hazardous Waste Manager's phone number on the label.

Marking Full Containers

Write the ASD below the container number when the container is considered full, or an expired shelf life item is determined to be a waste. The format for this date is "ASD: 22 June 2009" or "ASD: 22 Jun 2009." Do not use another format for the date. When the ASD is put on a container, it must be moved to the CAP within 3 calendar days. Contact the CAP at 723-4074 or 723-4733 to arrange for curbside pickup of the waste.

Universal Waste Container Markings

- Use a copy of the labels shown in Appendix C, as appropriate, on each container of universal waste. The container label shall be marked with the words "Universal Waste" followed by a category (for example, Battery, Mercury-Containing Device) and the associated nomenclature (for example, NiCad, Mercuric Oxide, Thermometer, or Fluorescent Lamp). Information on the label must be legibly written in permanent ink using a fine- point marker. Each container shall be assigned a unique container number by the Hazardous Waste Manager.
  Each universal waste may only be accumulated in the shop for up to 270 days (9 months). Thus, the label must include the date when the first item of universal waste is placed in the container.
- Mark each universal waste container with the name of the shop that generated the universal waste and the phone number of the Hazardous Waste Manager.

Marking Containers Awaiting Analysis

- **Containers awaiting analytical results must always be marked.**
- Write on the side of the container with an appropriate/color contrasting paint pen that the contents are awaiting analysis and the date the samples were sent to the laboratory.
- If the waste is being re-sampled because the analysis is old, or the process chemicals have changed, label the container with the appropriate label used for the previous analytical results. For example, if a container contains waste paint previously characterized as RCRA hazardous for chromium (D007), affix the RCRA hazardous waste container label (Appendix C) on the container while awaiting results from the new sampling effort.
- If the waste is new or unknown, use the Non-RCRA Regulated Waste container label (See Appendix C). Change the label as appropriate when the sample results are received.

Container Markings for Unidentified Waste

If an unlabeled container is discovered and the contents of a container are unknown, the discovering activity must notify Environmental Management or the CAP immediately so requirements for sampling and analysis of the contents may be determined. The container must be stored in the CAP until the analysis/characterization is complete—normally within 4 weeks. The container should be marked with all the information known about the container and its contents including:

- Mark with the words "Awaiting Analysis" and note the sample number and date the sample was drawn;
- Date and location where it was discovered;
- Office symbol of the owning organization.
- Name and phone number of the point of contact;
- Appropriate nomenclature (for example, paint chips); and
- Appropriate container number.

Non-Hazardous Waste Container Markings

Use the "Non-Hazardous Waste" label (Appendix C) to distinguish drums/containers containing waste that is not regulated by RCRA or by North Dakota from drums/containers that do contain regulated waste. The generator will assign a container number to drums or containers containing non-hazardous waste.

Container Markings for Unidentified Waste

If an unlabeled container is discovered and the contents of a container are unknown, the discovering activity must notify Environmental Management or the CAP immediately so requirements for sampling and analysis of the contents may be determined. The container must be stored in the CAP until the analysis/characterization is complete—normally within 4 weeks. The container should be marked with all the information known about the container and its contents including:

- Mark with the words "Awaiting Analysis" and note the sample number and date the sample was drawn;
- Date and location where it was discovered;
- Office symbol of the owning organization.
- Name and phone number of the point of contact;
- Appropriate nomenclature (for example, paint chips); and
- Appropriate container number.

Non-Hazardous Waste Container Markings

Use the "Non- Hazardous Waste" label (Appendix C) to distinguish drums/containers containing waste that is not regulated by RCRA or by North Dakota from drums/containers that do contain regulated waste. The generator will assign a container number to drums or containers containing non-hazardous waste.

Accumulation Storage Log

Container numbers must be recorded on an Accumulation Storage Log sheet. One log sheet should be used for each waste stream. For example, containers for waste streams BA-0101, BA- 0102, and BA-0103 would be listed on separate sheets. Enter the waste stream number on the line marked "Waste Stream" followed by the waste name (for example, "waste solvent"). Complete the remaining columns as follows:

- Container Number: Enter a unique number for each container, such as -001, - 002, -003, etc.
- Container Size/Type and Specification: Enter the size and type of container, such as "steel 55-gal drum, open head."
- EPA Code: Enter the EPA waste code(s) from the Waste Profile Sheet. If the waste is non-RCRA, Industrial, or universal waste, this column is to be left blank.
- Waste Start Date: Enter the date waste is first put into the container. It is Minot AFB policy that waste should not be accumulated in a shop for more than one year. (The only exception is LLRW.) If the container is not full at the end of one year, use a smaller container for future accumulations. This date is NOT the ASD.
- Date Container is Full: Enter the date the container is declared "full." Liquid waste should be accumulated to about 90 percent of the volume of the container, or about 4 inches from the top. Solid wastes can be accumulated to about 95 percent full or about 2 inches from the top. When full, containers shall be taken to the CAP within 3 calendar days. If the container is an expired shelf-life item, it should be moved to the CAP within one week of the determination that it is a waste. It is the responsibility of every SAP Manager to work with HAZMART to prevent expired shelf-life items from being wasted out. However, expired shelf-life items that are wasted out must never be stockpiled or put into closets, flammable lockers, or remote storage areas to delay the proper disposal of the item. Expired shelf life items are reported to the installation Hazardous Waste Program Manager for review. The installation Hazardous Waste Program Manager will analyze the data and report to the shops specific squadron commander if the quantities of expired shelf life items are deemed extensive enough to constitute a Pollution Prevention initiative.

DOT Hazard Labels

In addition to the container markings required by RCRA and this plan (Section 7.4), DLA-DS requires that hazardous waste containers be labeled with DOT hazard labels to identify the potential hazards posed by the contents. DOT labels will be added to SAP containers when they are moved to the CAP. Note that the DOT labels may not be altered or modified. The DOT hazard label must be positioned within 6 inches of the Hazardous Waste Label on the upper one-third of the container. When two or more DOT hazard labels apply, they are to be positioned adjacent to each other. Containers should be positioned so that the DOT hazard labels are readily visible in the event of an emergency.

7.5 Accumulation Area Management

Accumulation area management procedures are as follows:

- IAPs are used to accumulate up to 55 gallons of HW or 1 quart of acute HW
- If HW or acute HW are accumulated in excess of these amounts, the generator marks the container with the date the amount was exceeded and transfers the container to a HWAS or TSDF within 3 days
- HW is accumulated in a HWAS for up to 90 days for a large quantity generator (LQG), or 180 days (270 days if waste has to be shipped over 200 miles) for a small quantity generator (SQG)
- HWASs comply with all applicable federal, state, and local accumulation requirements, including proper waste segregation

Installation Supplement

Satellite Accumulation Points

A SAP is a storage area located at or near a point of generation where waste associated with a process is initially accumulated. SAPs are allowed to (1) accumulate not more than 55 gallons of a hazardous waste or not more than one quart of an acutely (P-listed) hazardous waste and (2) extend the waste storage TIME BEYOND THE 270 DAY LIMIT. It is emphasized that a SAP shall only accumulate and store RCRA and state regulated (industrial) waste generated at the shop where it is located. RCRA and stateregulated (industrial) wastes may not be consolidated from different work centers even if they belong to the same squadron, group, or wing, unless prior approval is obtained in writing from the installation Hazardous Waste Program Manager. Shops may consolidate used oil and universal waste from different work centers within the same squadron. Shops operating or wanting to operate a SAP shall adhere to the following procedures for establishing, operating, and closing a SAP.

Types of SAPs

Minot AFB accumulates various types of wastes, including RCRA regulated, RCRA-exempt, state regulated (industrial), recyclable, universal, and non- hazardous wastes. Whether a work center should establish a SAP is based on several factors including the hazardous materials used, the type of hazardous waste(s) generated, and the frequency and the volume generated. The following guidance should be used to determine when a SAP is needed. Given that not all situations concerning waste generation can be addressed here, contact the installation Hazardous Waste Program Manager (723-4870) with any questions about establishing a SAP.

- Shops Generating RCRA Regulated Wastes. These wastes are identified in 40 CFR 261, Subparts C. A shop generating hazardous wastes shall establish a SAP for initial waste accumulation regardless of the volume or frequency of generation.
- Shops Generating RCRA-Exempt Wastes. These wastes are specified in 40 CFR 261.4(b) and Subpart E. Exempt wastes currently generated at Minot AFB, include universal waste (discussed below), lead-acid batteries (when recycled), used oil, used oil filters (when recycled), and mixed low-level radioactive waste (LLRW). Shops generating LLRW must establish a SAP, obtain training, keep records, handle the wastes, and maintain the SAP in the same manner as a shop generating RCRA-regulated wastes.
- Shops Generating State-Regulated/Industrial Wastes. Minot AFB refers to state-regulated wastes as "industrial wastes." North Dakota bans the disposal of wastes containing certain organic or inorganic constituents and liquid waste into municipal landfills. These types of waste include oily, greasy, and other petroleum-contaminated wastes, aerosols, paints, solvents, alcohols, antifreeze, dilute acids, dilute bases, and wipes, applicators, and other debris contaminated with these chemicals. Shops generating these types of wastes in any volume greater than once per year must establish a SAP. Shops generating these wastes on a less- frequent basis do not have to establish a SAP, but must appoint Hazardous Waste Managers and adhere to the same training and record- keeping requirements as shops with an established SAP.
- Shops Generating Recyclable Hazardous Wastes. Minot AFB recycles many types of hazardous and state-regulated/industrial wastes including paint, antifreeze, used oil, used oil filters, parts washer solvent, and lead-acid batteries. These items are managed as hazardous waste until they are recycled. Shops generating only these items do not have to establish a SAP, but must have a designated area to store these items unless they are turned into the CAP as soon as they are generated. The shop supervisor shall also designate Hazardous Waste Managers who are responsible for proper storage, management, and record keeping requirements.
- Shops Generating Universal Wastes. Universal wastes are items specifically designated by the EPA that can be managed under less restrictive waste management rules. Universal wastes include fluorescent and other specialty lamps, batteries, some pesticides, and mercury containing devices (such as, mercury thermometers, switches). Shops generating only these items do not have to formally establish a SAP, but must have a designated area to store these items and designate Hazardous Waste Managers who are responsible for proper storage, management, and record keeping.
- Shops Generating Non-Hazardous Wastes. Shops generating only non- hazardous wastes do not have to establish a SAP. By definition, non-hazardous wastes are solid waste and can be put into the installation trash, with the following exceptions:
    - Powders and fine particles should be disposed of in their original container wrapped in a plastic bag. If the original container is not available, damaged, or leaking, the powder should be double-bagged and securely closed.
    - **North Dakota prohibits the disposal of liquid wastes of any kind in dumpsters or installation trashcans.** Generators of non-hazardous liquids should contact the Hazardous Waste Program manager (723-6380) for direction.
    - Large volumes of non-hazardous wastes may be collected for disposal. A large volume is greater than 1 quart of liquid or 1 pound of solid chemical waste. Generators of this waste should call the installation Hazardous Waste Manager (723-6380) for guidance.
- Special Project Generators. Special projects are self-help projects, quality-of- life initiatives, and commander-directed projects that are accomplished by installation personnel to improve their surroundings. Typically, such projects will produce wastes that may be hazardous or not, depending on the materials used. Wastes generated from these projects do not have to be accumulated in a SAP, but they must be properly managed nonetheless. The project lead must designate in writing an individual who is responsible for collecting the wastes, ensuring the wastes are properly managed and disposed of, and maintaining records in accordance with this section. At least 1 week prior to the project start, the name of the designated

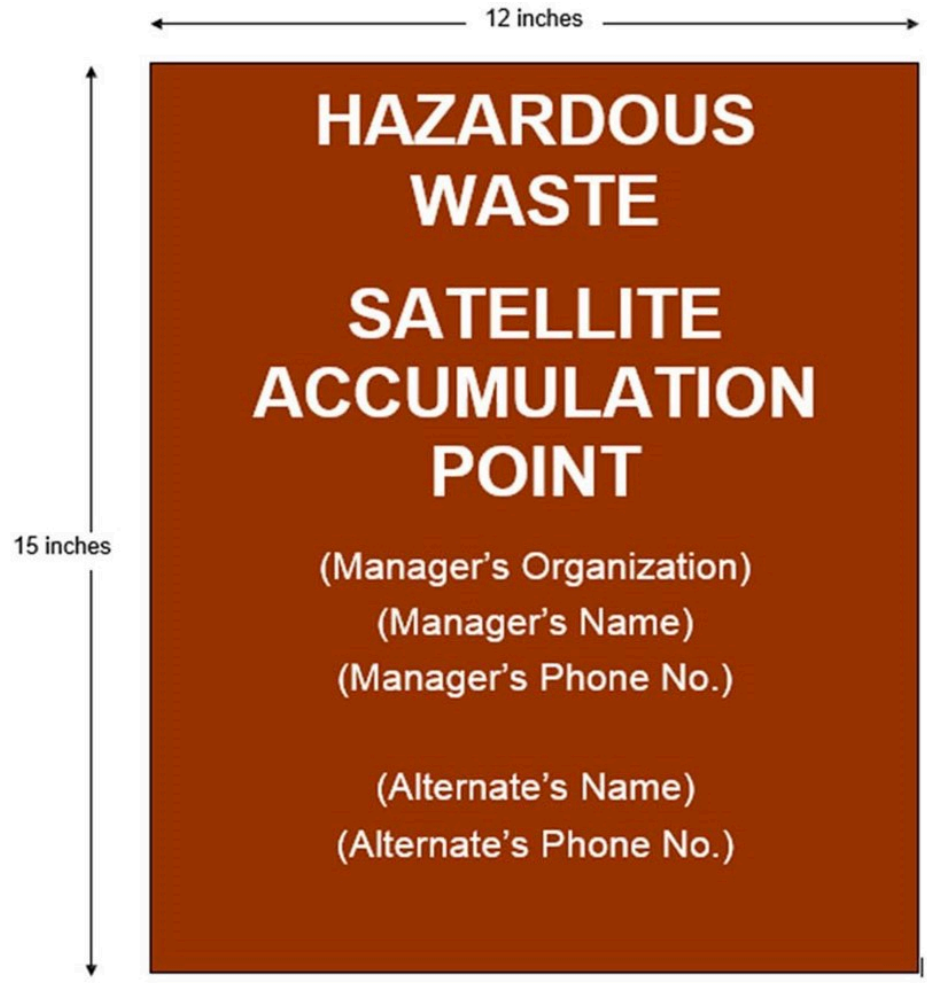
person, the project name and location, a list of hazardous materials procured, and an estimate of the waste volume that will be generated shall be sent to the installation Hazardous Waste Program Manager.

Establishing a SAP

Shops wanting to establish a SAP shall follow the guidelines below or contact the installation Hazardous Waste Program Manager (723-6380) for guidance before disposing of any wastes.

- Approvals. Organizations that want to establish a new SAP must contact Environmental Management, the Fire Protection Flight, and Bioenvironmental Engineering and obtain written approval for the siting of the SAP. Written approval for establishing a SAP must be obtained **prior to generating and accumulating any hazardous wastes.**
- Siting Criteria. Specific criteria apply to the location of a SAP on Minot AFB. In particular, a SAP must be:
    - At or near the point of waste generation; this is interpreted by North Dakota to be within the building of the process generating the waste. Ideally, the collection point should be within line-of-sight of the process. o Under direct control of the operator(s) of the waste generating process (that is, operator(s) must be in the same building as the process generating the waste). If the SAP is not located within line-of-sight of the process, it must be secured to prevent unauthorized personnel from accessing waste in the SAP container.
    - Conspicuously posted with a SAP-designation sign, which is illustrated in Hazardous Waste Accumulation Point Sign.
    - Posted with warning signs indicating "No Smoking or Ignition Sources within 50 Feet" from all approachable directions.
    - Located away from any floor drains, incompatible materials, and ignition sources.
    - The SAP should be on a hard surface, such as concrete. Where the SAP is on an absorbent surface, such as carpet, all waste containers must be placed in/on secondary containment or within a flammable locker.
    - SAPs may be located exterior to the building with the prior written approval of the installation Hazardous Waste Program Manager, Bioenvironmental Engineering, Ground Safety, and the Fire Protection Flight. o Exterior SAPs must be located as close to the waste generating process as possible, fully protected from the weather, and have secondary containment. No waste containers may be placed on the ground. Wastes must be protected from freezing during cold weather and have sufficient headspace to allow for waste expansion during hot weather. The SAP must also be secured to prevent unauthorized personnel from adding or removing waste to the SAP container.

Hazardous Waste Satellite Accumulation Point Sign



Color: Standard white letters on standard brown background.

Layout: Spacing as indicated on the illustration.

Heading: Uppercase Arial regular, 1-inch capital letter height.

Emblem: Emblems are not authorized.

Management of a SAP

- Regulatory Requirements. When RCRA was first enacted, there were no provisions for SAPs. Later, RCRA was amended to allow for SAPs, which are called "Initial Accumulation Points" (IAPs) in the regulation, and "Satellite Accumulation Points" (SAPs) by Minot AFB. The regulation requires hazardous waste containers to be in good condition, compatible with the waste they contain, and closed at all times. The regulation also requires the containers be marked, dated when full, and moved to the CAP within 3 calendar days of becoming full (40 CFR 262.34(c)). RCRA does not require SAP inspections, impose limits on accumulation times, mandate personnel training, require compliance with air emission standards, or prescribe emergency management and closure procedures.
- Minot AFB Requirements. Minot AFB requires SAPs be managed per more stringent procedures than required by federal and state regulations to ensure consistent compliance and generator accountability. Specifically, SAPs on Minot AFB must be managed as follows:
    - Containers and the SAP must be inspected monthly by the shop Hazardous Waste Manager, with each inspection not more than 30 days apart.
    - SAP containers shall be placed in a secure area and labeled to prevent accidental mixing of incompatible or hazardous with non-hazardous wastes, and to maintain the integrity of the contents without requiring analysis.
    - Containers holding liquid wastes must be provided secondary containment and not be filled to more than 90 percent.
    - Containers holding solid wastes must not be filled to more than 95 percent, but such containers do not have to be placed on secondary containment when stored inside a building with a hard floor.

- Containers stored outside or on carpeting must be placed on secondary containment and protected from precipitation and extreme temperatures.
- Personnel generating or handling hazardous wastes and their supervisors must obtain initial and annual refresher hazardous waste training.
- Ignitable waste containers must be grounded at all times.
- Incompatible wastes must not be mixed and must be stored separately.
- Wastes should ideally be containerized so they are disposed of approximately annually, except for mixed LLRW, which is disposed of as needed.
- Shops shall use a Site-Specific Spill Plan to detail emergency management procedures appropriate to their wastes and shop practices; the plan must be updated as managers, locations, and wastes change.
- SAPs must be established and closed by written acknowledgment from Environmental Management, Bioenvironmental Engineering, and Ground Safety.

- Delegation of Hazardous Waste Managers
  - The responsible unit Commander or delegated official shall appoint at least two Hazardous Waste Managers, one primary and one alternate, for each SAP. If the Hazardous Waste Manager will be absent for an extended amount of time (that is, greater than 3 weeks), an additional manager shall be delegated. Names must be submitted in writing to the installation Hazardous Waste Program Manager within 14 calendar days of appointment. The delegation letter shall be signed by the responsible Squadron Commander or delegated official.
  - One Hazardous Waste Manager should be present on each work shift. Exceptions are made on a case-by-case basis by the installation Hazardous Waste Program Manager (723-4825/6380).
  - It is the responsibility of the primary Hazardous Waste Manager to ensure all alternate Hazardous Waste Managers are trained and thoroughly understand their duties.
  - Shop Hazardous Waste Managers are responsible for ensuring the SAP and any associated hazardous wastes are managed in accordance with this plan and all applicable federal and state regulations. In addition, Hazardous Waste Managers must ensure all shop personnel generating or handling hazardous wastes and their supervisors are trained in accordance with Section 5.0 of this plan.
- Container Management
  - In accordance with 40 CFR 262.34(c), a SAP may accumulate no more than 55 gallons of a hazardous waste or 1 quart of an acutely P-listed hazardous waste. If incompatible, or both hazardous and non-hazardous wastes are generated, separate containers must be used to store the wastes.
  - All hazardous waste must be accumulated in DOT approved containers. Note that federal regulations prohibit the use of tanks to accumulate hazardous waste at SAPs. Every effort shall be made to use a container whose size ensures the turn-in of wastes at least annually— radioactive and mixed radioactive-hazardous wastes are the only exceptions.
  - The hazardous waste containers shall be at or near the point of generation (40 CFR 262.34(c)(1). The NDDEQ defines "at or near" as being within the same building or structure. Ideally, the SAP should be in line of sight of the process generating the waste, but exceptions can be made by the Hazardous Waste Program Manager.
  - Hazardous waste containers must be under the direct control of the operator of the process generating the waste. That is, the Hazardous Waste Managers must work in the same building as the SAP. This approach ensures controlled access and consistent management of the waste.
  - When the container is full or ready for transfer to the CAP, the Hazardous Waste Manager must mark the date on the hazardous waste label. This date serves as the accumulation start date (ASD) for the container. The container must be moved to the CAP within 3 calendar days of the ASD (40 CFR 262.34(c)(2). Managers may use containers smaller than 55 gallons to ensure at least annual turn-ins of shop wastes and prompt transfer to the CAP.
  - The Hazardous Waste Manager shall determine when a container is 90 percent full for liquids or 95 percent full for solids. The generator then has three days to transfer the container to the CAP. The Hazardous Waste Manager may contact the CAP contractor (723-4074/4733) to arrange for curbside pickup of the container.
  - The Hazardous Waste Manager determines when hazardous material (HAZMAT) becomes waste, but must make every reasonable effort to use or recycle the item, or donate it to Free Issue. If an item has an expired shelf life, contact the HAZMART (723-4143/4145) to determine whether the shelf life can be extended. If an item is no longer needed, does not meet specifications, cannot go to Free-Issue, or the shelf life cannot be extended, the item is a waste. At that determination, the shop Hazardous Waste Manager must mark the container with the waste ASD. For containers less than 55 gallons, the manager has 1 week to move it to the CAP. For containers of 55 gallons, the manager must move it to the CAP within 3 days. When a shop generates an item determined to be a hazardous waste, the waste must go into an established waste stream accumulation container, or taken to the CAP for disposal. No exceptions will be made, and no stockpiling is allowed.
- Inspections

- Shop Hazardous Waste Managers shall self-inspect their SAP every 30 days to ensure they are in compliance with environmental regulations and the provisions of this plan. The inspection may be noted on a log sheet or on the inspection checklist, which is shown in the Hazardous Waste Satellite Accumulation Point Inspection Checklist. The log or checklist must be dated to confirm compliance with the 30-day inspection interval.
- During SAP inspections, the Shop Hazardous Waste Manager shall also assess spill kits, shop personnel training records, the Site-Specific Contingency Plan, warning sign, and access to waste containers and emergency equipment. Discrepancies shall be noted on the SAP inspection log sheet or checklist, and each discrepancy must be resolved before the next SAP inspection. Repeat discrepancies shall be reported to the installation Hazardous Waste Program Manager for resolution.
- Shop Hazardous Managers shall be present and assist in assessments and compliance inspections conducted by other Air Force personnel, federal and state agency personnel, and DoD contractors. Repeated cancellations or avoidance of inspections shall be reported to the squadron commander.

- Closure of a SAP. To close a SAP, the shop Hazardous Waste Manager shall submit a written request to Environmental Management. After all hazardous waste has been removed, the SAP area must be cleaned by the owning organization and inspected by the installation Hazardous Waste Program Manager who will subsequently provide the owning activity a letter confirming closure. The closure letter must be retained in the shop Hazardous Waste Management binder for 3 years. This time limit is extended in the event the SAP is subject to an Enforcement Action.

Shops Without a SAP

At Minot AFB, some shops are allowed to generate hazardous wastes without accumulating waste in a SAP. These shops typically generate very small amounts or intermittent amounts of industrial and non-hazardous wastes, used oil, and/or universal waste. Accordingly, these shops shall appoint at least one primary and one alternate Hazardous Waste Manager who will be responsible for assuring training and record keeping requirements are met. The hazardous waste may be accumulated throughout the day or work shift, but must be brought to the CAP at the end of each work day/shift. Waste generated during swing and night shifts shall be turned-in to the CAP by 0800 hours the following day. **Shops generating characteristic and/or listed (RCRA) hazardous wastes shall operate as a SAP regardless of the quantity generated and/or accumulated.**

Hazardous Waste Satellite Accumulation Point Inspection Checklist

Satellite Accumulation Point (SAP) Inspection Checklist

Shop Name/Site #: _____ Inspection Date: _____

HW Manager(s): _____ Managers on current delegation letter? [Tab A] Y N

Waste Type Collected at Shop (*circle all that apply*): HW IW UW Scrap PCBs ICBM Batt LLRW

Waste Stream(s) Accumulated: _____

SAP Site: 1-12 Book: 12-18		Yes	No	N/A	Problem(s) Observed	Date of and Corrective Action
1	Are containers in good condition: not leaking, rusted, bulging or dented? (40 CFR 264.171; HWMP Chap 7.3 pg 21)					
1a	Are the containers compatible with the waste being accumulated in them? (40 CFR 264.172; HWMP Chap 4 pg 12)					
1b	Are containers securely closed except when adding/removing waste? (40 CFR 264.173; HWMP Chap 7.3 pg 22)					
1c	Are containers properly marked: waste name, codes, cont. #, DOT Haz Label, etc.? (40 CFR 261.20 (b) Subpart C, HWMP Chap 7.4 pg 24)					
2	Is more than 55 gallons of any one non-acute waste stream or 1 quart of P-listed waste being accumulated? (40 CFR 262.34(c)(1); HWMP Chap 7.5 pg 33					
3	If yes, has the excess been marked with Accumulation Start Date? (HWMP Chap 7.5, pg 31)					
4	Has Accumulation Start Date exceeded 72 hours (3 days)? (HWMP Chap 7.5 pg 33)					
5	Is Universal Waste accumulated in the shop for less than 9 months (270 days)? (HWMP Chap 7.11 pg 47)					
6	Are drums containing ignitable wastes (D001) properly grounded? (HWMP Chap 7.3 pg 22)					
7	Is there sufficient aisle space between containers for spill & fire control equipment? (HWMP Chap 7.3 pg 22)					
8	Is SAP sited correctly: at/near point of generation, under control of HW Mgr, warning signs posted? (HWMP Chap 7.5 pg 31)					
9	Are all hazardous wastes properly managed [i.e. no HW in trash cans, OWS, Used Oil, etc.]? (HWMP Chap 4 pg 11)					
10	Is stored hazardous waste collected from a nearby generation source? (HWMP Chap 7.5 pgs 30-33)					
11	Are emerg. response eq. (fire extinguishers, spill kits, etc.) in good condition & accessible? (HWMP Chap 7.10 pgs 42-46)					
12	Is Site Specific Contingency Plan current and posted near SAP & in HW Management binder? (HWMP Chap 7.10 pgs 44-45) [Tab D]					
13	Is there an Accumulation Log for each waste stream being accumulated? (HWMP Chap 6 pg 19 Example pg 27) [Tab E]					
14	Was last SAP inspection completed within 30 days and inspection log completed/signed? (HWMP Chap 7.5 pg 34) [Tab F]					
15	Is the HW Management binder maintained IAW HWMP? (Chap 4 pg 12, Chap 6 pg 18)					
16	Is there a shop HW Handler Training Plan that includes shop specific hazards? (HWMP Chap 5 pg 14) [Tab C]					
17	Have all HW Managers and Handlers received training within past 12 months? (HWMP Chap 5 pg 14) [Tab B]					
18	Who is the SAPs UEC? UEC will be emailed with Assessment results.					

Notes:

Sign: *Print:* *Date:*

Version: B.Hawkins/AFCEC/CZOM/HAZMART/723-4074/2November2023 **Supercedes all previous versions**

2024 UPDATED SAP ASSESSMENT (002).j

Central Accumulation Point

The main temporary storage facility for hazardous waste at Minot AFB is the CAP, which is located with the HAZMART in Building 525. The CAP is a contractor- operated facility serving as the central collection point where hazardous waste from outlying accumulation sites is staged and processed prior to transport. The hazardous waste is subsequently picked up by a DLA-DS contracted waste transporter who delivers it to an EPA-permitted TSDF. The waste transporter must be EPA-permitted and licensed by the NDDH to operate in the state. The transfer of hazardous waste off Minot AFB is overseen by a DLA-DS Contracting Officer's Representative (COR). Formerly, a hazardous waste storage facility at DLA-DS Minot was operated under a RCRA Part B permit. When Minot AFB's status changed to that of an LQG in 2021, the permit was no longer required. The facility was closed and cleaned in accordance with the RCRA Part-B permit closure plan. In June 2004, DLA-DS initiated "receipt-in-place" storage of Minot AFB hazardous wastes, which resulted in all hazardous wastes being collected and processed in the CAP. Minot AFB maintains the same strict operating procedures that existed under the RCRA Part-B permit to ensure environmental safety and compliance. Environmental Management oversees the performance of the CAP contractor, and ensures the CAP contractor complies with all the provisions of this plan.

CAP Maintenance and Operation

The CAP must be maintained and operated in a manner that minimizes the possibility of a fire, explosion, or any accidental release of hazardous waste to the air, soil, or surface water and or that could endanger human health or the environment. Accordingly, the following procedures have been implemented.

- Hazardous Waste Managers. The CAP contractor shall delegate two Hazardous Waste Managers (one primary and one alternate) in writing as specified in its contract. The delegation letter shall be forwarded to Environmental Management through the Minot AFB COR within 5 days of receipt by the COR. The delegation letter shall be updated as specified in the CAP contract, when the CAP contractor personnel no longer have Hazardous Waste Manager duties, and at the beginning of new CAP contracts.
- Security. The CAP contractor will control entry to Building 525 at all times. Specifically, the entrance to the building will be kept locked when the facility is not operational. Signs with the legend "DANGER—UNAUTHORIZED PERSONNEL KEEP OUT" must be posted on each side of the CAP to be seen from any approach.
- Equipment. The CAP must maintain the following equipment in appropriate quantities:
    - Drums: 55-gallon and smaller open-head drums; 55- gallon and smaller bung- type drums, and overpack drums;
    - Neoprene gloves and protective clothing;
    - Chemical splash goggles;
    - Rubber boots;
    - Shop brush;
    - Non-sparking shovel and hammer;
    - 18-inch push broom with synthetic fibers;
    - Absorbent materials and/or absorbent boom;
    - Wrenches: bung and 15/16-inch open-end wrenches;
    - Bungs: 2½-inch with gasket and 3/4-inch with gasket;
    - First aid kit;
    - Blank labels; and
    - Permanent ink and paint marking pens.

CAP Container Staging Requirements

- The CAP can store hazardous waste for up to 270 days from the accumulation start date on the container because Minot AFB is an LQG and the nearest TSDF is more than 200 miles away. To ensure strict compliance with environmental regulations, Minot AFB requests DLADS remove hazardous waste from the installation every 70 to 90 days.
- The CAP contractor shall ensure all received hazardous waste is properly segregated by compatibility and hazard class.
- All hazardous waste containers shall be stored on wooden pallets, metal racks or within secondary containment inside the CAP.
- Storage of hazardous waste shall adhere to the CAP floor plan. Placement of containers must allow for inspection of each container for leaks or deterioration and allow for the unobstructed movement of personnel, fire protection equipment, spill control equipment, and/or decontamination equipment.

7.6 Transportation

The HW Program Manager has overall responsibility for the transportation of HW from an IAP to an HWAS, and from an HWAS to the disposal facility. The HW Program Manager ensures:

- All transportation over public highways is conducted IAW applicable Department of Transportation (DOT) requirements
- Containers are DOT approved
- Transporters have the appropriate training

- Uniform HW manifests are prepared for offsite transportation (electronically or paper)
- All necessary documentation has been completed and records are maintained IAW all applicable federal, state, and local requirements and the AF Records Disposition Schedule

Installation Supplement

7.7 Turn In/Disposal

The turn in procedures contained in DoDM 4160.21, Volumes 1 - 4. DLA Disposition Services are followed. In the event an alternate route for disposal is needed, a waiver will be obtained with proper justification and approval.

Containers are inspected prior to turn-in to ensure that container management procedures have been followed and that containers are properly labeled and in good condition. If the container is not in good condition, contents are transferred to a container that is in good condition.

Installation Supplement

Waste Turn-In Procedures

This section describes the procedures and associated responsibilities for the transfer of waste from the generator to the CAP. It also establishes procedures for the on-base transportation of hazardous waste from a SAP to the CAP. **It is the shop Hazardous Waste Manager's responsibility to ensure that all wastes are turned in to the CAP (1) before exceeding associated storage time limits and (2) in compatible containers with required markings.** When requested, each generator should provide accurate forecasts of hazardous waste turn-in requirements to Environmental Management to enable proper planning and budgeting for the disposal of hazardous waste generated on Minot AFB and its associated GSUs.

Coordination

The transfer of waste from the CAP to DLA-DS for disposal must be coordinated and scheduled. The CAP contractor will arrange for curbside pickup of the containers, if requested, and will coordinate the transfer to DLA-DS.

Container Requirements

Prior to transfer to the CAP, the shop Hazardous Waste Manager will ensure each container satisfies the following requirements:

- Each container must have its contents properly identified and the waste characterized as hazardous, industrial, universal, DOT-regulated, or non- hazardous.
- Each container must be inspected to ensure it is closed securely and in good condition. The container shall have no leaks and no accumulation of liquid on the top. There should be no serious corrosion, dents, or sharp creases, and the container must not be bulging. If the container is leaking or in otherwise unsatisfactory condition, the waste must be transferred to a container in good condition, or the bad container must be overpacked in a salvage drum. Contact the CAP contractor before transferring the contents and to obtain an overpack drum.
- Containers must be labeled and marked per Section 7.3 of this plan. In particular, ensure the following:
    - If an accumulation start date was not previously marked on a SAP container, enter the turnin date on the label.
    - Add the EPA waste code(s) from the Hazardous Waste Profile Sheet (Part 1, block B.2) to the RCRA Hazardous Waste container label (see Appendix C) as soon as the waste is put into the container.
    - The CAP will accurately weigh the container to include overpacks, containers, and/or boxes. All weights will be rounded up to the nearest whole pound.

Documentation Requirements

On Minot AFB, a computerized database is used to track and report hazardous waste turn-ins, and print the documents needed to dispose of the waste through DLA-DS. To help complete the documents, the generator must provide all necessary information to the CAP, to include the container number, associated analytical results, and applicable MSDS or SDS. The generator will use email whenever possible. The shop Hazardous Waste Manager must keep copies of the turn- in documentation for 3 years. The installation Hazardous Waste Manager must keep copies up to 50 years in accordance with the AFRIMS T32-01 R17.00.

Analytical Results/MSDS or SDS

The Hazardous Waste Manager must provide the CAP with MSDS or SDS and/or analytical results used to identify and characterize the waste. The shop Hazardous Waste Manager ensures the analytical results and MSDS documentation provided matches the specific container by checking NSN, part number, mil- spec, sample number, batch number, size, ingredients, manufacturer, and other identifying markings or labels as appropriate. The MSDS or SDS shall not be more than 3 years older than manufacturing date of the waste material.

Hazardous Waste Profile Sheet

The Hazardous Waste Profile Sheet, DRMS Form 1930, identifies the chemical constituents and properties of a waste. The generator is responsible for assisting the CAP contractor in completing a Hazardous Waste Profile Sheet prior to disposal of the waste. The subsequent turn-in of identical waste does not require a new profile sheet. However, an updated profile is required when the composition of the waste changes. The shop Hazardous Waste Manager should contact the CAP and Environmental Management with any new information relating to a particular waste stream so that the database can be updated and allow for printing of the most current Hazardous Waste Profile Sheet. The waste profile sheet is created and maintained within the EESOH-MIS Hazardous Waste Module. After the generator has notified the CAP contractor of the required waste profile, the CAP contractor generates Hazardous Waste Profile Sheet, which is assigned a unique turn-in identification number.

Land Disposal Restriction Notification Form

The Land Disposal Restriction (LDR) Notification, DRMS Form 1851, is used to notify DLA-DS and the disposal contractor of applicable landfill restrictions before the hazardous waste is disposed of. The LDR Notification is based upon EPA waste codes and the chemical constituents of the waste. The DLA-DS currently generates the LDR Notification from the Hazardous Waste Profile Sheet.

Issue Release/Receipt Document

The DD Form 1348-1A, *Issue Release/Receipt Document*, is used to transfer accountability and funds to DLA-DS for waste disposal/recycling. To initiate the process, the generator contacts the CAP contractor to turn-in for each waste. The CAP contractor enters information provided by the shop Hazardous Waste Manager into the EESOH-MIS Hazardous Waste Module. When the Hazardous Waste Profile Sheet is generated, the container is assigned a unique turn-in identification number. The turn-in identification number will also appear on the Issue Release/Receipt Document DD Form 13481A. The Hazardous Waste Manager will certify the information on the form, including the container size/type and the container number, is correct. Then Environmental Management certifies the availability of funds and the information is transmitted to DLA- DS. After funding is approved, DLADS will transmit a delivery order to the CAP, which will provide the CAP with the necessary information to prepare the container for shipping to a disposal facility.

Hand Receipts

Hand receipts are required for universal waste turn-ins, and from units not designated as a type of accumulation site. The receipt is filled out by the generator, and signed by the CAP contractor and the generator. The generator will retain a copy for one year after the waste is turned in to the CAP. The CAP contractor will assign a suitable container number and attach the receipt to the appropriate

Hazardous Waste Profile Sheet and/or the DD Form 1348-1A.

Example Hand Receipt

W A S T E R E C E I P T **Date:**

Environmental Management Element
5 CES/CEIE
320 Peacekeeper Place Minot AFB, ND 58705

Generator Name, Squadron, Office Symbol, Duty Phone:

Shop #

***WARNING*! Federal law prohibits the improper disposal of hazardous waste. Hazardous wastes are regulated in accordance with 40 CFR 260 through 299.**

Waste Description	Qty	Weight
Universal Waste – Battery (*Type*)		
Universal Waste – Mercury Containing Device(s) (*Type*)		

Universal Waste – Lamps
(*Type*)
Ballasts / Capacitors: [ ] PCBs [ ] **No** PCBs [ ] Unknown
Other PCB item:
Used Oil / Off-Spec Fuel
Other *(Specify Type of Waste & Qty*)

Generator Signature CAP Signature

On-Base Waste Transfer

The Hazardous Waste Manager for the generating activity is responsible for the safe transfer of hazardous waste to the CAP. Specifically, the shop Hazardous Waste Manager must ensure the following:

- Hazardous waste must be accumulated and subsequently transported in the proper UN Performance Oriented Packaging (POP) specification containers as provided by the CAP. The installation Hazardous Waste Program Manager (723-6380) can be contacted to provide assistance on selection of the proper container.
- Hazardous waste containers removed from the SAP by the CAP contractor must be transported in a safe and secure manner. The SAP Manager must ensure the containers are in good condition and have the prescribed labeling, marking, and documentation before transport. Contact the CAP contractor (723-4145) for curbside pickup.

Containers brought to the CAP by the SAP Manager must be in good condition, have the required labeling and/or markings, and the appropriate MSDS or SDS.

7.8 Inspection

Inspection processes fulfill the "Check" function of the EMS "Plan, Do, Check, Act" cycle. HWASs are inspected at least weekly to ensure proper accumulation and container management. Resource Conservation and Recovery Act (RCRA) Part B permitted storage facilities are inspected according to the inspection schedule established in the permit. All other inspections occur IAW DAFI 90-302, *The Inspection System of the Department of the Air Force*, and the Commander's Self Inspection Program. Inspection records are maintained IAW the Recordkeeping and Reporting section of this plan.

7.9 Waste Minimization/HW Recycling

HW manifests include certification that a waste minimization program is in place. Below are key activities and processes that are performed as part of waste minimization and pollution prevention efforts:

- Hazardous material process authorization and hazardous materials management processes – Each process involving use of hazardous materials and generation of waste streams is evaluated and authorized. Process authorization is performed through EESOH-MIS. The HW Program Manager, Hazardous Materials Management Program (HMMP) Team, and the generating activity make a final determination whether or not the results of the process authorization effort are sufficient to reduce waste toxicity and volume
- Procurement and use of minimal quantities – When a material with environmental risk must be used, minimal quantities are procured to minimize surplus quantities and shelf life exceedances
- Recycling – When the use of hazardous materials is unavoidable, excess or waste material is evaluated for reuse or recycling
- Environmental action planning – Environmental action plans (EAPs) are developed and maintained as part of the overall EMS. EAPs are management plans that translate environmental objectives and targets into actionable plans. Waste minimization efforts are considered during development of EAPs
- Recycle or reclaim RCRA Subtitle C HW as appropriate, maximize RCRA authorized exclusions, variances, or exemptions to reduce the amount of HW and other regulated waste generated, avoid disposal costs, and improve solid waste diversion rates
- Specific installation procedures are outlined below

Installation Supplement

7.10 Preparedness and Prevention

Preparedness and prevention practices are described in emergency prevention and response plans available through the references section of this plan and are maintained IAW the installation EMS framework. A HW Contingency Plan is specifically required for Large Quantity Generators. The HW Contingency plan can be included in the Installation Spill Prevention, Control, and Countermeasure (SPCC) Plan (or equivalent) or a specific HW Contingency Plan.

7.11 Waste Specific Procedures

Waste-specific procedures are included in the installation supplement below or maintained as separate operational controls outside of this plan. As applicable, this may include unique processes or procedures to deal with a specific waste not already addressed elsewhere in the HWMP, such as acutely hazardous wastes, pharmaceutical hazardous waste, used oil, etc..

Installation Supplement

State- Regulated/Industrial Wastes Sites

Minot AFB refers to non-RCRA, state-regulated waste as industrial waste. Industrial wastes include, but are not limited to, state-regulated liquids, oily/greasy wastes, non-RCRA regulated waste, and/or RCRA- excluded waste. Shops generating industrial wastes must have a storage area located at or near a point of generation where the waste is initially accumulated. Per North Dakota environmental regulations, industrial wastes may not be disposed in municipal landfills. It is the policy of Minot AFB to manage industrial wastes in the same manner as hazardous wastes, and associated accumulation sites must be managed the same as SAPs.

Universal Waste Sites

A universal waste site is a storage area where only universal waste (batteries, lamps, pesticides, and mercury containing devices) is accumulated. Container marking, management, and handling criteria are described in Section 7.4 of this plan.

Special Management Items

This section describes the proper management of waste commodities that require special attention and/or that typically generate the most questions from Hazardous Waste Managers.

Aerosol Cans

Empty aerosol cans are collected for recycling. Waste aerosol cans will be disposed of as follows:

- If the aerosol can spray nozzle is inoperable, replace the nozzle, and use the material.
- If the material is no longer needed, turn in the material to HAZMART free Issue and zero out the barcodes.
- If the item is no longer usable, contact the HAZMART to clear the bar code and turn it in to the shop Hazardous Waste Manager for disposal.
- If the aerosol can is empty of all product and propellant, take to the CAP for recycling. The shop Hazardous Waste Manager can accumulate empty aerosol cans before taking them to the CAP for recycling or turn them in at the end of each duty day. The shop Hazardous Waste Manager must contact the installation Hazardous Waste Program Manager (723-4870) before accumulating the cans.

Empty aerosol shall be taken to the CAP where they are punctured and the aerosol can is recycled. The liquid contents of the cans are collected and have been characterized as a hazardous waste.

Batteries

The proper disposal of a battery is dependent on the type of battery. The following disposal procedures apply on Minot AFB.

- General Requirements. Rechargeable batteries must have at least the positive terminal taped over to prevent shorting when the batteries are accumulated in a container, unless the batteries are returned to their original packaging. Batteries that are palletized, such as lead/acid batteries, do not need their positive terminals taped. However, palletized batteries must be stored and transported in such a way as to prevent damage to their casing and shortouts from terminals touching each other.
- Lead-Acid and Silver Batteries. When properly recycled, lead-acid and silver batteries are exempt from hazardous waste regulations. Because these batteries have commodity value, the generator must recycle them through the CAP. If the battery cannot be recycled, it must be managed as a universal waste or a hazardous waste.
- Nickel Cadmium (NiCad), Mercuric Oxide, Lithium Sulfur Dioxide, Lithium Thionyl Chloride, Lithium Polycarbonate, and Magnesium Batteries. These types of batteries will be recycled as universal waste per Section 7.3 of this plan.
- Carbon-Zinc, Lithium Ion, Lithium Manganese Dioxide, Nickel Metal Hydride, and singlecell Lithium Sulfur Dioxide Batteries. These batteries have been shown to be non-hazardous under current hazardous waste regulations1. However, to ensure compliance and protection of the environment, all such batteries on Minot AFB, regardless of quantity, will be recycled and

are not to be thrown into the general refuse. These batteries must be separated by type when delivered to the CAP for recycling.

- Alkaline Batteries. Must be collected for proper disposal.
- Other Battery Types. Contact the CAP for disposal directions of other types of batteries not specifically addressed in this plan. If a battery cannot be recycled and is a hazardous waste, it must be marked and managed as a hazardous waste and disposed of properly through the CAP.
- Leaking Batteries. Batteries typically contain acid either as a liquid or as a gel. Therefore, leaking batteries must be stored, labeled, and managed as hazardous waste.

Certain commonly generated wastes are managed as universal wastes in accordance with 40 CFR 273, *Standards for Universal Waste Management*, which provides relief from the full regulatory aspects of RCRA. Specifically, the objective of the universal waste rule is to simplify collection and management requirements while promoting recycling. The EPA has designated the following types of waste that exhibit one or more of the characteristics of a hazardous waste as universal waste:

(1)batteries; (2) lamps (fluorescent and other mercury vapor, neon, high pressure sodium, and metal halide bulbs); (3) some pesticides; and (4) mercury-containing devices (thermostats, switches, and other devices containing mercury). The EPA leaves implementation of the universal rule up to the individual states.

- General Policies for Managing Universal Waste. The following general policies regarding the management of universal waste apply on Minot AFB:
    - All individuals that generate or manage universal waste must complete training prior to generating or managing any universal waste. This training encompasses proper handling and emergency procedures, and is available from The Environmental Awareness Course Hub (TEACH) (https://usaf.learningbuilder.com/). Contact the installation Hazardous Waste Program Manager (723-4870) for the initial registration password. Personnel will complete their training before accumulating or managing universal waste, and shall renew this training annually in accordance with Section 5.0 of this plan.
    - Any residues generated from the cleanup of spilled universal waste will be managed as hazardous waste. For health concerns, contact Bioenvironmental Engineering at 723-5151. For spill and disposal information, contact the base Hazardous Waste Program Manager (723-4870) or the CAP (723-4074/4733).
    - Universal waste may be accumulated at any convenient and secure location, but it does not have to be stored at a SAP.
    - Accumulate universal waste in a designated container by type of material for a maximum of nine months (270 days). The container must be properly labeled, secured so it cannot be knocked over, have a lid or other means to completely contain the waste, and only opened when adding or removing waste. Alternately, universal waste may be turned-in the same day it is generated. Coordinate turn-in of universal waste with the CAP.
    - Organizations may consolidate like universal wastes into a single container provided the start date on the container reflects the date the first item was put into the container and the 270-day limitation is not exceeded.
    - Each universal waste container must be labeled as universal waste with the type of waste (for example, batteries, lamps, etc.), the container number, and the accumulation start date, which is the date the first waste item is added to the container. See Appendix C for the universal waste labels needed.
    - Universal waste containers must have a lid and must be kept closed except when adding or removing waste. For boxes, folded or taped flaps will satisfy this requirement. When storing fluorescent lamps, do not store unused lamps in the same container as spent lamps.
    - Universal waste batteries or mercury-containing equipment/devices that **do not** show evidence of leaks, spills, or damage are not required to be stored in a closed container.
- Batteries Managed as Universal Waste On Minot AFB, rechargeable batteries will be recycled as universal waste. Lithium sulfur dioxide batteries must be fully discharged before being accumulated in the shop or turned in to the CAP. All batteries accumulated in a container must have the positive terminal securely taped over prior to being collected and/or turned in to the CAP.
    - Universal Waste batteries must be transferred to the CAP at least every 9 months (270 days). Universal waste batteries may also be turned in to the CAP by 4 p.m. of each workday, depending on Hazmart personnel availability (Hazmart operates by appointment only, however, exceptions may be made for certain circumstances). If universal waste batteries are accumulated longer than 1- d a y , the storage area and containers must comply with the provisions of this plan. Battery types should be segregated, but may be consolidated from multiple shops within the same squadron.
    - If a shop chooses to accumulate batteries, it must mark each item or container as “Universal Waste – Battery(ies)” and complete the label shown in Appendix C before securely affixing to the container. The SAP/Waste Manager is responsible for turning in the accumulation container within 270 days (9 months). Batteries must be collected separately, though some types of lithium batteries may be comingled. The positive terminals of each battery must be taped to prevent a short circuit when the waste is in transit.

- Light bulbs/Lamps Managed as Universal Waste. Light bulbs/lamps may contain hazardous constituents, such as mercury vapor, making them subject to state regulations. North Dakota specifically regulates wastes with a mercury concentration of more than 0.2 ppm (0.00002%) as a hazardous waste. However, under the universal waste regulations, light bulbs/lamps that are recycled are exempt from hazardous waste regulations. These lamps/bulbs include, but are not limited to, regular and compact fluorescent lamps, high intensity discharge lamps, and neon, mercury vapor, high-pressure sodium, and metal halide lamps. In all cases, great care should be taken to protect light bulbs/lamps from breakage during handling, storage, and shipping by using the original packaging of the replacement bulb/lamp whenever possible.

  Spent light bulbs/lamps must be managed as follows:

  - Recycle unbroken fluorescent bulbs as universal waste at the CAP. Whenever possible, the spent bulbs should be turned-in the same day they are changed out. A shop may choose to accumulate the spent fluorescent bulbs for no more than 270 days as "Universal Waste— Lamps" in boxes that are taped closed and are labeled.
  - Broken fluorescent lamps are managed separately as hazardous waste. A broken lamp should be securely contained and brought to the CAP as soon as it is cleaned up. Contact Bioenvironmental Engineering (723-5151) for safe clean-up directions.
  - High intensity lights, such as mercury vapor, high/low pressure sodium vapor, and tungsten metal-halide bulbs, should be brought to the CAP the day they are changed out. A shop may choose to accumulate the spent high intensity lights for no more than 270 days as "Universal Waste—Lamp(s)" in containers labeled with the notation indicating the specific type of bulb. Each type of high intensity light must be accumulated separately. Protect all bulbs from breakage using the manufacturer's packaging for the replacement bulb.
  - Broken high intensity lights will be accumulated separately as "Universal Waste—Lamps— Mercury Containing Devices." A shop may choose to accumulate broken high intensity lights for no more than 30 days from the date the first broken lamp is placed in the container, and must be labeled as "Broken Lamps."
  - Incandescent light bulbs may be discarded into general base refuse.
- Mercury-Containing Devices Managed as Universal Waste. The North Dakota Department of Environmental Quality regulates the disposal of all wastes with greater than 0.2 ppm (0.00002 percent) mercury as a hazardous waste. However, North Dakota universal waste regulations exempt mercury containing devices for example, a thermometer or switch) that are recycled from hazardous waste regulations. On Minot AFB, mercury- containing devices are managed as universal waste in the following manner:
  - All mercury thermometers, switches, or other articles containing liquid mercury may be managed as universal waste, and may be accumulated no more than 30 days in a secure and closed container.
  - If the device is broken, the entire article (glass, mercury liquid and cleanup debris) must be properly managed as hazardous waste. It must be containerized along with any contaminated items that cannot effectively be cleaned and turned in to the CAP within three days of clean up. Contact Bioenvironmental Engineering (723-5151) for safe cleanup directions.
  - The container must be marked as "Universal Waste—Mercury Containing Device(s)" and have the start date for the container written on the label. Also, include the shop name, point of contact, and associated phone number on the container.
  - On Minot AFB, all mercury-containing articles that are not recycled will be managed as hazardous waste.

Used Oil/Fuel Filters

On Minot AFB, used, non-terne[2], metal filters contaminated with oil, hydraulic fluid, diesel fuel, and/or JP-8 must be recycled as scrap metal. Per federal and North Dakota regulations, filter recyclers may only accept used metal filters in drums containing no free liquid. Plastic, glass, and paper oil/fuel filters must be collected separately and turned in as industrial waste. These filters may be crushed if crushing does not cause the filter to disintegrate. Used oil/fuel filters are to be managed as follows:

- Used Oil/Non-MOGAS Fuel Filters. The following procedure applies to used oil and fuel filters, except gasoline (MOGAS) filters.
  - Used oil filters must be hot-drained (that is, at a temperature above 60°F) of all free flowing oil for a period of at least twelve hours, but no more than 24 hours, to be considered drained.
  - A canister-type filter with a single opening must be punctured to promote complete draining. Puncture a hole on the top and the side (near the bottom) of the filter. Invert the filter on a grated drip pan over a drum labeled "**Used Oil**." Any drum that previously contained oil or hydraulic fluid may be used.
  - Shops with the appropriate equipment may crush drained metal filters to reduce the accumulated volume and increase recovery of used oil. Plastic filters may not be crushed if it causes the filter to disintegrate.
  - Prepare an open head drum to accumulate the drained, used filters. Any such drum that previously contained oil or hydraulic fluid may be used. Label the drum "Scrap Metal – Used (*type*) Filters." The drum must be closed securely

except when adding or removing waste. Cut a hole on the side near the base of the drum to drain any free flowing oil. Prop the drum slightly so it is slanted towards the hole to allow drainage of oil collected in the bottom. Allow the drum to drain for no more than 12 hours. When the drum is finished draining, place a piece of duct tape over the hole and contact the CAP without delay for curbside pick-up service.

- Drained oil shall be collected in a separate oil container. The shop may choose the size of the container, but it must be closed securely when oil is not being added or removed from it, and the collected oil must be removed at least annually. Contact Environmental Management before accumulating used oil in the shop to determine the appropriately sized container.

Used MOGAS (Gasoline) Filters Used MOGAS (gasoline) filters require special handling because of the ignitability of gasoline remaining within the filter. Used fuel filters contaminated with gasoline, whether drained or undrained, will exhibit the ignitability characteristic (D001) and typically the toxicity characteristics for benzene (D018).

- Used MOGAS (gasoline) filters must not be commingled with any other filters.
- Used MOGAS (gasoline) filters will be containerized in non-leaking, approved, open head drums with tight fitting lids. All used MOGAS (gasoline) filters will be marked, managed, and disposed of as a hazardous waste. Containers used to store ignitable/flammable waste must be grounded whenever waste is being added or removed from the container.

Unserviceable Hazardous Materials

- When a hazardous material is no longer needed, it should be returned to the HAZMART or to the appropriate Source of Supply (SOS) for re-use through the Free- Issue program. In general, the HAZMART will accept any returned hazardous material for the Free-Issue program if the material has 6 months or more of shelf-life remaining or the shelf-life can be extended and there is at least one other authorized user for that type of material at Minot AFB.
- If the HAZMART cannot accept the hazardous material for the Free-Issue program, but the material is unopened and in good condition, the HAZMART will work with DRMS and the CAP to see if the material can be resold, recycled, re-used, or donated.

Alternately, if the hazardous material must be disposed of as hazardous waste, the shop Hazardous Waste Manager will dispose of the material using the procedures detailed in this plan (see Section 7.7). Hazardous material determined to be a hazardous waste will be transferred to the CAP and managed accordingly.

Two-Part Sealing Compounds, Adhesives, and Paints

- Two-part sealing compounds, adhesives, and paints must be carefully managed so only enough of the ingredients are mixed to complete the task. Any waste residues, including applicators, non- empty containers, and dried product, will be managed together as a common waste stream. In particular, residues of two-part sealing compounds, adhesives, and paints that contain chemicals specified in 40 CFR 261, Subparts C and D (methyl ethyl ketone [MEK] or lead) must be managed as hazardous waste until a sample analysis determines whether it can be managed differently.
- A partial list of these chemicals is in P and U-Listed Wastes Excerpts: Discarded Commercial Chemical Products table of Appendix A.

Lead-Based Paint Waste

Construction debris determined to be hazardous due to lead-based paint (LBP) is regulated under RCRA and must be managed accordingly. Wastes that can be contaminated with LBP include construction debris, paint chips or dust, clean-up equipment, rinse water, sampling equipment, plastic sheeting used to protect floors and other surfaces, personal protective equipment, work clothes, and any material that becomes heavily contaminated during a LBP abatement process. Minot AFB policy does not exempt LBP-contaminated construction debris from testing prior to disposal based on history. Accordingly, LBP- contaminated construction debris from non- housing areas must be analyzed to determine if it is hazardous. Contact the installation Toxics Program Manager at 723-1964 to obtain a copy of the Lead-Based Paint Management Plan before beginning on any project that will disturb painted surfaces.

Used Oil

Used oil is defined as any synthetic oil or oil refined from crude that has been used and, as a result of such use, is contaminated by physical or chemical impurities.3 Containers or tanks holding used oil must be labeled "USED OIL" and inspected at least monthly to ensure they are in good physical condition (not leaking, deteriorating, nor damaged). Manage used oil as follows:

- The CAP contractor will coordinate the pickup of used oil as it is generated if the quantity is 55 gallons or less.
- If a shop chooses to accumulate used oil, the used oil must be collected in a tank/container that is either double-walled or stored on secondary containment large enough to hold the entire contents of the tank/container plus 10 percent of the single largest volume. The shop must notify the CAP when the used oil tank/container is filled to 75 percent of its capacity so that transfer to an oil reclaiming/recycling firm may be arranged. One-week's-advanced notice before the tank/container is declared "full" is the minimum acceptable time to ensure proper management of the used oil.

- All spills or leaks shall be immediately cleaned up, and all residual waste that is generated shall be managed as industrial waste.
  If used oil is accidentally mixed or contaminated with another hazardous material, contact the installation Hazardous Waste Program Manager (723-4870) immediately for guidance. Do not add this mixture to the used oil tank/container unless authorized by the installation Hazardous Waste Program Manager.
- Refrigerant oil recovered from refrigeration systems is managed separately as "Used Oil from Refrigerant Systems" and must be kept separate from other used oil until picked up by the CAP.
- To preclude contaminating PCB-free used oils with PCBs used transformer oils (electrical or dielectric oils) shall not be mixed with other types of used oil and shall be managed separately.

Unexploded Ordnance

Unexploded ordnance/munitions are sometimes discovered during excavation of an old firing range or by a citizen who discovers unexploded munitions on private or public lands. These munitions pose a threat to people and the environment due to their potential instability. In accordance with North Dakota law, emergency treatment of unexploded ordnance by the Explosive Ordnance Disposal (EOD) Flight is allowed. If EOD must detonate unexploded ordnance, Environmental Management will advise the North Dakota Department of Environmental Quality of the situation and obtain verbal permission for emergency treatment.

Waste Military Munitions

The EPA has delineated when unused munitions become a solid waste potentially subject to RCRA. Given the complexity of the regulations regarding this determination, questions regarding the proper handling of waste military munitions should be referred to Environmental Management for resolution. On Minot AFB, the Conventional Munitions Flight, 5 MUNS/MXWC, will normally assume custody of waste munitions.

Contaminated Rags

Cloth rags that are used and then laundered for reuse do not constitute a solid or hazardous waste. Specifically, the North Dakota Department of Environmental Quality does not consider laundering to constitute recycling or disposal, and, therefore, reusable rags are not considered a hazardous waste. Reusable rags shall be managed as follows:

- Hazardous waste or other debris **shall not be** mixed with launderable rags.
- No liquids shall be poured onto the rags while in the collection container. Likewise, containers holding rags shall be managed as waste if there is free liquid in the bottom of the container.
- Rags shall not be saturated with any product. Rags saturated with oil may be wrung out and reused. When too dirty to be used, these rags should be wrung out and collected for laundering. Rags saturated with other products must be collected separately, managed as waste, and properly disposed of.
- Used rags should be taken to the Hazmart for laundering.
- Used rags must be collected in a container lined with 2 plastic bags. The container must have a secure lid and be labeled "DIRTY RAGS."
- Place the soiled rags inside the inner plastic bag lining the container.
- Ensure no free liquid and/or debris is commingled with the rags. If the bag/container has free liquid in the bottom or contains debris, the bag must be removed immediately and managed as a waste.
- When full, use duct tape to securely close the bag for transport and mark it as requiring special handling, such as return to a specific work center. Note that rags contaminated with sodium chromate MUST be bagged separately from other rags. In a dissolvable bag which is supplied by the linen exchange contractor. The rags are put through a wringer to capture/dispose of any liquid chromate. The dissolvable bag containing "Chromate Rags" are placed in a red barrel that is provided by the linen exchange contractor.
- Transport the bagged rags to the Hazmart facility, remove the bag and weigh it, and then place it into the receiving can.
- Disposable or non-reusable rags (for example, paper towels, and tech-wipes) must be accumulated and properly characterized. Coordinate this determination through Environmental Management as necessary to resolve any uncertainty.
- Hazardous waste rags will be managed at a SAP by the associated Hazardous Waste Manager. Non- hazardous rags that cannot be laundered must be turned in to the CAP as industrial waste per the turn-in procedures described in Section 7.7.

Fluorescent Light Ballasts

Depending on when they were manufactured, fluorescent light ballasts may contain PCBs, which are typically present in the small capacitors or associated potting compound that comprise the ballasts. The handling and disposal of fluorescent light ballasts may be subject to regulation under the Toxic Substances Control Act (TSCA) as implemented by 40 CFR 761, *Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution In Commerce, and Use Prohibitions*. Use the following guidelines for the disposal of these ballasts.

- Fluorescent light ballasts that are factory labeled with a "No PCBs" mark are not regulated by TSCA or RCRA and may be recycled as scrap metal through the CAP.
- Specific disposal requirements apply for fluorescent light ballasts either that are known to contain PCBs or for which a determination as to PCB content cannot be made conclusively. Normally, non- leaking ballasts that are known to contain PCBs or that *may* contain PCBs are not regulated as a TSCA or RCRA waste and are therefore not subject to special storage and handling requirements. **On Minot AFB, non-leaking ballasts—unless they are marked as not containing PCBs—will be managed, as a TSCA waste to ensure environmental compliance, but the TSCA standards for storage and handling will be eased.**
  Specifically, non-leaking fluorescent light ballasts that contain or that may contain PCBs will be collected and stored in open head drums meeting UN 1A2 specifications. These drums will be clearly marked with the words "NON-LEAKING LIGHTING BALLASTS—Potential to contain either PCB or DEHP" and labeled with a Large PCB Mark (ML) sticker (Appendix C). These labels may be obtained from the CAP.
- Non-leaking light ballasts must never be mixed with leaking ballasts. Leaking light ballasts that contain PCBs or that *may* contain PCBs are to be collected in a separate container from non-leaking ballasts to avoid cross contamination.
- By law, absorbent must be placed in this container to preclude the presence of any free liquid, and the container must be secured except when adding or removing waste.
- The container must be labeled with the Large PCB Mark that must be dated with the same date that the first leaking item was placed in the container. This date is termed "Date Removed from Service" and is required under 40 CFR 761.
- The container should be clearly labeled with the contents. For example, "LEAKING LIGHT BALLASTS, SWEEPINGS, ABSORBENT, CONTAMINATED SOIL," is applicable for a container holding these materials.
- PCB items may be accumulated for a 30-day maximum accumulation period, which begins from the time the first leaking item is placed in the container. At the end of 30 days, the container must be transferred to a TSCA conforming storage site, which includes the CAP. Consult with Environmental Management about where leaking transformers that contain or may contain PCBs may be stored.
- The PCB shipping label depicted in Appendix C will be completed and affixed to the shipping container.

Mixed Radioactive and Hazardous Waste

Mixed radioactive and hazardous waste, generally referred to as "mixed waste," meets both the EPA's definition of a hazardous waste and the Nuclear Regulatory Commission's definition of radioactive waste. Accordingly, mixed waste on Minot AFB will be managed in accordance with this plan to ensure compliance with hazardous waste regulations and with any requirements posed by the Radiation Protection Officer of the Bioenvironmental Engineering Element. Key reference material regarding the proper handling of mixed waste can be obtained from AFPD 32-70, *Environmental Quality*; AFI 10- 2501, *Air Force Emergency Management (EM) Program Planning and Operations*; AFMAN 32-7002 *Environmental Compliance and Pollution Prevention*; and AFPD 48-1, *Aerospace Medicine Program*. In particular, the following requirements apply to the handling of mixed waste on Minot AFB:

- All requests for the disposition of radioactive mixed waste will be coordinated through the Bioenvironmental Engineering.
- The Environmental Management will review all pick-up documentation, including the manifest, analytical results, land disposal restriction notifications, and any other applicable documents. On Minot AFB, the installation Hazardous Waste Program Manager will sign mixed waste manifests as the designated representative of the Wing Commander. Manifests for only radioactive wastes will be signed by the base Radiation Officer, Bioenvironmental Engineering.

Dental Amalgam

The North Dakota Department of Environmental Quality regulates the disposal of all wastes with greater than 1.2 ppm (0.00002%) mercury as a hazardous waste. Dental amalgam contains varied concentrations of mercury and silver. These variations will cause some samples to fail the RCRA Toxicity Characteristic Leaching Procedure analysis. Minot AFB does not recycle dental amalgam, and therefore dental amalgam must be managed as hazardous waste per the following:

- All types of amalgam will be considered a RCRA hazardous waste unless the generator provides analytical evidence to the contrary. In those instances where analytical proof is available, the amalgam will be stored and disposed of as an industrial waste in accordance with this plan.
- Contact amalgam, non-contact amalgam, and teeth with fillings must be collected separately.
- All **contact** amalgam, **teeth** with fillings, and **mixtures** of contact and non-contact amalgam will require a signed statement certifying, "All items have been disinfected and dried, and they contain no biological hazards."
- Collection of ***only*** **non-contact** amalgam must be separate from contact amalgam, whether it is used, and unused (expired shelf-life) non-contact amalgam.
- All amalgam will be contained in tightly closed unbreakable containers. Small plastic containers with secure lids may be used to collect amalgam within the dental suite. When full, the containers must be placed inside a drum with a secure lid inside the

dental SAP. No loose amalgam may be stored within the dental suite, the drum, or in the SAP.

- Containers holding amalgam for disposal must be clearly marked. The small plastic containers used within the dental suites must be marked "Mercury Containing Article—Used Amalgam" with the date that the first article was placed in the container. This information must be legibly printed with a permanent marker. The drum within the SAP must be labeled as hazardous waste in accordance with this section. The hazardous waste label will contain the nomenclature "Mercury Containing Articles—Dental Amalgam."
- The small plastic containers used within the dental suite to collect amalgam may be used until they are full or for no longer than 9 months from the date the first article is placed into the container, whichever is shorter. When the container is determined to be full, it must be immediately placed within the specified drum inside the dental SAP. The drum will be managed in accordance with this section and will be turned-in for disposal at least once a year. The required certifications from Bioenvironmental Engineering and appropriate MSDSs or SDSs will accompany the drum at the time of turn in.

Industrial Waste

- Wastes that are non-hazardous but contain or are contaminated with certain organic/inorganic chemicals are designated as industrial wastes. In accordance with North Dakota environmental regulations, certain restrictions apply to the types and quantities of industrial wastes that can be managed at municipal landfills. For this reason, Minot AFB manages industrial waste as a Non- RCRA Regulated Waste. Industrial waste typically includes items such as paper or cloth contaminated with certain sealants, paints, adhesives, greases, and oils. Procedures for making this determination are contained in Appendix A of this plan.
- Labeling of industrial waste containers is described Section 7.4 of this plan. Hazardous Waste Managers are responsible for managing these containers. The Environmental Management will assist as required.
- Turn-in of industrial waste is coordinated through the CAP as described in Section 7.7 of this plan.

Contractor-Generated Waste

Normally, contract specifications detail the requirements for the management and disposal of contractor- g e n e r a t e d waste. At no time shall the contractor mix non-hazardous wastes with RCRA hazardous wastes to evade federal, state, and local regulations. The contractors shall provide their own containers for accumulating project-generated wastes. The containers shall be compatible with the wastes generated, secured at all times, and marked and labeled in accordance with federal and state regulations. Warning signs for RCRA hazardous wastes shall be posted as required. The following procedures must be used by contractors to manage and dispose of their wastes:

- **RCRA Hazardous Wastes**
    - Contractors on Minot AFB who generate hazardous waste from the use of their hazardous materials are responsible for the proper management and disposal of that waste in accordance with all applicable federal, state and Air Force regulations, and the contract specifications.
    - The government will dispose of hazardous wastes that are left behind by facility personnel after the contract is let. This can include cleaning supplies or fluorescent lamps hidden in a closet, but does not include fluids contained in equipment designated as real property (for example, HVAC fluid or tank sludge).
    - Hazardous wastes that are specifically to be removed by contract or generated during demolition (for example, lead-based paint) will be disposed of in accordance with contract specifications.
    - RCRA hazardous wastes will not be accumulated on Minot AFB for more than 90 days without express written permission from the installation Hazardous Waste Program Manager.
    - Management of RCRA hazardous waste containers shall be in accordance with Section 7.3 of this plan. The contractors do not have to create waste stream and container numbers but must be able to internally track their waste containers and comply with the portions of the RCRA and DOT regulations that govern container selection, management, and shipping.
    - The contractor shall properly characterize generated hazardous waste and send a copy of the characterization paperwork to the installation Hazardous Waste Program Manager via the Contract Administrator no less than 1 week before the waste is shipped off Minot AFB. o The contractor will send a copy of the signed Hazardous Waste Uniform Manifest (manifest) to the Contract Administrator and Environmental Management should receive it no less than 7 calendar days after the contractor-generated hazardous waste is shipped off base. No shipments off base shall be made on a Friday or weekend day.
    - The contractor must have personnel on-site during the performance of the contract that are certified to properly characterize hazardous waste, and are DOT-certified to sign the manifest. These certifications must be available on the work site during the performance of the contract and available for inspection as requested.
    - The contractor will properly segregate incompatible wastes, and ensure they are fully contained within compatible storage containers (for example, acid solution not stored in a steel drum).

- The contractor is responsible for cleaning up spills of hazardous waste and properly characterizing, managing, and disposing of the spill cleanup materials.

- **Non-RCRA Wastes**
  - Wastes that are characterized as non-hazardous waste, not regulated under RCRA, and construction and demolition (C&D) debris must be properly disposed of by the contractor in accordance with all applicable federal, state and Air Force regulations, and the contract specifications.
  - The contractor will properly segregate non-RCRA wastes, which can include stateregulated wastes, wastes regulated by federal regulations other than RCRA (for example, TSCA or FIFRA), C&D debris, and general refuse.
  - The contractor will properly characterize generated wastes and send a copy of the characterization paperwork to the Contract Administrator and Environmental Management receives it no less than 1 week before the waste is shipped off Minot AFB.
  - The contractor must have personnel on-site during the performance of the contract who are certified to properly characterize wastes.
  - Wastes disposed of by the contractor must be segregated and shipped in accordance with all applicable federal, state, and local regulations.
  - A copy of all waste shipping papers must be sent to the Contract Administrator and received by Environmental Management within 1 business day of the waste leaving Minot AFB.
- **Waste Metrics**
  - The Contractor must collect and report waste data in accordance with the following guidelines:
  - Provide a quarterly report that shows:
    - The weight of RCRA-hazardous wastes disposed of and/or recycled during a calendar month; this information should correspond to RCRA wastes being manifested off Minot AFB.
    - The weight and type of non-RCRA wastes disposed of and/or recycled during a calendar month
    - The weight of C&D wastes disposed of and/or recycled during a calendar month.
    - The name of the disposal and/or recycling facilities accepting the waste, the months the wastes were disposed of and/or recycled, the facility address and phone number, and the EPA identification number, if applicable.
  - Reports should be submitted to the Contract Administrator, and Environmental Management must receive the report no less than 10 days after the end of a calendar quarter. One report may be submitted by the contractor, but it should be clear which type of waste went to the facilities named, whether the waste was disposed of or recycled, and the weight of each type of waste.
  - The report must be on company letterhead paper and signed by the project manager or a company representative who is responsible for the performance of the contract.
  - Prime contractors must submit reports for sub-contractors. Reporting may be delegated to the sub-contractor, but the person responsible for reporting these metrics must be named in writing. The delegation of reporting waste metrics by a sub-contractor must be sent to Environmental Management via the Contract Administrator no less than 2 weeks before the sub-contractor begins work on the contract.
- **Waste Minimization.** The contractor and sub-contractors must make every effort to use nonhazardous materials, chemicals and products, prevent spills, and recycle wastes in the performance of the contract. Unused chemicals purchased for the contract must be removed from Minot AFB at the end of the contract, and may be used on other projects. Contractors must request in writing permission from Environmental Management to leave chemicals on Minot AFB. Requests must be submitted at least 3 weeks before the end of the contract, written on company letterhead paper, and signed by project manager or another company representative who is responsible for the performance of the contract. The letter must state what chemicals will be left, the volume/weight of the chemical, and the name and phone number of the government agent accepting the chemicals.

Household Hazardous Materials

Per 40 CFR 261.4(b)(1), most household waste (that is, waste from a house, apartment, or dorm) that is generated as a result of normal maintenance performed by the owner/occupant is exempt from regulation under RCRA. Notable exceptions include lead acid batteries, fluorescent lamps, used oil, and spent antifreeze.

- Residents of military family housing on Minot AFB should exchange depleted lead-acid batteries through the vendor from whom they purchase the replacement battery. The AFFES Service Station will exchange depleted lead-acid batteries at the time a replacement battery is purchased.
- Used oil and used antifreeze generated by military family housing residents during do-it-yourself oil changes should be taken to the Auto Skill Center.

- Regular and compact fluorescent lamps should be taken to the Military Family Housing maintenance contractor office. No fluorescent lamps may go in the trash.

8 REFERENCES

Standard References *(Applicable to all AF Installations)*

- DAFI 32-7001, *Environmental Management*
- DAFMAN 32-7002, Environmental Compliance and Pollution Prevention
- DAFI 90-302, The Inspection System of the Department of the Air Force
- AFLOA HW Legal and Other Requirements – The Air Force Legal Operations Agency (AFLOA) legal registry lists and provides access to federal (e.g., CFR, U.S. Code), DoD, AF, and other legal requirements
- ARCNet – Training resource for Air Force Reserve Command
- DoD 4160.21 (all volumes), Defense Materiel Disposition
- EASIER Database
- eDASH HW Environmental Action Plans (EAPs)
- eDASH HW Home Page
- eDASH HW Training Matrix
- EESOH-MIS Application Login
- EESOH-MIS Support Portal
- The Environmental Awareness Course Hub (TEACH)
- HW Playbook

9 ACRONYMS

Standard Acronyms (Applicable to all AF Installations)

- eDASH Acronym Library
- HW Playbook – Acronym Section
- U.S. EPA Terms & Acronyms

Installation Supplement

ASD	Accumulation Start Date
IW	Industrial Waste
LF	Launch Facility
LLRW	Low-Level Radioactive Waste
NDAC	North Dakota Administrative Code
NDDEQ	North Dakota Department of Enviornmental Quality

10 DEFINITIONS

Standard Definitions (Applicable to all AF Installations)

- HW Playbook – Definitions Section

11 INSTALLATION-SPECIFIC CONTENT

Installation Supplement

Hazardous waste regulatory requirements should be applied in the context of the installation's status as an SQG, which provides administrative and procedural advantages that significantly reduce the installation's regulatory burden for handling and managing hazardous waste. Additionally, it should be noted that the 91st Missile Wing's missile alert and launch facilities—which constitute GSUs under the purview of Minot AFB—are regulated by the U.S. Environmental Protection Agency (EPA) and North Dakota as Conditionally Exempt Small Quantity Generators (CESQGs). This status allows Minot AFB to manage the hazardous waste generated at these locations under less stringent regulatory requirements even though the GSU wastes count toward the installation's overall generator status. Significantly, each GSU has its own state/EPA Generator Identification Number, and wastes generated by the GSUs are annually reported to the EPA and North Dakota regulators.

Management of 91st Missile Wing Wastes

Background. Personnel of the 91st Missile Wing who perform maintenance at GSUs in the surrounding missile field typically obtain a hazardous material on Minot AFB, transport it to the missile fieldwork site, and generate a waste at the work site. The waste is then transported back to Minot AFB where it is consolidated and processed for turn-in to the CAP. This established process and generation of low waste volumes resulted in NDDEQ classifying the missile flights as a CESQG, each with its own EPA identification number.

Generator Status Determination

- Each missile flight (Alpha through Oscar) is identified as a CESQG. Federal and state regulations do not require a CESQG to have an EPA identification number. However, to track wastes specifically generated by a missile flight, the NDDEQ Division of Waste Management gave each its own EPA identification number in the following table.

EPA Identification Numbers for 91st Missile Wing Missile Flights

Missile Flight	EPA Number	Owning Organization
Alpha	ND8570090110	740th Missile Squadron
Bravo	ND8570090094	740th Missile Squadron
Charlie	ND4570090106	740th Missile Squadron
Delta	ND5570090105	740th Missile Squadron
Echo	ND3570090099	740th Missile Squadron
Foxtrot	ND6570090104	741st Missile Squadron
Golf	ND0570090100	741st Missile Squadron
Hotel	ND6570090096	741st Missile Squadron
India	ND7570090095	741st Missile Squadron
Juliet	ND9570090101	741st Missile Squadron
Kilo	ND3570090107	742d Missile Squadron
Lima	ND2570090108	742d Missile Squadron
Mike	ND7570090103	742d Missile Squadron
November	ND8570090102	742d Missile Squadron
Oscar	ND1570090109	742d Missile Squadron

- Wastes generated from each missile flight must be characterized as RCRA hazardous, state regulated, universal, or non-hazardous, and must be counted on a monthly basis. If the volume of RCRA hazardous wastes for that month is greater than 220 lbs., the specific missile flight must follow the regulations governing an SQG. All wastes for that month from that missile flight must be handled in the same manner as waste generated on Minot AFB, which includes manifesting the wastes from the field to the installation, labeling and marking waste containers as "Hazardous Waste" while in transit to the installation, and moving the wastes to the CAP within 3 days of arriving on the installation.
- Currently, wastes from the missile field are well within the CESQG requirements. It is the responsibility of all 91st Missile Wing hazardous waste generators to correctly characterize the wastes generated in the field and to know how to properly transport them. The 91st Missile Wing Hazardous Waste Managers shall count monthly waste volumes and report the information to the installation Hazardous Waste Program Manager via email or phone (723-4870). The Hazardous Waste Managers shall also notify the installation Hazardous Waste Program Manager of all changes in their waste streams and any project that would increase their waste generation volume, which could potentially change generator status to a higher level of compliance.

91st Missile Wing Waste Manager Delegations

Each 91st Missile Wing organization that generates waste must delegate at least one primary and one alternate Hazardous Waste Manager and one UEC to be responsible for their generated waste. The delegated personnel may be at the squadron or group level, but must have at least one year left on station and be trained in accordance with Section 5.0 of this plan. Squadron level managers shall be trained as Hazardous Waste Managers, but group level managers shall be trained as UECs in order to serve as focal points for environmental compliance within their group. 91st Missile Wing UECs shall assist in assessments and inspections of the waste generating areas under their responsibility. Delegation of 91st Missile Wing Hazardous Waste Managers and UECs shall be in writing and signed by the appropriate commander. Delegation letters shall be sent to Environmental Management in accordance with this plan. In addition, when a shop Hazardous Waste Manager or UEC separates or PCS's from Minot AFB, they shall notify the Hazardous Waste Program Manager (723-4870) at least 30 days prior to their leaving.

91st Missile Wing Waste Container Management

Waste containers from the 91st Missile Wing may hold a single product or be a consolidation of wastes. Regardless, waste generated in the missile field shall have the missile flight(s) identified on the waste container label or on a waste log sheet that is turned in with the waste. The information shall be presented as follows.

- Wastes from several places within one missile flight shall show the flight designation. For example, used solvent collected from E-7, E-10 and E-12 would be written on the container label as "Waste Solvent, E Flight."
- Wastes from different missile flights shall show each flight designation. For example, glycol brine from C-1, F-8, and N-7 should be written on the container label as "Waste glycol brine, C, F, N Flights."
- Wastes generated by 91st Missile Wing contractors shall also be identified by missile flight(s). For example, waste lead-based paint generated by ABC Paint Company from B-1, C-1, K-1, and L-1 shall be identified on the container label as "Waste lead-based paint and debris, B, C, K, L Flights (ABC Paint Co)."
- In lieu of writing the different missile flights on the container label, a waste log sheet may accompany the waste container when turned in to the CAP. The log sheet shall show the dates, volumes, and locations of the consolidated wastes. This is especially helpful for large volume collections (that is, greater than 55 gallons) or when wastes come from a large number of sites.

APPENDIX A WASTE ANALYSIS PLAN

Installation Supplement

Introduction

Background

This appendix comprises the Minot AFB Waste Analysis Plan (WAP) and describes the procedures used on the installation to (1) identify all potentially hazardous waste streams and (2) assess whether waste streams require additional evaluation and analysis. While the EPA requires WAPs for treatment, storage and disposal facilities (TSDFs) and generators who treat hazardous waste to meet land disposal restriction (LDR) standards, the Air Force requires WAPs for all installations per AFMAN 32-7002, Environmental Compliance and Pollution Prevention. In particular, this AFMAN stipulates that the waste stream inventory (WSI) and the WAP be included in the installation's Hazardous Waste Management Plan. This WAP applies to all hazardous waste streams generated on Minot AFB and its associated GSUs that are subsequently transferred to the CAP at Building 531 for disposal.

Contents

The WAP is intended to comply with applicable EPA, North Dakota Department of Environmental Quality (NDDEQ), and Air Force requirements and guidance relating to the identification and evaluation of hazardous wastes generated at Minot AFB and its associated GSUs. Accordingly, this WAP includes the following:

- An outline of the procedures used for the identification, sampling, and analysis of known and suspected hazardous waste to ensure proper handling and disposal.
- A description of the methods used to characterize wastes stored on Minot AFB to ensure sufficient information is available for safe handling and storage.
- Detailed procedures for obtaining physical and chemical analyses of wastes, which includes selecting parameters for analysis, selecting sampling and analytical methods, documenting samples, selecting laboratories, and re- evaluating hazardous waste streams.

Roles and Responsibilities

- Generators of hazardous waste are responsible for identifying and characterizing all waste streams within their area of responsibility. The shop Hazardous Waste Manager is responsible for informing the Environmental Management Element, 5 CES/CEIE, (Environmental Management) of any new waste streams and anticipated amounts. When required, the generating activity may request a hazardous waste determination from Environmental Management. Subsequently, the shop Hazardous Waste Manager must ensure that all wastes are managed in accordance with applicable regulations and Section 7.5, Accumulation Area Management, of this plan.
- As required, Environmental Management will help waste generators properly characterize their waste streams as described in 40 CFR 262.11.
- Bioenvironmental Engineering visits on-base shops in accordance with the Environmental Safety and Occupational Health (ESOH) Assessment schedule to confirm that all waste streams have been properly identified and characterized.

Hazardous Waste Characterization Process

The first step in characterizing a hazardous waste stream is determining conclusively whether the material is a hazardous waste subject to regulation under RCRA and/or associated North Dakota regulations. This determination starts by asking three questions:

- Is the material a waste?
- Is the waste a solid waste?
- Is the solid waste a hazardous waste?

Each generating activity is responsible for making this determination. If it is determined that the material to be discarded is a hazardous waste, then the material must be described and the amount quantified. Accordingly, the process of characterizing a hazardous waste stream includes three sequential steps: (1) confirm that the material is a hazardous waste; (2) describe the waste per applicable regulations; and (3) determine the amount of waste generated annually. Waste streams on Minot AFB and its associated GSUs are characterized using the generator's knowledge of the waste, associated SDSs or MSDSs, and by analytical testing if required. The following paragraphs detail the hazardous waste characterization process that applies to Minot AFB.

Waste Determination

A material is a waste if it meets any of the following criteria:

- The material can no longer be used for its intended purpose. This could be due to the material being contaminated, no longer meeting specifications, or comprising a spill residue. Examples include contaminated solvent, dried epoxy resins or sealant, spent bead blast media, and spent grease.
- The material loses its usefulness due to age or poor storage, such as expired shelf-life items or paint that was not sealed properly and dried out.
- The material is an unintended or unusable byproduct, such as spill residue.
- The material is discarded for eventual treatment, storage, recycling, or disposal, such as used antifreeze or used oil.
- The material is a product of the cleanup of a previously uncontrolled waste site, such as contaminated soil from a storage yard.
- A hazardous material that has exceeded its shelf life is not considered a waste until:
    - The shelf life cannot be extended;
    - No other uses are available for the material in another process or application;
    - The material cannot be recycled or returned to the manufacturer; and
    - The material cannot be sold through DLA.

Solid Waste Determination

After determining that the material is a waste, it must be determined whether the waste is a “solid waste.” Solid wastes can be solid, liquid, gas or a mixture of these physical states as described by EPA regulations and associated definitions. In the context of this WAP, a solid waste is generally defined as any discarded material that is abandoned, recycled, or considered inherently waste-like regardless of the physical state. Thus, a waste is a solid waste if it meets any of the following criteria:

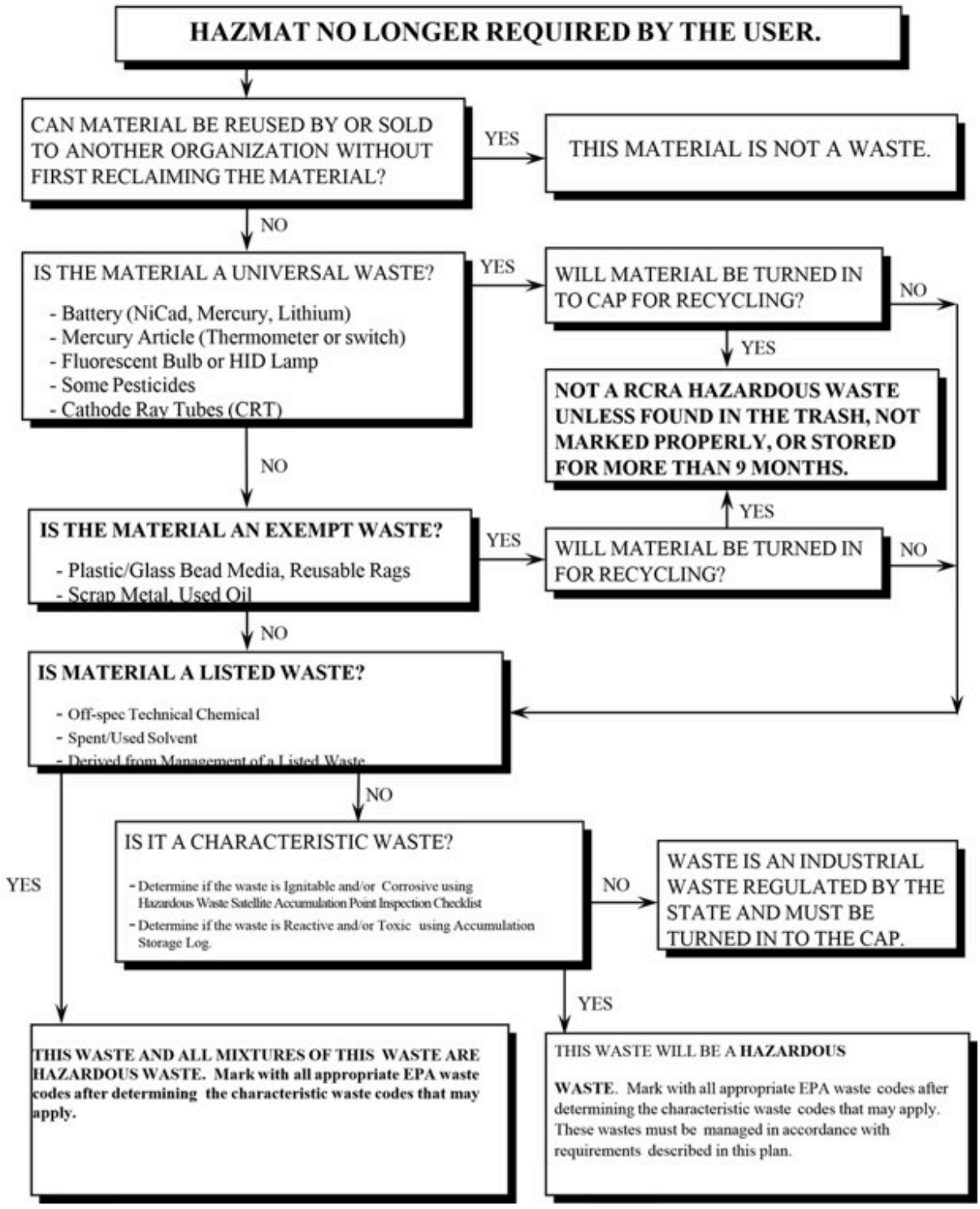
- The waste material is being abandoned through disposal, burning, or incineration.
- The waste material is accumulated, stored or treated, but not recycled, before or instead of being disposed of, burned, or incinerated.
- The waste material is a secondary material and is recycled or is accumulated, stored, and treated before recycling.
- The material is inherently waste-like and is recycled.

Certain exemptions may apply, but these are not particularly relevant to Minot AFB. If generators have questions regarding solid waste exemptions, they should contact the installation Hazardous Waste Program Manager at 723-4870.

Hazardous Waste Determination

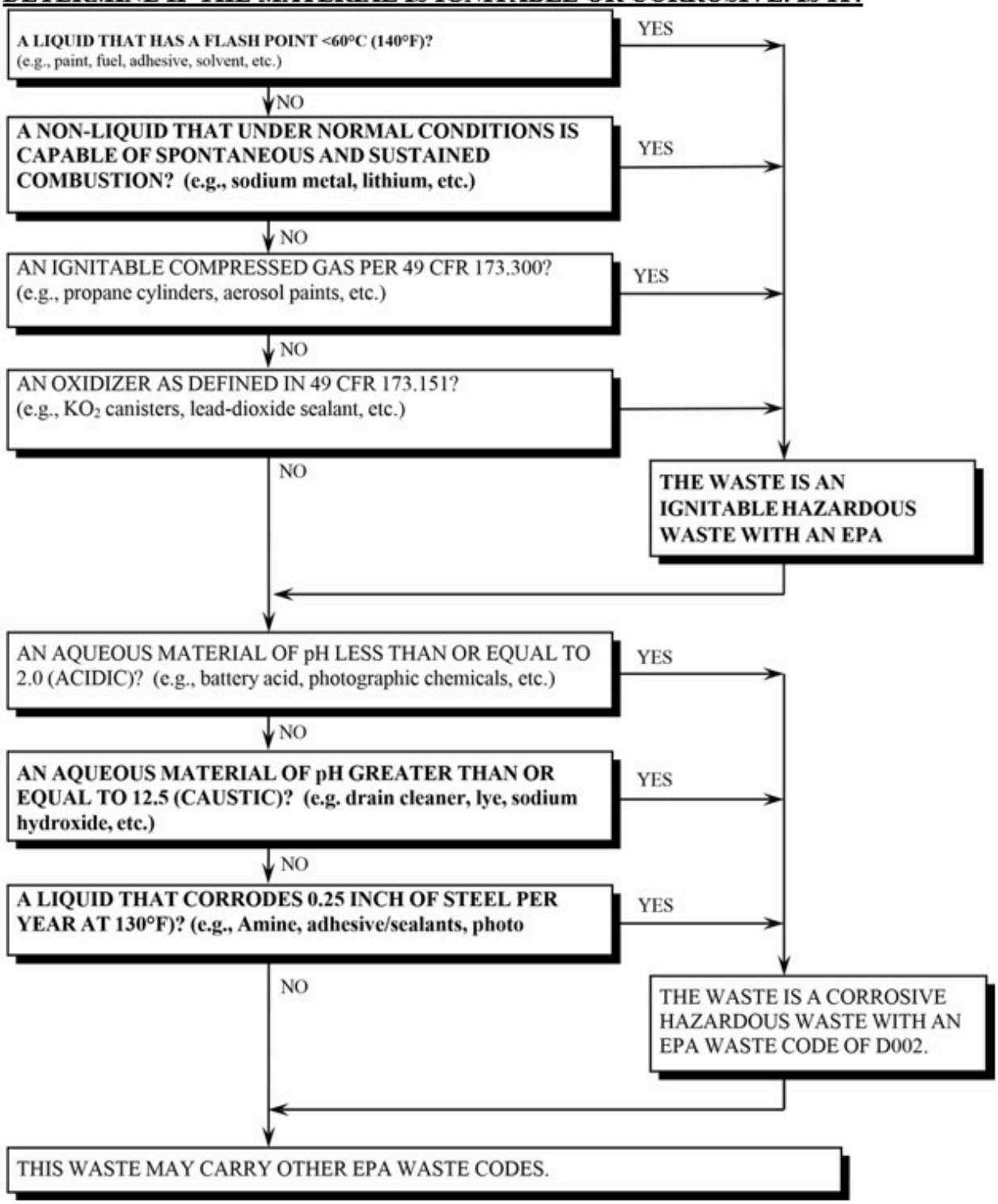
After determining that the waste material is a solid waste per the RCRA definition1, the generator must determine whether the solid waste is a hazardous waste. The EPA considers a solid waste to be hazardous if (1) it possesses certain characteristics, such as ignitability, corrosivity, reactivity, or toxicity or (2) it is on a list of specific wastes determined by the EPA to be hazardous. Mixing nonhazardous and hazardous wastes creates more hazardous waste. Hazardous Waste Managers should use the decision trees on the following pages to characterize hazardous waste in accordance with EPA criteria.

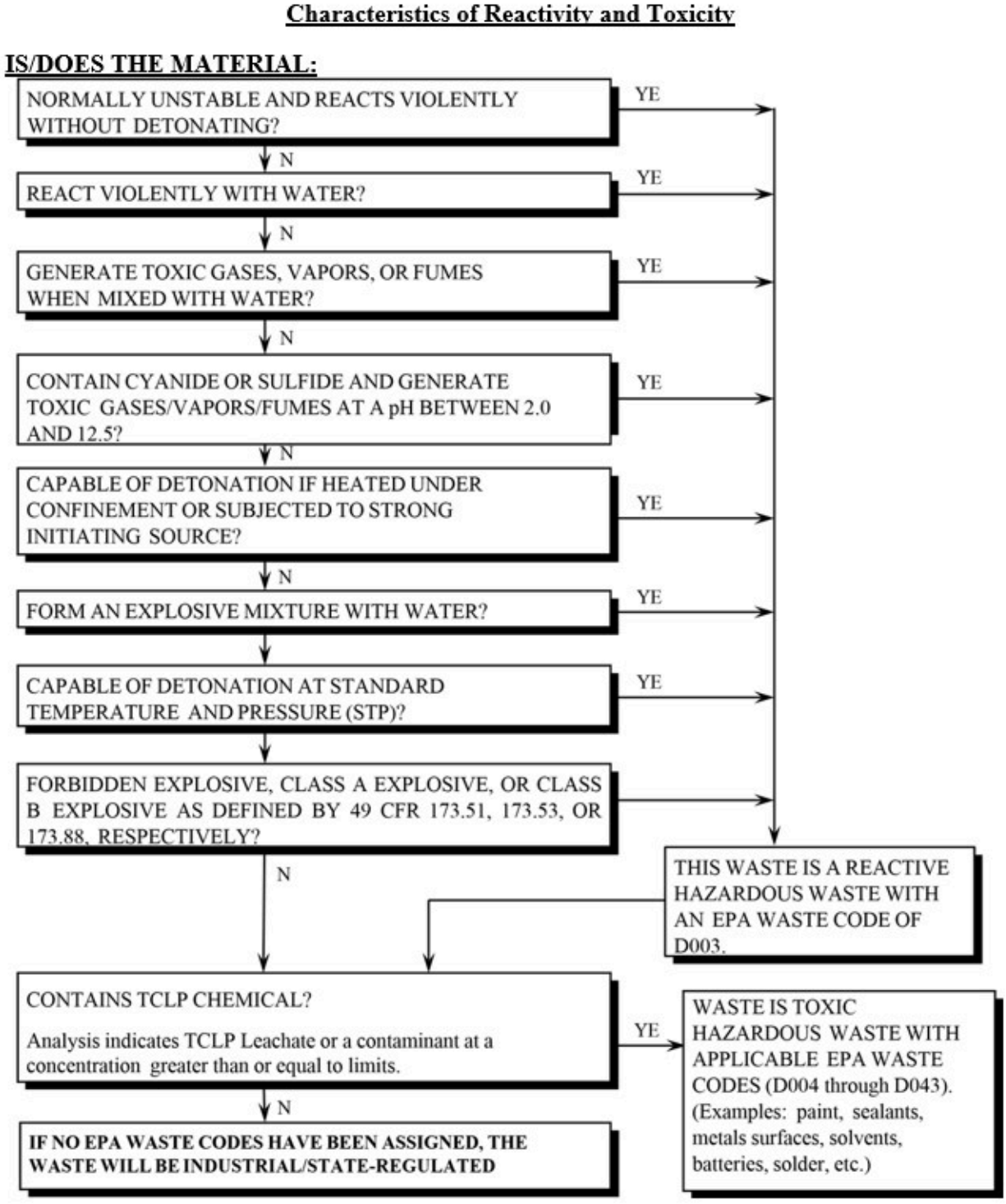
Waste Stream Characterization



Characteristics of Ignitability and Corrosivity

DETERMINE IF THE MATERIAL IS IGNITABLE OR CORROSIVE. IS IT:





Characteristic Hazardous Waste

A solid waste with the characteristic of ignitability, corrosivity, reactivity and/or toxicity per EPA criteria, is described as a characteristic hazardous waste. Some characteristic hazardous wastes may also be listed hazardous wastes. All characteristic wastes shall be managed in accordance with federal, state, and local regulations, and this plan. The following paragraphs describe the criteria for determining whether a solid waste is a characteristic hazardous waste.

- Ignitability Criteria. A solid waste that meets any of the following ignitability criteria is considered an ignitable hazardous waste and is assigned the EPA hazardous waste number D001:
    - A liquid that has a flash point of less than 140°F.
    - A non-liquid that (1) under normal ambient conditions can cause a fire through friction, absorption of moisture, or spontaneous chemical changes and (2) when ignited burns so vigorously and persistently that it creates a hazard.

    - A compressed gas defined as ignitable by DOT regulation 49 CFR 73.300, *Exceptions for Class 4*.
    - An oxidizer as defined in DOT regulation 49 CFR 173.151.
- Corrosivity Criteria. A solid waste that meets any of the following corrosivity criteria is considered a corrosive hazardous waste and is assigned the EPA hazardous waste number D002:
    - An aqueous solution having a pH greater than *or* equal to 12.5 (i.e., strong base) or less than or equal to 2.0 (i.e., strong acid).
    - A liquid that corrodes steel faster than 0.25-inch per year at 130°F.
- Reactivity Criteria. A solid waste that meets any of the following reactivity criteria is considered a reactive hazardous waste and is assigned the EPA hazardous waste number D003:
    - It is normally unstable and readily undergoes violent change without detonating.
    - It reacts violently with water.
    - It forms potentially explosive mixtures with water
    - When mixed with water, it generates toxic gases, vapors, or fumes in a quantity sufficient to endanger human health or the environment.
    - Cyanide or sulfide bearing waste that, when exposed to pH conditions between 2.0 and 12.5, can generate toxic gases, vapors, or fumes in a quantity sufficient to endanger human health and the environment.
    - It is capable of detonation, explosive decomposition, or reaction at standard temperature and pressure.
    - A forbidden explosive per 49 CFR 173.51, a Class A explosive per 49 CFR 173.53, or a Class B explosive per 49 CFR 173.88
- Toxicity Criteria. The EPA's toxicity criteria is designed to encompass those solid wastes that (1) are harmful or fatal when ingested or come in contact with the skin and/or (2) when disposed of on land may produce liquids that could pollute the groundwater. Toxicity is determined using the Toxicity Characteristic Leaching Procedure (TCLP), also known as EPA Test Method 1311. A solid waste that meets the EPA's toxicity criteria is considered a characteristic hazardous waste and is assigned an EPA hazardous waste number in the range of D004 to D043 (40 CFR 261.24, Table 1, *Maximum Concentration of Contaminants for the Toxicity Characteristic*). TCLP leachate concentrations used to define toxicity are contained in the Toxicity Characteristic Leachate Procedure Regulatory Concentration Levels table. Examples of this waste are lead solder, cadmium paint, and sodium chromate.

Listed Hazardous Waste

The EPA specifically lists certain hazardous wastes within RCRA because they are considered hazardous to humans and the environment. These wastes are listed in 40 CFR 261, Subpart D, *Lists of Hazardous Wastes*, and are referred to as listed hazardous wastes. Listed hazardous waste may be derived from:

- Nonspecific sources (e.g., spent solvents like methanol, toluene and MEK), which are F-listed wastes,
- Specific sources (e.g., wastewater treatment sludge from the processing of explosives), which are K-listed wastes, or
- From off-specification commercial chemicals, which are P-listed (acutely toxic), or U-listed (toxic) wastes.

Hazardous wastes from these sources are listed in 40 CFR parts 261.31 through 261.33. Note that each listed hazardous waste is assigned its own EPA hazardous waste number that must be used to comply with notification, record keeping, and reporting requirements described in 40 CFR 262 through 265, 268, and 270.

The F-Listed Non-Specific Source Waste, Spent Solvents and P and U–Listed Wastes Excerpts: Discarded Commercial Chemical Products tables list nonspecific and off-specification waste streams applicable to Minot AFB. Air Force installations generally do not generate wastes from specific sources, so the K-listed wastes are not included. Listed hazardous waste may also be a characteristic hazardous waste (see the Toxicity Characteristic Leachate Procedure Regulatory Concentration Levels table). For example, MEK is D001 (ignitable) and F005 when spent or U159 when unused. All applicable hazardous waste identification numbers are to be assigned to a hazardous waste to ensure full compliance with federal and North Dakota disposal requirements.

Universal Waste

The EPA and North Dakota Department of Environmental Quality regulations allow less stringent management standards for batteries, lamps/bulbs, mercury-containing articles, and certain recalled/unused pesticides. Refer to Section 7.11, of this plan for further details regarding the management of Universal Waste.

Excluded Waste

A number of wastes are excluded from classification as a hazardous waste by the EPA, and several of these exclusions may apply to Minot AFB. Excluded wastes include the following:

- Used oil filters that are not manufactured of terne-plate, are gravity hot- drained at 60°F or higher for a minimum of 12 hours, and are not mixed with a listed hazardous waste per 40 CFR 261.4(b)(13). (Terne-plate is an alloy coating of lead and tin used to inhibit corrosion of steel.)
- Used chlorofluorocarbon (CFC) refrigerants from totally enclosed heat transfer equipment (such as mobile air conditioning systems and commercial and industrial air conditioning and refrigeration systems) provided the refrigerant is reclaimed for further use per 40 CFR 261.4(b)(12).
- Household waste from residential sources per 40 CFR 261.4(b)(1).

If generators have questions regarding the applicability of such exclusions, they should contact the Hazardous Waste Program Manager at 723-4870.

Toxicity Characteristic Leachate Procedure Regulatory Concentration Levels (D- List/Characteristic Wastes)

EPA HW No.	Contaminant	CAS No.	Regulatory Level (mg/L)[1]	Regulatory Level (%)[1]
D004	Arsenic[2]	7440-39-2	5.0	0.0005
D005	Barium[2]	7440-39-3	100	0.01
D018	Benzene	71-43-2	0.5	0.00005
D006	Cadmium[2]	7440-43-9	1.0	0.0001
D019	Carbon tetrachloride	56-23-5	0.5	0.00005
D020	Chlordane	57-74-9	0.03	0.000003
D021	Chlorobenzene	108-90-7	100	0.01
D022	Chloroform	67-66-3	6.0	0.0006
D007	Chromium[2]	7440-47-3	5.0	0.0005
D023	o-Cresol	95-48-7	200.0 [3]	0.02
D024	m-Cresol	108-39-4	200.0 [3]	0.02
D025	p-Cresol	106-44-5	200.0 [3]	0.02
D026	Cresol		200.0 [3]	0.02
D016	2,4-D	94-75-7	10.0	0.001
D027	1,4-Dichlorobenzene	106-46-7	7.5	0.00075
D028	1,2-Dichloroethane	107-06-2	0.5	0.00005
D029	1,1-Dichloroethylene	75-35-4	0.7	0.00007
D030	2,4-Dinitrotoluene	121-14-2	0.13 [4]	0.000013
D012	Endrin	72-20-8	0.02	0.000002
D031	Heptachlor (and its epoxide)	76-44-8	0.008	0.0000008
D032	Hexachlorobenzene	118-74-1	0.13 [4]	0.000013
D033	Hexachlorobutadiene	87-68-3	0.5	0.00005
D034	Hexachloroethane	67-72-1	3.0	0.0003
D008	Lead[2]	7439-92-1	5.0	0.0005
D013	Lindane	58-89-9	0.4	0.00004
D009	Mercury[2]	7439-97-6	0.2	0.00002
D014	Methoxychlor	72-43-5	10.0	0.001
D035	Methyl ethyl ketone	78-93-3	200.0	0.02
D036	Nitrobenzene	98-95-3	2.0	0.0002
D037	Pentrachlorophenol	87-86-5	100.0	0.01
D038	Pyridine	110-86-1	5.0 [4]	0.0005
D010	Selenium[2]	7782-49-2	1.0	0.0001
D011	Silver[2]	7440-22-4	5.0	0.0005
D039	Tetrachloroethylene	127-18-4	0.7	0.00007
D015	Toxaphene	8001-35-2	0.5	0.00005
D040	Trichloroethylene	79-01-6	0.5	0.00005

D041	2,4,5-Trichlorophenol	95-95-4	400.0	0.04
D042	2,4,6-Trichlorophenol	88-06-2	2.0	0.0002
D017	2,4,5-TP (Silvex)	93-72-1	1.0	0.0001
D043	Vinyl chloride	75-01-4	0.2	0.00002

NOTES:

•Regulatory Level mg/L = milligrams per liter, which is approx. equal to parts per million (ppm) or 0.0001% • Compounds and mixtures of D004, D005, D006, D007, D009, D010, and D011 can also be toxic.

•If o-, m-, and p-Cresol concentrations cannot be differentiated, then the total cresol (D026) concentration is used. The regulatory level of total cresol is 200 mg/L.

•Quantitation limit is greater than the calculated regulatory level. The quantitation limit, therefore, becomes the regulatory level.

This table is adapted from EPA Publication SW-846, Test Methods for Evaluating Solid Waste, Physical/Chemical Methods, which is incorporated into 40 CFR 260.11 by reference.

F-Listed Non-Specific Source Waste, Spent Solvents [a]

EPA Hazardous Waste No.	Hazardous Waste	Constituent	Hazard Code
F001	• Spent halogenated solvents listed at right that are used in degreasing. • All spent solvent mixtures/blends used in degreasing containing, before use, a total of ten percent or more (by volume) of one or more of the halogenated solvents listed at right or those solvents listed in F002, F004, and F005. • Still bottoms from the recovery of these spent solvents and spent solvent mixtures. • Spent halogenated solvents listed at right.	Tetrachloroethylene Trichloroethylene Methylene chloride 1,1,1-Trichloroethane Carbon tetrachloride Chlorinated fluorocarbons	**(T)**
F002	• All spent solvent mixtures/blends containing, before use, a total of ten percent or more (by volume) of one or more of the above-halogenated solvents or those listed in F001 F004, or F005. • Still bottoms from the recovery of these spent solvents and spent solvent mixtures • Spent halogenated solvents listed at right.	Tetrachloroethylene Methylene chloride Trichloroethylene 1,1,1-Trichloroethane Chlorobenzene 1,1,2-Trichloro-1,2,2- trifluoroethane Ortho-dichlorobenzene Trichlorofluoromethane 1,1,2- Trichloroethane	**(T)**
F003	• All spent solvent mixtures/blends containing, before use, only the spent non-halogenated solvents listed at right. • All spent solvent mixtures/blends containing, before use, one or more of the non-halogenated solvents listed at right, and, a total of ten percent or more by volume) of one or more of those solvents listed in F001, F002, F004, and F005. • Still bottoms from the recovery of these spent solvents and spent solvent • The spent non-halogenated solvents listed at right	Xylene Acetone Ethyl acetate Ethyl benzene Ethyl ether Methyl isobutyl ketone N-Butyl alcohol Cyclohexanone Methanol	**(I)**

F004	• All spent solvent mixtures/blends containing, before use, a total of ten percent or more (by volume) of one or more of the non-halogenated solvents listed at right or those solvents listed in F001, F002 and F005 • Still bottoms from the recovery of these spent solvents and spent solvent mixtures • The spent non-halogenated solvents listed at right.	Cresols and cresylic acid Nitrobenzene	**(T)**
F005	• All spent solvent mixtures/blends containing, before use, a total of ten percent or more(by volume) of one or more of the non-halogenated solvents listed at right or those solvents listed in F001, F002, or F004 • Still bottoms from the recovery of these spent solvents and spent solvent mixtures	Methyl Ethyl Ketone Carbon Disulfide Isobutanol Pyridine Benzene 2-Ethoxyethanol 2-Nitropropane	**(I, T)**

a.
I = ignitable

This table is adapted from 40 CFR 261.31, *Hazardous Wastes from Non-Specific Sources*.

P and U–Listed Wastes Excerpts: Discarded Commercial Chemical Products

EPA HW No.	Substance	CAS No.	EPA HW No.	Substance	CAS No.
P042	Epinephrine	51-43-4	**U129**	Lindane	58-89-9
P078	Nitrogen Dioxide	10102-44-0	**U151**	Mercury	7439-97-6
P081	Nitroglycerine (R)	55-63-0	**U029**	Bromomethane	74-83-9
P115	Sulfuric Acid, Thallium(I) Salt	7446-18-6	**U075**	Dichlorodifluoromethane	75-71-8
P001	Warfarin	81-81-2	**U121**	Trichlorofluoromethane	75-69-4
U002	Acetone	67-64-1	**U154**	Methanol (I)	67-65-1
U003	Acetonitrile (I,T)	75-05-8	**U226**	Methyl Chloroform	71-55-6
U020	Benzenesulfonyl Chloride (C,R)	98-09-9	**U080**	Methylene Chloride	75-09-2
U211	Carbon Tetrachloride	56-23-5	**U159**	Methyl Ethyl Ketone (I,T)	78-93-3
U036	Chlordane	57-74-9	**U165**	Naphthalene	91-20-3
U044	Chloroform	67-66-3	**U169**	Nitrobenzene (I,T)	98-95-3
U057	Cyclohexanone	108-94-1	**U188**	Phenol	108-95-2
U058	Cyclophosphamide	50-18-0	**U170**	p-Nitrophenol	100-02-7
U061	DDT	50-29-3	**U202**	Saccharin, & Salts	81-07-2
U070	O-Dichlorobenzene	95-50-1	**U103**	Sulfuric Acid, Dimethyl Ester	77-78-1
U072	P-Dichlorobenzene	106-46-7	**U210**	Tetrachloroethylene	127-18-4
U108	1,4-Dioxane	123-91-1	**U213**	Tetrahydrofuran (I)	109-99-9
U117	Ethyl Ether (I)	60-29-7	**U220**	Toluene	108-88-3
U122	Formaldehyde	50-00-0	**U227**	1,1,2-Trichloroethane	79-00-5
U132	Hexachlorophene	70-30-4	**U228**	Trichloroethylene	79-01-6
U140	Isobutyl Alcohol (I,T)	78-83-1	**U239**	Xylene (I)	1330-20-7

Note: This is only a partial list of Off-Specification Commercial Chemical Listed Wastes. Complete lists are found in 40 CFR 261.33. The underlined EPA HW numbers identify waste codes for items that have been generated in the past on Minot AFB.

I = Ignitable T = Toxic C = Corrosive R = Reactive

Exempt Substance

Certain substances meeting the definition of a hazardous waste may be exempt from hazardous waste regulations for a variety of reasons, which include (1) encouraging collection for recycling, reuse, and energy recovery; (2) other laws that have primary regulation of the waste (e.g., the Toxic Substance Control Act and the Atomic Energy Act); or (3) other practical considerations. The management of these materials and items is further detailed in Section 7.1, of this plan. Among the exempt substances applicable to Minot AFB are the following:

- Samples collected for testing while in transport or in storage prior to transport to the laboratory. Once the sample is returned, it is no longer exempt.
- Waste generated by a Conditionally Exempt Small Quantity Generator, such as a missile alert facility (MAF) or a launch facility (LF), is exempt while in transit per 40 CFR 261.5, *Special Requirements for Hazardous Waste Generated by Conditionally Exempt Small Quantity Generators*. When the waste is to be disposed of, it is no longer exempt.
- Certain hazardous wastes that are recycled (i.e., recyclable materials) are exempt. At Minot AFB, this includes spent lead-acid batteries, scrap metal, and used oil per 40 CFR 261.6, *Requirements for Recyclable Materials*.
- Residues of hazardous waste in empty containers or inner liners provided the waste is not a listed acutely hazardous waste (i.e., P-listed waste) and the amount of the residue does not exceed the amount stipulated in 40 CFR 261.7, *Residues of Hazardous Waste in Empty Containers*.
- PCB wastes regulated under the Toxic Substances Control Act (TSCA), which exempts them from RCRA hazardous waste regulations, per 40 CFR 261.8, *PCB Waste Regulated under the Toxic Substance Control Act*.

If generators have questions regarding the applicability of such exemptions, they should contact the Hazardous Waste Program Manager at 723-6509/6380.

Non-RCRA Regulated Waste

In accordance with North Dakota Department of Environmental Quality regulations, waste that is non-hazardous but contains or is contaminated with certain organic or inorganic chemicals is a regulated State Waste and is also known as an Industrial Waste. North Dakota regulations restrict the types and quantities of Industrial Waste disposed of at municipal landfills. Minot AFB manages this waste as a Non-RCRA Regulated Waste. Shops generating this type of waste are managed as a Satellite Accumulation Point (SAP).

Other Waste

- Some wastes are regulated under other regulations, such as the TSCA for asbestos and PCBs and the Atomic Energy Act for radioactive waste from nuclear sources including nuclear munitions. These wastes are known as "other wastes" and are not addressed in this plan.
- All conventional explosive ordnance, whether usable or waste, will be managed in accordance with DoD 6055.9-STD, DoD *Ammunition and Explosive Safety Standards*. This standard mandates the waste military munitions be managed in accordance with the EPA's Military Munitions Rule (see below).
- Mixed wastes that are characterized as both "other waste" and hazardous waste will be managed in accordance to both RCRA and any other applicable regulation.

Military Munitions Rule

The Military Munitions Rule (MMR) was created by the EPA in consultation with the DOD, and published in 1997 in the Federal Register, Volume 62, Number 29, *Military Munitions Rule: Hazardous Waste Identification and Management; Explosives Emergencies; Manifest Exemption for Transport of Hazardous Waste on Right-of-Ways on Contiguous Properties*. The MMR details the proper management of waste military munitions and identifies when munitions become a solid waste subject to RCRA regulations. MMR defines the term "military munitions" to include ammunition products and components used by U.S. Armed Services for national defense and security. Military munitions do not include nuclear weapons, wholly inert items, and improvised explosive devices.

- Unused Munitions. Unused military munitions are considered a solid waste when any of the following conditions apply:
    - The unused munition is abandoned by being disposed of, burned, detonated (except during intended use), incinerated, or otherwise treated prior to disposal.
    - The unused munition is removed from storage for the purpose of being disposed of, burned, incinerated, or treated prior to disposal.
    - The unused munition is deteriorated to the extent it can no longer be returned to a serviceable condition, and it cannot be reasonably recycled or used for other purposes.
    - The unused munition has been declared a solid waste by an authorized military official.
- Munitions Used for Intended Purpose. The MMR, as detailed in 40 CFR 266.202(a), states that for regulatory purposes used munitions are not a solid waste under the following circumstances:
    - The munition is used for training of military personnel, EOD personnel, and emergency response specialists.

- The munition is used in weapons research, development, testing, and evaluation.
- The recovery and collection for on-range treatment and/or destruction of unexploded ordnance (UXO) and munitions fragments during range clearance activities.
- Disassembly operations; that is, unused munitions/components being repaired, reused, recycled, reclaimed, disassembled, reconfigured, or otherwise subject to material recovery activities.

- Used or Fired Munitions. As described in 40 CFR 266.202(c) and (d), used or fired munitions are potentially subject to RCRA regulation when:
  - The munition is removed or managed (i.e., stored, treated, or disposed of) off-range.
  - The munition is recovered, collected, and then disposed by being buried or landfilled on-range.
  - The munition lands off-range.
- Stored Waste Military Munitions. In accordance with 40 CFR 266.205, stored waste military munitions are conditionally exempt from RCRA regulation only if the following conditions are met:
  - The waste munitions are not chemical agents or chemical munitions.
  - The waste munitions are subject to the jurisdiction of the Department of Defense Explosives Safety Board (DDESB).
  - The waste munitions are stored in accordance with DDESB storage standards.
  - Access to waste munition storage units is restricted, and the EPA must be notified of the location of any new waste munitions storage units within 90 days.

Municipal Waste

Municipal waste is not regulated as hazardous, industrial, or other waste, and is not addressed in this plan.

Waste Awaiting Analysis

Unidentified and uncharacterized wastes waiting for analysis is not considered a hazardous waste by the North Dakota Department of Environmental Quality until the analytical results are returned, providing the following occurs:

- Before requesting characterization from Environmental Management the waste container number, nomenclature describing the waste, and the words "Awaiting Analysis" are marked on each container by the generator.
- The generator submits a sample request to Environmental Management as soon as possible after the waste is discovered, but no longer than three days.
- From the time of the request, Environmental Management samples the waste within 15 working days.
- Each container of sampled waste is marked with the sample date and number.
- Waste awaiting analysis must be stored at the CAP, or at a SAP. Waste containers must be in good condition and kept closed. Additional wastes generated by the SAP require the container awaiting analysis to be moved immediately to the CAP.
- Waste analytical results are typically returned within 25 working days after the sample is sent for analysis. However, analytical testing can sometimes take longer depending on the type of testing that must be performed. If the analytical results confirm the waste is hazardous, the generator is responsible for labeling the drum properly. If the container is full, it should be turned in to the CAP within three days.

Sampling and Analysis

- Sampling Protocol. Environmental Management shall make the arrangements for sample collection as needed to determine whether a solid waste is a hazardous waste; the following guidelines apply:
  - Sampling and analysis must be conducted in accordance with the requirements specified in EPA publication SW-864, *Test Methods for Solid Waste Analysis*, and this WAP.
  - Detailed instructions regarding quality assurance/quality control (QA/QC), personal protective equipment, and sampling equipment/procedures are contained in this WAP.
  - Testing is not required for waste that is an unused commercial product or a laboratory chemical in the original, marked container.
  - Upon receipt of the analytical results, the generator will properly label the container and turn it in to the CAP. The generator will assist Environmental Management to complete the Hazardous Waste Profile Sheet (DRMS Form 1930), as needed.
- Frequency of Analysis. Waste streams must be periodically re-sampled to ensure no changes have gone unnoticed. In general, large volume waste streams (i.e., three drums or more per year) will be sampled annually. If the process and materials producing the large volume waste streams remains unchanged, this type of waste may be analyzed at a longer time interval, such as every 3 years. Smaller quantity waste streams will be sampled every third year. Other types of waste streams will be analyzed as follows:

- When a new waste stream is generated, the shop Hazardous Waste Manager will request analysis if user knowledge cannot confirm whether the waste is hazardous or not.
- When the process or chemicals generating a waste are changed, the waste stream will be re- characterized prior to the disposal of any new waste. This may include lab analysis.

Documenting the Type of Hazardous Waste

Once a waste stream is determined to be hazardous, the generating activity is responsible for documenting the properties and chemical constituents of the waste. This may be accomplished by reviewing the associated MSDSs for the source materials or by chemical analysis. Specifically, the following must be identified:

- The physical state of the waste (solid, liquid, or gas)
- The physical properties of the waste (ignitability, corrosivity, reactivity, and toxicity)
- The chemical components of the waste and their respective proportion
- Other information as required; e.g., total petroleum hydrocarbons (TPH), halogenated organic compound (HOC) content, etc.

Identified wastes will be characterized by the generator and Environmental Management into one of the following categories:

- Universal Waste
- Exempt/Excluded Waste
- Listed Hazardous Waste
- Characteristic Hazardous Waste
- Contaminated Industrial/State-regulated Waste
- Generators should use the decision trees in Hazardous Waste Satellite Accumulation Point Sign, Hazardous Waste Satellite Accumulation Point Inspection Checklist, and Example Accumulation Storage Log to determine whether a waste is hazardous and the waste's characteristics.

Recordkeeping

- The generator is responsible for maintaining copies of all documentation, including analytical results and associated SDSs or MSDSs, that are used for characterizing wastes generated by their shop regardless of whether the waste is turned into the CAP or not.
- The generator shall retain this documentation for five (5) years from the date the last of that waste is turned in to the CAP.
- The generator will provide copies of associated MSDSs to the CAP at the time the waste is turned in to the CAP.
- Environmental Management will provide a copy of the analytical results to the generator.

Quantifying the Amount of Hazardous Waste Generated

Per EPA and North Dakota regulations, generator status is based on the amount of hazardous waste generated each calendar month. Regulatory compliance becomes stricter with the changes in generator status. Therefore, generators must be thorough in characterizing and counting their wastes.

- Countable Waste
  - To determine the generator status for Minot AFB, all hazardous wastes generated on installation and at MAFs/LFs must be counted together. That is, all hazardous wastes generated within one calendar month determine the generator status for Minot AFB. The countable hazardous wastes include all listed and characteristic hazardous wastes as previously defined that satisfy any of the following conditions:
    - Accumulated on-site for any period of time prior to subsequent management
    - Packaged and/or containerized and then transported off site
    - Placed directly into a regulated on-site treatment or disposal unit
    - Generated as still bottoms or sludge and removed from product storage containers or tanks for disposal
    - Placed into satellite accumulation containers
    - Are Universal Wastes managed as hazardous wastes
    - These wastes are counted on a calendar month basis. Thus, changes in hazardous waste volumes can change the generator status for Minot AFB and the associated regulatory requirements that apply.
- Non-countable Waste
  - In determining the amount of hazardous waste generated by Minot AFB, Environmental Management does not count wastes that are:
    - Specifically exempt from counting per EPA and North Dakota regulations,
    - Left as residue in thoroughly emptied containers or in the bottom of product storage tanks,
    - Reclaimed continuously without storing prior to being reclaimed,

        - Managed in an elementary neutralization unit, totally enclosed treatment unit, or a wastewater treatment unit,
        - Discharged directly to a federally owned treatment facility without first being stored or accumulated,
        - Already counted once in a calendar month, treated on site, or reclaimed and reused, and/or
        - Managed as a Universal Waste.
    - Non-countable wastes include spent parts washing solvent that is recycled, lead- acid batteries sent off base for recycling, and spent sulfuric acid that is neutralized before discharge to the sanitary sewer.

Biennial Report

As required by the North Dakota Department of Environmental Quality, Environmental Management must complete a biennial hazardous waste report by 1 March of every even calendar year covering the preceding odd calendar year. The biennial report includes information on the quantity of hazardous waste generated as documented in hazardous waste manifests covering the reporting period.